

Nonlinear Binscatter Methods

Supplemental Appendix

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Abstract

This supplement collects all technical proofs for more general theoretical results than those reported in the main paper. Several of our new theoretical results for nonlinear partitioning-based series estimation may be of independent interest. More details on methodological aspects of nonlinear binscatter are also provided. Companion general-purpose software and replication files are available at <https://github.com/nppackages/binsreg>.

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SA-1 Introduction

This supplemental appendix is a comprehensive collection of all our new theoretical results for nonlinear binscatter estimators with semi-linear covariate-adjustment and random partitioning. Many of our results contribute to the broader literature on nonparametric estimation and inference, particularly when using series estimators, and are thus of independent interest outside of binned scatter plots. To help place our results in the literature, we include a remark labelled “Improvements over literature” at the end of each technical subsection that discusses in detail the technical improvements presented in that subsection and gives related references.

Here we give a brief summary of this appendix, including pointers to some of the major new results. We proceed as follows. The next subsection lists notation used throughout; further notation is defined throughout Section SA-2 and at the outset of Section SA-3. Then, Section SA-2 describes the setup for nonlinear binscatter methods, including the statistical model, parameters of interest, and assumptions, as well as the (random) partitioning and estimation details. Specifically, Assumption SA-DGP imposes some basic conditions on the data generating process. Assumption SA-SM imposes some technical conditions that characterize and restricts the statistical model of interest. The loss function specified there is general enough to cover many practically important examples such as mean regression, quantile regression, logit/probit estimation, and Huber regression. Assumption SA-HLE imposes some mild high-level conditions on the estimation procedure. Assumption SA-RP summarizes the key conditions on the partitioning scheme used in our theory. We allow for a large class of random partitions. Importantly, the “convergence” of the random partition (Assumption SA-RP(ii)) is not necessary for most of our main theoretical results, thereby allowing for flexible data-driven partitioning methods, including certain recursive adaptive partitioning methods: see Devroye et al. (2013), Zhang and Singer (2010), and references therein.

Section SA-3.1 presents some preliminary technical lemmas for analyzing nonlinear binscatter (and thus also partitioning-based estimators more broadly). New results include precise non-asymptotic concentration results related to Gram matrices (Lemma SA-3.1), asymptotic variances (Lemmas SA-3.2 and SA-3.3), approximation errors (Lemma SA-3.4), and uniform convergence (Lemma SA-3.5). Sharp control of these objects is a crucial ingredient for obtaining results under weak conditions, as below.

Section [SA-3.2](#) presents a tight uniform (in x) Bahadur representation for nonlinear binscatter (Theorem [SA-3.1](#)). This is our first main result. We allow for random partitions and much weaker rate restrictions (on the tuning parameter J) than previously imposed in the literature, in addition to additional controls. The data-dependent partitioning means our series estimator uses random basis functions, and this is entirely new. In terms of tuning parameter rate restrictions, when the number of additional control variables is fixed, previous results required $J^4/n \rightarrow 0$ (up to $\log(n)$ terms) or something stricter, while our main moment-driven restrictions are $\frac{J^{\frac{\nu-2}{\nu-1}} \log n}{n} = o(1)$ and $\frac{J^{\frac{2\nu}{\nu-1}} (\log n)^{\frac{\nu}{\nu-1}}}{n} = o(1)$, with $\nu > 2$ denoting the number of finite moments of the “score”; in particular, when $\nu \geq 3$, the second restriction is the binding polynomial rate and may be substantially weaker than $J^4/n \rightarrow 0$. Furthermore, our theory can allow the number of covariates to grow with the sample size, if the control coefficient estimates converge faster than the nonparametric components. We also provide primitive sufficient conditions to verify the required fast convergence rates (Lemma [SA-3.6](#)). Note that our class of models is broader than prior work in other aspects as well. Importantly, our results can now allow for piecewise constant binscatters, i.e., with degree $p = 0$, which is excluded by prior results in the literature (i.e., for previous technical results there was no sequence $J \rightarrow \infty$ such that the bias and variance are simultaneously controlled). In addition, employing our novel uniform Bahadur representation, we can establish the uniform convergence rates of nonlinear binscatter (Corollary [SA-3.1](#)) and variance estimators (Theorem [SA-3.2](#)) under similarly weak restrictions.

Section [SA-3.3](#) studies the pointwise distributional approximation for nonlinear binscatter estimators. These results are omitted from the main paper to save space, but are standard properties of interest in the nonparametrics literature and thus are included for completeness. The main result is Theorem [SA-3.3](#), which establishes pointwise asymptotic normality for our point estimators, again allowing for random (and possibly “non-convergent”) partitions, and under mild rate restrictions similar to those for the (uniform) Bahadur representation.

Section [SA-3.4](#) presents a new Nagar-type approximate IMSE expansion for nonlinear binscatter estimators with semi-linear covariate-adjustment and random partitions (Theorem [SA-3.4](#)), which has no antecedent in the literature. Our results can be used to design data-driven procedures for selecting IMSE-optimal choices of tuning parameters for nonlinear binscatter. Again, these results are novel in their breadth, the weakness of the assumptions, and the conditions on the partitioning.

Here we do require an extra assumption on the partitioning in order to characterize the leading terms in the expansion: intuitively, the random partitioning must “settle” to a population partition so that the leading constants of the expansion can be expressed. For example, sample quantiles converge to population quantiles, so this assumption is satisfied.

Uniform inference is dealt with in the next several sections of this appendix. First, Section [SA-3.5](#) establishes a uniform (in x) distributional approximation for nonlinear binscatter estimators. The two main results, which are combined into one in the main text (Theorem 2), are the (conditional) strong approximation in Theorem [SA-3.5](#) and the feasible implementation thereof in Theorem [SA-3.6](#). Again, we allow for a large class of random partitions, a broad class of (possibly) nonlinear and nonsmooth models, and additional controls. Here the partitions do not need to be “convergent” in any sense. These results are obtained under weak assumptions, including in particular mild rate restrictions, as in the case of the uniform Bahadur representation, all of which improve on the literature in various directions as explained in the text below. Finally, Theorem [SA-3.7](#) shows a distributional approximation for the suprema of the t -statistic processes in the case of the convergent partition (as in the previous paragraph).

Sections [SA-3.6–SA-3.8](#) employ the strong approximation results to study uniform inference for various parameters of interest in the specific context of nonlinear binscatter. These results rely on, and inherit the novelty of, Theorems [SA-3.5](#) and [SA-3.6](#). New results include valid uniform confidence bands (Theorem [SA-3.8](#)), consistent hypothesis tests about parametric specification (Theorem [SA-3.9](#)) and tests for shape restrictions (Theorem [SA-3.10](#)). All these results explicitly account for the possibly random partitioning scheme and semi-linear covariate-adjustment with random evaluation points.

Section [SA-4](#) discusses implementation details for nonlinear binscatter, including standard error computation, feasible data-driven number of bins selector, and choices of polynomial orders given the number of bins. For a more explicit treatment of the package `binsreg` per se, see [Cattaneo et al. \(2025\)](#) and <https://github.com/nppackages/binsreg>.

Finally, Section [SA-5](#) contains the proofs for all the technical results in Section [SA-3](#).

SA-1.1 Notation

See [van der Vaart and Wellner \(1996\)](#), [Bhatia \(2013\)](#), [Giné and Nickl \(2016\)](#), and references therein, for background definitions.

Matrices and Norms. For (column) vectors, $\|\cdot\|$ denotes the Euclidean norm, $\|\cdot\|_1$ denotes the L_1 norm, $\|\cdot\|_\infty$ denotes the sup-norm, and $\|\cdot\|_0$ denotes the number of nonzeros. For matrices, $\|\cdot\|$ is the operator matrix norm induced by the L_2 norm, and $\|\cdot\|_\infty$ is the matrix norm induced by the supremum norm, i.e., the maximum absolute row sum of a matrix. For a square matrix $\mathbf{A}$, $\lambda_{\max}(\mathbf{A})$ and $\lambda_{\min}(\mathbf{A})$ are the maximum and minimum eigenvalues of $\mathbf{A}$, respectively. $[\mathbf{A}]_{ij}$ denotes the (i, j) th entry of a generic matrix $\mathbf{A}$. We will use $\mathcal{S}^L$ to denote the unit sphere in $\mathbb{R}^L$, i.e., $\|\mathbf{a}\| = 1$ for any $\mathbf{a} \in \mathcal{S}^L$. For a real-valued function $g(\cdot)$ defined on a measure space $\mathcal{Z}$, let $\|g\|_{\mathbb{Q},2} := (\int_{\mathcal{Z}} |g|^2 d\mathbb{Q})^{1/2}$ be its L_2 -norm with respect to the measure $\mathbb{Q}$. In addition, let $\|g\|_\infty = \sup_{z \in \mathcal{Z}} |g(z)|$ be L_∞ -norm of $g(\cdot)$, and if g is a univariate function, let $g^{(v)}(z) = d^v g(z)/dz^v$ be the v th derivative for $v \geq 0$.

Asymptotics. For sequences of numbers or random variables, we use $l_n \lesssim m_n$ to denote that $\limsup_n |l_n/m_n|$ is finite, $l_n \lesssim_{\mathbb{P}} m_n$ or $l_n = O_{\mathbb{P}}(m_n)$ to denote $\limsup_{\varepsilon \rightarrow \infty} \limsup_n \mathbb{P}[|l_n/m_n| \geq \varepsilon] = 0$, $l_n = o(m_n)$ implies $l_n/m_n \rightarrow 0$, and $l_n = o_{\mathbb{P}}(m_n)$ implies that $l_n/m_n \rightarrow_{\mathbb{P}} 0$, where $\rightarrow_{\mathbb{P}}$ denotes convergence in probability. Accordingly, we write $l_n \gtrsim m_n$ if $m_n \lesssim l_n$, and $l_n \gtrsim_{\mathbb{P}} m_n$ if $m_n \lesssim_{\mathbb{P}} l_n$. $l_n \asymp m_n$ implies that $l_n \lesssim m_n$ and $m_n \lesssim l_n$.

Empirical Process. We employ standard empirical process notation: $\mathbb{E}_n[g(\mathbf{v}_i)] = \frac{1}{n} \sum_{i=1}^n g(\mathbf{v}_i)$, and $\mathbb{G}_n[g(\mathbf{v}_i)] = \frac{1}{\sqrt{n}} \sum_{i=1}^n (g(\mathbf{v}_i) - \mathbb{E}[g(\mathbf{v}_i)])$ for a sequence of random variables $\{\mathbf{v}_i\}_{i=1}^n$. In addition, we employ the notion of covering number extensively in the proofs. Specifically, given a measurable space $(A, \mathcal{A})$ and a suitably measurable class of functions $\mathcal{G}$ mapping A to $\mathbb{R}$ equipped with a measurable envelope function $\bar{G}(z) \geq \sup_{g \in \mathcal{G}} |g(z)|$, the *covering number* $N(\mathcal{G}, L_2(\mathbb{Q}), \varepsilon)$ is the minimal number of $L_2(\mathbb{Q})$ -balls of radius ε needed to cover $\mathcal{G}$ for a measure $\mathbb{Q}$. The covering number of $\mathcal{G}$ relative to the envelope is denoted as $N(\mathcal{G}, L_2(\mathbb{Q}), \varepsilon \|\bar{G}\|_{\mathbb{Q},2})$.

Other. $\lceil z \rceil$ outputs the smallest integer no less than z and $a \wedge b = \min\{a, b\}$. “w.p.a. 1” means “with probability approaching one”.

SA-2 Setup

For each n , suppose that $\{(y_i, x_i, \mathbf{w}_i') : 1 \leq i \leq n\}$ is i.i.d. across i , where $y_i \in \mathcal{Y}$ is a scalar response variable, $x_i \in \mathcal{X}$ is a scalar covariate, and $\mathbf{w}_i \in \mathcal{W}_n \subset \mathbb{R}^{d_\gamma}$ is a vector of additional control variables, where d_γ may increase with n . Population objects may depend on n , with this dependence suppressed. To simplify notation, we write $\mathcal{W} = \mathcal{W}_n$ below. Let $\mathbf{D} = [(y_i, x_i, \mathbf{w}_i')' : i = 1, 2, \dots, n]$.

For a loss function $\rho(\cdot; \cdot)$ and a strictly monotonic transformation function $\eta(\cdot)$, define

$$(\mu_0(\cdot), \gamma_0) = \arg \min_{\mu \in \mathcal{M}, \gamma \in \mathbb{R}^{d_\gamma}} \mathbb{E} \left[\rho \left(y_i; \eta(\mu(x_i) + \mathbf{w}_i' \gamma) \right) \right], \quad (\text{SA-2.1})$$

where $\mathcal{M}$ is a space of functions satisfying certain smoothness conditions to be specified later.

Equation (SA-2.1) assumes that the minimizers $\mu_0(\cdot)$ and γ_0 are unique. For example, this holds when the index $\theta_0(x, \mathbf{w}) = \mu_0(x) + \mathbf{w}' \gamma_0$ is the unique solution to the population optimization criterion, almost surely, and

$$c_w \mathbf{I}_{d_\gamma} \leq \mathbb{E} \left[\{\mathbf{w}_i - \mathbb{E}[\mathbf{w}_i \mid x_i]\} \{\mathbf{w}_i - \mathbb{E}[\mathbf{w}_i \mid x_i]\}' \mid x_i = x \right] \leq C_w \mathbf{I}_{d_\gamma},$$

for some $0 < c_w < C_w < \infty$, where the intercept is absorbed into $\mu_0(\cdot)$, and therefore $\mathbf{w}_i$ does not contain a constant. These conditions rule out any nonzero vector $\mathbf{a}$ such that $\mathbf{w}_i' \mathbf{a}$ is a function of x_i almost surely and, therefore, identify $\mu_0(\cdot)$ and γ_0 separately.

This setup is general. For example, consider $\gamma_0 = \mathbf{0}$. If $\rho(\cdot; \cdot)$ is a squared loss and $\eta(\cdot)$ is the identity function, $\mu_0(x)$ is the conditional expectation of y_i given $x_i = x$. Let $\mathbb{1}(\cdot)$ denote the indicator function. If $\rho(y; \eta) = (q - \mathbb{1}(y < \eta))(y - \eta)$ for some $0 < q < 1$ and $\eta(\cdot)$ is an identity function, then $\mu_0(x)$ is the q th conditional quantile of y_i given $x_i = x$. Introducing a transformation function $\eta(\cdot)$ is useful. For instance, it may accommodate logistic regression for binary responses. When $\gamma_0 \neq \mathbf{0}$, the parametric and the nonparametric components are additively separable, and thus (SA-2.1) becomes a generalized partially linear model.

Binscatter estimators are typically constructed based on a (possibly random) partition of the support of the covariate x_i . Specifically, the relevant support of x_i is partitioned into J disjoint

intervals, leading to the partitioning scheme $\widehat{\Delta} = \{\widehat{\mathcal{B}}_1, \widehat{\mathcal{B}}_2, \dots, \widehat{\mathcal{B}}_J\}$, where

$$\widehat{\mathcal{B}}_j = \begin{cases} [\widehat{\tau}_{j-1}, \widehat{\tau}_j) & \text{if } j = 1, \dots, J-1 \\ [\widehat{\tau}_{J-1}, \widehat{\tau}_J] & \text{if } j = J \end{cases},$$

One popular choice in binscatter applications is the quantile-based partition: $\widehat{\tau}_0 = \inf \mathcal{X}$, $\widehat{\tau}_J = \sup \mathcal{X}$, and $\widehat{\tau}_j = \widehat{F}_X^{-1}(j/J)$ for $j = 1, \dots, J-1$, with $\widehat{F}_X(u) = n^{-1} \sum_{i=1}^n \mathbb{1}(x_i \leq u)$ the empirical cumulative distribution function and $\widehat{F}_X^{-1}$ its generalized inverse. Our theory is general enough to cover other partitioning schemes satisfying certain regularity conditions specified below. An innovation herein is accounting for the additional randomness from the partition $\widehat{\Delta}$. The number of bins J plays the role of the tuning parameter for the binscatter method, and is assumed to diverge: $J \rightarrow \infty$ as $n \rightarrow \infty$ throughout the supplement, unless explicitly stated otherwise.

The piecewise polynomial basis of degree p , for some choice of $p = 0, 1, 2, \dots$, is defined as

$$\begin{bmatrix} \mathbb{1}_{\widehat{\mathcal{B}}_1}(x) & \mathbb{1}_{\widehat{\mathcal{B}}_2}(x) & \cdots & \mathbb{1}_{\widehat{\mathcal{B}}_J}(x) \end{bmatrix}' \otimes \begin{bmatrix} 1 & x & \cdots & x^p \end{bmatrix}',$$

where $\mathbb{1}_{\mathcal{A}}(x) = \mathbb{1}(x \in \mathcal{A})$ and $\otimes$ is the Kronecker product operator. For convenience of later analysis, we use $\widehat{\mathbf{b}}_{p,0}(x)$ to denote a *standardized rotated* basis, the j th element of which is given by

$$\sqrt{J} \times \mathbb{1}_{\widehat{\mathcal{B}}_j}(x) \times \left(\frac{x - \widehat{\tau}_{\bar{j}-1}}{\widehat{h}_{\bar{j}}} \right)^{j-1-(\bar{j}-1)(p+1)}, \quad j = 1, \dots, (p+1)J,$$

where $\bar{j} = \lceil j/(p+1) \rceil$, $\lceil \cdot \rceil$ is the ceiling operator, and $\widehat{h}_{\bar{j}} = \widehat{\tau}_{\bar{j}} - \widehat{\tau}_{\bar{j}-1}$. Thus, each local polynomial is centered at the start of each bin and scaled by the length of the bin. $\sqrt{J}$ is an additional scaling factor which helps simplify some expressions of our results. The standardized rotated basis $\widehat{\mathbf{b}}_{p,0}(x)$ is equivalent to the original piecewise polynomial basis in the sense that they represent the same (linear) function space.

To impose the restriction that the estimated function is $(s-1)$ -times continuously differentiable for $1 \leq s \leq p$, we introduce the following basis

$$\widehat{\mathbf{b}}_{p,s}(x) = \left(\widehat{b}_{p,s,1}(x), \dots, \widehat{b}_{p,s,K_{p,s}}(x) \right)' = \widehat{\mathbf{T}}_s \widehat{\mathbf{b}}_{p,0}(x), \quad K_{p,s} = (p+1)J - s(J-1),$$

where $\widehat{\mathbf{T}}_s := \widehat{\mathbf{T}}_s(\widehat{\Delta})$ is a $K_{p,s} \times (p+1)J$ matrix depending on $\widehat{\Delta}$, which transforms a piecewise polynomial basis into a smoothed binscatter basis. Some useful properties of $\widehat{\mathbf{T}}_s$ are given in Lemma SA-5.3 in Section SA-5, and the explicit representation of $\widehat{\mathbf{T}}_s$ is available in the proof of Lemma SA-3.2 in Cattaneo, Crump, Farrell and Feng (2024a). When $s = 0$, we let $\widehat{\mathbf{T}}_0 = \mathbf{I}_{(p+1)J}$, the identity matrix of dimension $(p+1)J$. When $s = p$, $\widehat{\mathbf{b}}_{p,s}(x)$ is the well-known B -spline basis of order $p+1$ with simple knots, which is $(p-1)$ -times continuously differentiable. When $0 < s < p$, they can be defined similarly as B -splines with knots of certain multiplicities. See Definition 4.1 in Section 4 of Schumaker (2007) for more details about spline functions. We require $s \leq p$, since if $s = p+1$, $\widehat{\mathbf{b}}_{p,s}(x)$ reduces to a global polynomial basis of degree p .

Given a choice of basis, we consider the following generalized binscatter estimator:

$$\widehat{\mu}_{p,s}^{(v)}(x) := \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\boldsymbol{\beta}}, \quad \begin{bmatrix} \widehat{\boldsymbol{\beta}} \\ \widehat{\boldsymbol{\gamma}} \end{bmatrix} = \arg \min_{\boldsymbol{\beta}, \boldsymbol{\gamma}} \sum_{i=1}^n \rho\left(y_i; \eta(\widehat{\mathbf{b}}_{p,s}(x_i)' \boldsymbol{\beta} + \mathbf{w}_i' \boldsymbol{\gamma})\right), \quad (\text{SA-2.2})$$

where $\widehat{\mathbf{b}}_{p,s}^{(v)}(x) = \frac{d^v}{dx^v} \widehat{\mathbf{b}}_{p,s}(x)$ for some $v \in \mathbb{Z}_+$ such that $v \leq p$. At interior partition knots, derivatives of the basis and resulting estimators are interpreted as right derivatives, while at the endpoints of $\mathcal{X}$ they are interpreted as inward derivatives; using left derivatives at interior knots instead leaves all results unchanged. This estimator can be written as:

$$\widehat{\mu}_{p,s}^{(v)}(x) = \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\boldsymbol{\beta}}, \quad \widehat{\boldsymbol{\beta}} := \widehat{\boldsymbol{\beta}}(\widehat{\boldsymbol{\gamma}}) := \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^{K_{p,s}}} \sum_{i=1}^n \rho\left(y_i; \eta(\widehat{\mathbf{b}}_{p,s}(x_i)' \boldsymbol{\beta} + \mathbf{w}_i' \widehat{\boldsymbol{\gamma}})\right). \quad (\text{SA-2.3})$$

Throughout, each displayed optimization problem is understood to admit a finite solution w.p.a. 1, with any measurable solution used when it is not unique. We take the representation (SA-2.3) as the starting point for theoretical analysis, which allows us to be more general and agnostic about the estimation of γ_0 , and also simplifies some of the proofs. More specifically, our theory requires only a sufficiently fast convergence rate of $\widehat{\boldsymbol{\gamma}}$ (see Assumption SA-HLE below), which, in nonlinear estimation models, can be justified in different ways, e.g., joint estimation, backfitting, profiling, and split-sampling, among other possibilities. For the special case of the joint estimator (SA-2.2), Lemma SA-3.6 below verifies the fast convergence of $\widehat{\boldsymbol{\gamma}}$ under intuitive primitive conditions, which allows d_γ to possibly grow. Our software implementation (Cattaneo, Crump, Farrell and Feng, 2025) relies on joint estimation, as done by the default base estimation packages in Python, R, and

Stata.

In this supplement, we focus on estimation and inference of the following three parameters:

- (i) the nonparametric component $\mu_0^{(v)}(x)$ for any $v \geq 0$,
- (ii) the level function $\vartheta_0(x, \mathbf{w}) = \eta(\mu_0(x) + \mathbf{w}'\gamma_0)$, and
- (iii) the marginal effect $\zeta_0(x, \mathbf{w}) = \frac{\partial}{\partial x}\eta(\mu_0(x) + \mathbf{w}'\gamma_0)$,

where $\mathbf{w} = \mathbf{w}_n \in \mathcal{W}$ is a user-chosen evaluation point of the control variables, and thus these parameters are viewed as functions of x only in our theory. Nevertheless, all our results are readily applied to other linear or nonlinear transformations of $\mu_0(x)$, such as the higher-order derivatives $\frac{\partial^v}{\partial x^v}\eta(\mu_0(x) + \mathbf{w}'\gamma_0)$. Given the binscatter estimates $\hat{\mu}_{p,s}(x)$ and $\hat{\gamma}$ in (SA-2.2), the estimators of the three parameters defined above are given by

$$\hat{\mu}_{p,s}^{(v)}(x), \quad \hat{\vartheta}_{p,s}(x, \hat{\mathbf{w}}) = \eta(\hat{\mu}_{p,s}(x) + \hat{\mathbf{w}}'\hat{\gamma}), \quad \text{and} \quad \hat{\zeta}_{p,s}(x, \hat{\mathbf{w}}) = \eta^{(1)}(\hat{\mu}_{p,s}(x) + \hat{\mathbf{w}}'\hat{\gamma})\hat{\mu}_{p,s}^{(1)}(x)$$

respectively, for some consistent estimate $\hat{\mathbf{w}}$ (non-random or generated based on $\{\mathbf{w}_i\}_{i=1}^n$) of the evaluation point $\mathbf{w}$. As a reminder, we need to require $p \geq v$ to get $\hat{\mu}_{p,s}^{(v)}(x)$, $p \geq 0$ to get $\hat{\vartheta}_{p,s}(x, \hat{\mathbf{w}})$, and $p \geq 1$ to get $\hat{\zeta}_{p,s}(x, \hat{\mathbf{w}})$.

Recall that in the main text we always set $s = p$ and omit the dependence of estimators on s . Thus, $\hat{\mu}_p^{(v)}(x) = \hat{\mu}_{p,p}^{(v)}(x)$, $\hat{\vartheta}_p(x, \hat{\mathbf{w}}) = \hat{\vartheta}_{p,p}(x, \hat{\mathbf{w}})$, and $\hat{\zeta}_p(x, \hat{\mathbf{w}}) = \hat{\zeta}_{p,p}(x, \hat{\mathbf{w}})$. In this supplement, however, all our results hold for a general choice of the degree and the smoothness of the basis. For ease of notation, the subscripts p and s of the above estimators are dropped hereafter:

$$\hat{\mu}^{(v)}(x) := \hat{\mu}_{p,s}^{(v)}(x), \quad \hat{\vartheta}(x, \hat{\mathbf{w}}) := \hat{\vartheta}_{p,s}(x, \hat{\mathbf{w}}), \quad \text{and} \quad \hat{\zeta}(x, \hat{\mathbf{w}}) := \hat{\zeta}_{p,s}(x, \hat{\mathbf{w}}).$$

Remark SA-2.1 (Smoothness and Bias Correction). This supplemental appendix presents *all* results under general choices of the number of bins J , the degree of the basis p , and the smoothness of the basis s . By contrast, for simplicity, the main paper employs the basis with the maximum smoothness, i.e. choosing $s = p$, and considers the special case in which J is taken to be the IMSE-optimal choice for a fixed p (see Theorem SA-3.4), and inference is conducted based on the binscatter basis of degree $(p + 1)$. Such a choice of J guarantees that the smoothing bias of the

binscatter estimator is negligible in inference under mild conditions and thus can be viewed as a bias correction strategy. $\lrcorner$

Throughout, n -dependent quantities (e.g., $\mathbf{w}_i$, γ_0 , and $\widehat{\mathbf{b}}_{p,s}$) are written without an n subscript for notational simplicity, whereas generic constants and upper and lower bounds are independent of n , unless explicitly stated otherwise. We first assume the following basic conditions on the data generating process.

Assumption SA-DGP (Data Generating Process).

- (i) For each n , $\{(y_i, x_i, \mathbf{w}'_i) : 1 \leq i \leq n\}$ are i.i.d. across i , satisfy (SA-2.1), and are supported on $\mathcal{Y} \times \mathcal{X} \times \mathcal{W}$, where $\mathcal{X}$ is a compact interval, $\mathcal{W} \subset \mathbb{R}^{d_\gamma}$ is compact for each n , and $\sup_n \sup_{\mathbf{w} \in \mathcal{W}} \|\mathbf{w}\|_\infty < \infty$.
- (ii) $F_X(x) := \mathbb{P}[x_i \leq x]$ has a Lipschitz continuous (Lebesgue) density $f_X(x)$ bounded away from zero on $\mathcal{X}$.
- (iii) Let $F_{Y|XW}(y|x, \mathbf{w}) := \mathbb{P}[y_i \leq y | x_i = x, \mathbf{w}_i = \mathbf{w}]$. If this conditional distribution is absolutely continuous with respect to Lebesgue measure, its conditional density $f_{Y|XW}(y|x, \mathbf{w})$ is uniformly bounded over its support and $x \in \mathcal{X}$, $\mathbf{w} \in \mathcal{W}$.

Next, we impose several technical conditions related to the statistical model of interest.

Assumption SA-SM (Statistical Model).

- (i) For some $\delta > 0$, there exists a compact interval $\mathcal{I}$ not depending on n and contained in the interior of the domain of η such that $[\theta_0(x, \mathbf{w}) - \delta, \theta_0(x, \mathbf{w}) + \delta] \subseteq \mathcal{I}$ uniformly over n , $x \in \mathcal{X}$, and $\mathbf{w} \in \mathcal{W}$. $\rho(y; \cdot)$ is absolutely continuous on an open set containing $\eta(\mathcal{I})$ and admits a derivative $\psi(y, \eta) := \psi^\dagger(y - \eta)\psi^\ddagger(\eta)$ almost everywhere on this set. $\psi^\ddagger(\cdot)$ is continuously differentiable and strictly positive or negative on this set. $\psi^\dagger(\cdot)$ is Lipschitz continuous if $F_{Y|XW}(y|x_i, \mathbf{w}_i)$ does not have a Lebesgue density, or piecewise Lipschitz with finitely many discontinuity points otherwise.
- (ii) $\rho(y; \eta(\theta))$ is convex with respect to θ . $\eta(\cdot)$ is strictly monotonic and three-times continuously differentiable.

(iii) $\mathbb{E}[\psi(y_i, \eta(\mu_0(x_i) + \mathbf{w}'_i \gamma_0)) | x_i, \mathbf{w}_i] = 0$. For $\sigma^2(x, \mathbf{w}) := \mathbb{E}[\psi(y_i, \eta(\mu_0(x_i) + \mathbf{w}'_i \gamma_0))^2 | x_i = x, \mathbf{w}_i = \mathbf{w}]$, $\inf_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \sigma^2(x, \mathbf{w}) \gtrsim 1$. $\mathbb{E}[\eta^{(1)}(\mu_0(x_i) + \mathbf{w}'_i \gamma_0)^2 \sigma^2(x_i, \mathbf{w}_i) | x_i = x]$ is Lipschitz continuous on $\mathcal{X}$, and $\sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \mathbb{E}[|\psi(y_i, \eta(\mu_0(x_i) + \mathbf{w}'_i \gamma_0))|^\nu | x_i = x, \mathbf{w}_i = \mathbf{w}] \lesssim 1$ for some $\nu > 2$. Let $\Psi(x, \mathbf{w}; \tau) := \mathbb{E}[\psi(y_i, \tau) | x_i = x, \mathbf{w}_i = \mathbf{w}]$. For every n and every $(x, \mathbf{w}) \in \mathcal{X} \times \mathcal{W}$, the map $\tau \mapsto \Psi(x, \mathbf{w}; \tau)$ is twice continuously differentiable on $\eta(\mathcal{I})$. For $j \in \{1, 2\}$, define $\Psi_j(x, \mathbf{w}; \tau) := \frac{\partial^j}{\partial \tau^j} \Psi(x, \mathbf{w}; \tau)$ and assume

$$\sup_{n \geq 1} \sup_{(x, \mathbf{w}, \tau) \in \mathcal{X} \times \mathcal{W} \times \eta(\mathcal{I})} |\Psi_j(x, \mathbf{w}; \tau)| \lesssim 1, \quad j \in \{1, 2\}.$$

(iv) $\inf_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \kappa(x, \mathbf{w}) \gtrsim 1$ and $\mathbb{E}[\kappa(x_i, \mathbf{w}_i) | x_i = x]$ is Lipschitz continuous on $\mathcal{X}$ where $\kappa(x, \mathbf{w}) := \Psi_1(x, \mathbf{w}; \eta(\mu_0(x) + \mathbf{w}' \gamma_0))(\eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0))^2$.

(v) $\mu_0(\cdot)$ is ς -times continuously differentiable for some $\varsigma \geq p + 1$, and

$$\sup_{n \geq 1} \sup_{x \in \mathcal{X}} |\mu_0^{(p+1)}(x)| \lesssim 1.$$

Our next assumption imposes mild high-level conditions on the estimator $\hat{\gamma}$ of the coefficient vector γ_0 , the estimator $\hat{\mathbf{w}}$ of the evaluation point $\mathbf{w}$ for control variables, and the estimator of the function Ψ_1 defined previously in Assumption SA-SM(iv).

Assumption SA-HLE (High-Level Estimation Conditions).

(i) $\|\hat{\gamma} - \gamma_0\| \lesssim_{\mathbb{P}} \mathbf{r}_\gamma$ for a deterministic sequence $\mathbf{r}_\gamma$ satisfying $\sqrt{d_\gamma} \mathbf{r}_\gamma = o(\sqrt{J/n} + J^{-p-1})$. Moreover, $\|\hat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(1)$, $\mathbb{P}[\hat{\mathbf{w}} \in \mathcal{W}] \rightarrow 1$, and $\sup_n \|\gamma_0\| < \infty$.

(ii) For some estimator $\hat{\Psi}_1$ of Ψ_1 , $\|\mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \hat{\mathbf{b}}_{p,s}(x_i)' (\hat{\kappa}(x_i, \mathbf{w}_i) - \kappa(x_i, \mathbf{w}_i))] \| \lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-2/\nu}}\right)^{1/2}$ where $\hat{\kappa}(x_i, \mathbf{w}_i) = \hat{\Psi}_1(x_i, \mathbf{w}_i; \eta(\hat{\mu}(x_i) + \mathbf{w}'_i \hat{\gamma}))(\eta^{(1)}(\hat{\mu}(x_i) + \mathbf{w}'_i \hat{\gamma}))^2$.

Note that $\Upsilon(x, \mathbf{w}) = \Psi_1(x, \mathbf{w}; \eta(\mu_0(x) + \mathbf{w}' \gamma_0))$ in the main paper to streamline the presentation. Part (i) is a high-level condition on the convergence of $\hat{\gamma}$ and $\hat{\mathbf{w}}$. The factor $\sqrt{d_\gamma}$ converts the Euclidean coefficient rate into a uniform fitted-index rate under the coordinatewise support bound in Assumption SA-DGP(i). Part (ii) is a high-level condition that ensures we have a valid feasible estimator of the Gram matrix ($\bar{\mathbf{Q}}$ or $\mathbf{Q}_0$ defined at the outset of Section SA-3 below). Note that the

convergence rate of $\eta^{(1)}(\hat{\mu}(x_i) + \mathbf{w}_i' \hat{\gamma})$ can be deduced from Corollary SA-3.1 below, which requires only part (i) of this assumption. Thus, part (ii) can be largely viewed as a restriction on $\hat{\Psi}_1$ only. Note that $\hat{\Psi}_1$ does not have to be consistent for Ψ_1 in any sense; it suffices that the estimator $\mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \hat{\mathbf{b}}_{p,s}(x_i)' \hat{\mathcal{Z}}(x_i, \mathbf{w}_i)]$ based on $\hat{\Psi}_1$ as a whole is consistent. See Section SA-4 for several examples of the estimator $\hat{\Psi}_1$.

SA-2.1 Partitions

We need some regularity conditions on the partitioning scheme, which can be verified on a case-by-case basis. We first define a family of “quasi-uniform” partitions for some absolute constant $C > 0$:

$$\Pi_C = \left\{ \Delta : \frac{\max_{1 \leq j \leq J} h_j(\Delta)}{\min_{1 \leq j \leq J} h_j(\Delta)} \leq C \right\}, \quad (\text{SA-2.4})$$

where $h_j(\Delta)$ denotes the length of the j th bin in the partition Δ . Roughly speaking, (SA-2.4) says that the bins in any $\Delta \in \Pi_C$ do not differ too much in length. Also, let $\mathbf{X} = [x_1, \dots, x_n]'$, $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_n]'$ and $\mathbf{Y} = [y_1, \dots, y_n]'$.

Assumption SA-RP (Random Partition).

- (i) $\hat{\Delta} \perp\!\!\!\perp \mathbf{Y} | (\mathbf{X}, \mathbf{W})$ and $\hat{\Delta} \in \Pi_C$ w.p.a. 1 for some absolute constant $C > 0$.
- (ii) There exists a non-random partition of $\mathcal{X}$, $\Delta_0 = \{\mathcal{B}_1, \dots, \mathcal{B}_J\}$ with $\mathcal{B}_j = [\tau_{j-1}, \tau_j)$ for $j \leq J-1$ and $\mathcal{B}_J = [\tau_{J-1}, \tau_J]$ such that $\frac{\max_{1 \leq j \leq J} h_j}{\min_{1 \leq j \leq J} h_j} \leq c_{\text{QU}}$ for some absolute constant $c_{\text{QU}} > 0$, and $|\hat{\tau}_0 - \tau_0| \lesssim_{\mathbb{P}} \mathbf{r}_{\text{RP}}$ and $\max_{1 \leq j \leq J} |\hat{h}_j - h_j| \lesssim_{\mathbb{P}} J^{-1} \mathbf{r}_{\text{RP}}$ for $\mathbf{r}_{\text{RP}} = o(1)$.

Part (i) is the key condition for our main results and will be imposed throughout. First, it requires that the possibly random partition $\hat{\Delta}$ be independent of the outcome $\mathbf{Y}$ given the covariates $(\mathbf{X}, \mathbf{W})$. This conditional independence assumption is trivially satisfied if $\hat{\Delta}$ is deterministic (e.g., equally-spaced partition) or depends on $\mathbf{X}$ and $\mathbf{W}$ only (e.g., quantile-spaced partition based on $\mathbf{X}$). It also holds if a sample splitting scheme is used: a subsample (including the information about the outcome) is used for constructing the partition, and the other is employed to construct the binscatter estimator (so that $\hat{\Delta}$ is independent of the data $(\mathbf{X}, \mathbf{W}, \mathbf{Y})$). Second, $\hat{\Delta}$ is required to be “quasi-uniform” with large probability. It is trivially true for equally-spaced partitions and can be verified for quantile-spaced partitions under the mild conditions on the covariate density

imposed before (see Lemma SA-5.2). However, this condition may be too restrictive for other modern machine-learning-based partitioning methods, in which case some additional regularization may be necessary to recover the quasi-uniformity property.

Part (ii) requires that the random partition $\hat{\Delta}$ “stabilizes” to a fixed one in large samples. This is true if the partition is deterministic or generated by sample quantiles (since sample quantiles converge to population quantiles), but more generally, it is not always possible. Fortunately, this “convergence” requirement is not necessary for most of our main results (except Theorem SA-3.4 and Theorem SA-3.7). Thus, we will always make it clear if part (ii) of Assumption SA-RP is imposed.

Given the random partition $\hat{\Delta}$, $\mathbb{E}_{\hat{\Delta}}[\cdot]$ denotes unconditional population expectation with $\hat{\Delta}$ held fixed (this is not a conditional expectation given $\hat{\Delta}$). We use $\mathcal{F}_{XW\Delta}$ to denote the σ -field generated by $(\mathbf{X}, \mathbf{W}, \hat{\Delta})$. To further simplify notation, let $\hat{h}_j = \hat{\tau}_j - \hat{\tau}_{j-1}$ be the width of the j th bin $\hat{\mathcal{B}}_j$, and when the “limiting” partition $\Delta_0 = \{\mathcal{B}_1, \dots, \mathcal{B}_J\}$ is defined (Assumption SA-RP(ii) holds), let h_j be the width of $\mathcal{B}_j$. Analogously to $\hat{\mathbf{b}}_{p,s}(x)$, $\mathbf{b}_{p,s}(x)$ denotes the binscatter basis of degree p that is $(s-1)$ -times continuously differentiable and is constructed based on the *nonrandom* partition Δ_0 . We sometimes write $\mathbf{b}_{p,s}(x; \Delta) = (b_{p,s,1}(x; \Delta), \dots, b_{p,s,K_{p,s}}(x; \Delta))'$ to emphasize a binscatter basis is constructed based on a particular partition Δ . Therefore, $\hat{\mathbf{b}}_{p,s}(x) = \mathbf{b}_{p,s}(x; \hat{\Delta})$ and $\mathbf{b}_{p,s}(x) = \mathbf{b}_{p,s}(x; \Delta_0)$. Accordingly, we use $\mathbf{T}_s$ to denote the transformation matrix based on the non-random partition Δ_0 (which transforms $\mathbf{b}_{p,0}(x)$ to $\mathbf{b}_{p,s}(x)$).

SA-3 Main Results

We introduce the following quantities that will be extensively used throughout the supplement:

$$\begin{aligned} \eta_i &= \eta(\mu_0(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_0), & \hat{\eta}_i &= \eta(\hat{\mu}(x_i) + \mathbf{w}_i' \hat{\boldsymbol{\gamma}}), \\ \eta_{i,1} &= \eta^{(1)}(\mu_0(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_0), & \hat{\eta}_{i,1} &= \eta^{(1)}(\hat{\mu}(x_i) + \mathbf{w}_i' \hat{\boldsymbol{\gamma}}), \\ \eta_{0,1}(x, \mathbf{w}) &= \eta^{(1)}(\mu_0(x) + \mathbf{w}' \boldsymbol{\gamma}_0), & \hat{\eta}_{0,1}(x, \hat{\mathbf{w}}) &= \eta^{(1)}(\hat{\mu}(x) + \hat{\mathbf{w}}' \hat{\boldsymbol{\gamma}}), \\ \hat{\mu}(x_i) &= \hat{\mathbf{b}}_{p,s}(x_i)' \hat{\boldsymbol{\beta}}, & \epsilon_i &= y_i - \eta_i, & \hat{\epsilon}_i &= y_i - \hat{\eta}_i, \\ \hat{\mathbf{Q}}_{p,s} &:= \hat{\mathbf{Q}}_{p,s}(\hat{\Delta}) := \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \hat{\mathbf{b}}_{p,s}(x_i)' \hat{\Psi}_1(x_i, \mathbf{w}_i; \hat{\eta}_i) \hat{\eta}_{i,1}^2], \end{aligned}$$

$$\begin{aligned}
\bar{\mathbf{Q}}_{p,s} &:= \bar{\mathbf{Q}}_{p,s}(\hat{\Delta}) := \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i)\hat{\mathbf{b}}_{p,s}(x_i)'\Psi_1(x_i, \mathbf{w}_i; \eta_i)\eta_{i,1}^2], \\
\mathbf{Q}_{0,p,s} &:= \mathbf{Q}_{p,s}(\Delta_0) := \mathbb{E}[\mathbf{b}_{p,s}(x_i)\mathbf{b}_{p,s}(x_i)'\Psi_1(x_i, \mathbf{w}_i; \eta_i)\eta_{i,1}^2], \\
\hat{\Sigma}_{p,s} &:= \hat{\Sigma}_{p,s}(\hat{\Delta}) := \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i)\hat{\mathbf{b}}_{p,s}(x_i)'\psi(y_i, \hat{\eta}_i)^2\hat{\eta}_{i,1}^2], \\
\bar{\Sigma}_{p,s} &:= \bar{\Sigma}_{p,s}(\hat{\Delta}) := \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i)\hat{\mathbf{b}}_{p,s}(x_i)'\sigma^2(x_i, \mathbf{w}_i)\eta_{i,1}^2], \\
\Sigma_{0,p,s} &:= \Sigma_{p,s}(\Delta_0) := \mathbb{E}[\mathbf{b}_{p,s}(x_i)\mathbf{b}_{p,s}(x_i)'\psi(y_i, \eta_i)^2\eta_{i,1}^2], \\
\hat{\Omega}_{\mu^{(v)},p,s}(x) &:= \hat{\Omega}_{\mu^{(v)},p,s}(x; \hat{\Delta}) := \hat{\mathbf{b}}_{p,s}^{(v)}(x)'\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\Sigma}_{p,s}\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}^{(v)}(x), \\
\bar{\Omega}_{\mu^{(v)},p,s}(x) &:= \bar{\Omega}_{\mu^{(v)},p,s}(x; \hat{\Delta}) := \hat{\mathbf{b}}_{p,s}^{(v)}(x)'\bar{\mathbf{Q}}_{p,s}^{-1}\bar{\Sigma}_{p,s}\bar{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}^{(v)}(x), \\
\Omega_{\mu^{(v)},p,s}(x) &:= \Omega_{\mu^{(v)},p,s}(x; \hat{\Delta}) := \hat{\mathbf{b}}_{p,0}^{(v)}(x)'\mathbf{T}'_s\mathbf{Q}_{0,p,s}^{-1}\Sigma_{0,p,s}\mathbf{Q}_{0,p,s}^{-1}\mathbf{T}_s\hat{\mathbf{b}}_{p,0}^{(v)}(x), \\
\hat{\Omega}_{\vartheta,p,s}(x) &:= \hat{\Omega}_{\vartheta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\hat{\mu}(x) + \hat{\mathbf{w}}'\hat{\gamma})]^2\hat{\mathbf{b}}_{p,s}(x)'\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\Sigma}_{p,s}\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}(x), \\
\bar{\Omega}_{\vartheta,p,s}(x) &:= \bar{\Omega}_{\vartheta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\mu_0(x) + \mathbf{w}'\gamma_0)]^2\hat{\mathbf{b}}_{p,s}(x)'\bar{\mathbf{Q}}_{p,s}^{-1}\bar{\Sigma}_{p,s}\bar{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}(x), \\
\Omega_{\vartheta,p,s}(x) &:= \Omega_{\vartheta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\mu_0(x) + \mathbf{w}'\gamma_0)]^2\hat{\mathbf{b}}_{p,0}(x)'\mathbf{T}'_s\mathbf{Q}_{0,p,s}^{-1}\Sigma_{0,p,s}\mathbf{Q}_{0,p,s}^{-1}\mathbf{T}_s\hat{\mathbf{b}}_{p,0}(x), \\
\hat{\Omega}_{\zeta,p,s}(x) &:= \hat{\Omega}_{\zeta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\hat{\mu}(x) + \hat{\mathbf{w}}'\hat{\gamma})]^2\hat{\mathbf{b}}_{p,s}^{(1)}(x)'\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\Sigma}_{p,s}\hat{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}^{(1)}(x), \\
\bar{\Omega}_{\zeta,p,s}(x) &:= \bar{\Omega}_{\zeta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\mu_0(x) + \mathbf{w}'\gamma_0)]^2\hat{\mathbf{b}}_{p,s}^{(1)}(x)'\bar{\mathbf{Q}}_{p,s}^{-1}\bar{\Sigma}_{p,s}\bar{\mathbf{Q}}_{p,s}^{-1}\hat{\mathbf{b}}_{p,s}^{(1)}(x), \quad \text{and} \\
\Omega_{\zeta,p,s}(x) &:= \Omega_{\zeta,p,s}(x; \hat{\Delta}) := [\eta^{(1)}(\mu_0(x) + \mathbf{w}'\gamma_0)]^2\hat{\mathbf{b}}_{p,0}^{(1)}(x)'\mathbf{T}'_s\mathbf{Q}_{0,p,s}^{-1}\Sigma_{0,p,s}\mathbf{Q}_{0,p,s}^{-1}\mathbf{T}_s\hat{\mathbf{b}}_{p,0}^{(1)}(x).
\end{aligned}$$

Recall that in the main text we always set $s = p$ and omit the dependence on s whenever there is no confusion. Thus,

$$\begin{aligned}
\hat{\mathbf{Q}}_p &= \hat{\mathbf{Q}}_{p,p}, \quad \bar{\mathbf{Q}}_p = \bar{\mathbf{Q}}_{p,p}, \quad \mathbf{Q}_{0,p} = \mathbf{Q}_{0,p,p}, \\
\hat{\Sigma}_p &= \hat{\Sigma}_{p,p}, \quad \bar{\Sigma}_p = \bar{\Sigma}_{p,p}, \quad \Sigma_{0,p} = \Sigma_{0,p,p}, \\
\hat{\Omega}_{\mu^{(v)},p}(x) &= \hat{\Omega}_{\mu^{(v)},p,p}(x), \quad \bar{\Omega}_{\mu^{(v)},p}(x) = \bar{\Omega}_{\mu^{(v)},p,p}(x), \quad \Omega_{\mu^{(v)},p}(x) = \Omega_{\mu^{(v)},p,p}(x), \\
\hat{\Omega}_{\vartheta,p}(x) &= \hat{\Omega}_{\vartheta,p,p}(x), \quad \bar{\Omega}_{\vartheta,p}(x) = \bar{\Omega}_{\vartheta,p,p}(x), \quad \Omega_{\vartheta,p}(x) = \Omega_{\vartheta,p,p}(x), \\
\hat{\Omega}_{\zeta,p}(x) &= \hat{\Omega}_{\zeta,p,p}(x), \quad \bar{\Omega}_{\zeta,p}(x) = \bar{\Omega}_{\zeta,p,p}(x), \quad \text{and} \quad \Omega_{\zeta,p}(x) = \Omega_{\zeta,p,p}(x).
\end{aligned}$$

In this supplement, however, all our results hold for a general choice of the degree and the smoothness of the basis. For ease of notation, the subscripts p and s of the above quantities are

dropped hereafter:

$$\begin{aligned}
\widehat{\mathbf{Q}} &= \widehat{\mathbf{Q}}_{p,s}, & \bar{\mathbf{Q}} &= \bar{\mathbf{Q}}_{p,s}, & \mathbf{Q}_0 &= \mathbf{Q}_{0,p,s}, \\
\widehat{\Sigma} &= \widehat{\Sigma}_{p,s}, & \bar{\Sigma} &= \bar{\Sigma}_{p,s}, & \Sigma_0 &= \Sigma_{0,p,s}, \\
\widehat{\Omega}_{\mu^{(v)}}(x) &= \widehat{\Omega}_{\mu^{(v)},p,s}(x), & \bar{\Omega}_{\mu^{(v)}}(x) &= \bar{\Omega}_{\mu^{(v)},p,s}(x), & \Omega_{\mu^{(v)}}(x) &= \Omega_{\mu^{(v)},p,s}(x), \\
\widehat{\Omega}_{\vartheta}(x) &= \widehat{\Omega}_{\vartheta,p,s}(x), & \bar{\Omega}_{\vartheta}(x) &= \bar{\Omega}_{\vartheta,p,s}(x), & \Omega_{\vartheta}(x) &= \Omega_{\vartheta,p,s}(x), \\
\widehat{\Omega}_{\zeta}(x) &= \widehat{\Omega}_{\zeta,p,s}(x), & \bar{\Omega}_{\zeta}(x) &= \bar{\Omega}_{\zeta,p,s}(x), & \text{and } \Omega_{\zeta}(x) &= \Omega_{\zeta,p,s}(x).
\end{aligned}$$

In addition, given the family Π_C of the quasi-uniform partitions defined in (SA-2.4), for any $\Delta \in \Pi_C$, we let $\beta_0(\Delta) \in \mathbb{R}^{K_{p,s}}$ be any vector such that for every $v \leq p$,

$$\sup_{x \in \mathcal{X}} \left| \mu_0^{(v)}(x) - \mathbf{b}_{p,s}^{(v)}(x; \Delta)' \beta_0(\Delta) \right| \lesssim J^{-p-1+v}.$$

Let $r_{0,v}(x; \Delta) = \mu_0^{(v)}(x) - \mathbf{b}_{p,s}^{(v)}(x; \Delta)' \beta_0(\Delta)$ denote the corresponding approximation error. Accordingly, given the random partition $\widehat{\Delta}$, we let $\widehat{\beta}_0 := \beta_0(\widehat{\Delta})$, and $\widehat{r}_{0,v}(x) = \mu_0^{(v)}(x) - \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\beta}_0$ denote the corresponding approximation error. The existence of such vectors is guaranteed by Assumptions SA-DGP and SA-SM(v), and is verified in Lemma SA-5.5 in Section SA-5.

SA-3.1 Preliminary Lemmas

Lemma SA-3.1 (Gram). *Suppose that Assumptions SA-DGP, SA-SM, and SA-RP(i) hold. If $\frac{J \log J}{n} = o(1)$, then*

$$1 \lesssim \lambda_{\min}(\bar{\mathbf{Q}}) \leq \lambda_{\max}(\bar{\mathbf{Q}}) \lesssim 1, \quad |[\bar{\mathbf{Q}}^{-1}]_{ij}| \lesssim \varrho^{|i-j|} \quad w.p.a. \ 1, \quad \text{and} \quad \|\bar{\mathbf{Q}}^{-1}\|_{\infty} \lesssim_{\mathbb{P}} 1,$$

where $\varrho \in (0, 1)$ is some absolute constant.

If, in addition, Assumption SA-RP(ii) holds. Then,

$$\begin{aligned}
1 &\lesssim \lambda_{\min}(\mathbf{Q}_0) \leq \lambda_{\max}(\mathbf{Q}_0) \lesssim 1, \\
\|\bar{\mathbf{Q}} - \mathbf{Q}_0\| &\lesssim_{\mathbb{P}} \left(\frac{J \log J}{n} \right)^{1/2} + \mathfrak{r}_{\text{RP}}, \quad \text{and} \quad \|\bar{\mathbf{Q}}^{-1} - \mathbf{Q}_0^{-1}\|_{\infty} \lesssim_{\mathbb{P}} \left(\frac{J \log J}{n} \right)^{1/2} + \mathfrak{r}_{\text{RP}}.
\end{aligned}$$

The next lemma shows that the limiting variance is bounded from above and below.

Lemma SA-3.2 (Asymptotic Variance). *Suppose that Assumptions SA-DGP, SA-SM, and SA-RP(i) hold. If $\frac{J \log J}{n} = o(1)$, then w.p.a. 1,*

$$\begin{aligned} J^{1+2v} &\lesssim \inf_{x \in \mathcal{X}} \bar{\Omega}_{\mu^{(v)}}(x) \leq \sup_{x \in \mathcal{X}} \bar{\Omega}_{\mu^{(v)}}(x) \lesssim J^{1+2v}, \\ J &\lesssim \inf_{x \in \mathcal{X}} \bar{\Omega}_{\vartheta}(x) \leq \sup_{x \in \mathcal{X}} \bar{\Omega}_{\vartheta}(x) \lesssim J, \\ J^3 &\lesssim \inf_{x \in \mathcal{X}} \bar{\Omega}_{\zeta}(x) \leq \sup_{x \in \mathcal{X}} \bar{\Omega}_{\zeta}(x) \lesssim J^3. \end{aligned}$$

If, in addition, Assumption SA-RP(ii) holds, then w.p.a. 1,

$$\begin{aligned} J^{1+2v} &\lesssim \inf_{x \in \mathcal{X}} \Omega_{\mu^{(v)}}(x) \leq \sup_{x \in \mathcal{X}} \Omega_{\mu^{(v)}}(x) \lesssim J^{1+2v}, \\ J &\lesssim \inf_{x \in \mathcal{X}} \Omega_{\vartheta}(x) \leq \sup_{x \in \mathcal{X}} \Omega_{\vartheta}(x) \lesssim J, \\ J^3 &\lesssim \inf_{x \in \mathcal{X}} \Omega_{\zeta}(x) \leq \sup_{x \in \mathcal{X}} \Omega_{\zeta}(x) \lesssim J^3. \end{aligned}$$

The next lemma gives a bound on the variance component of the nonlinear binscatter estimator.

Lemma SA-3.3 (Uniform Convergence: Variance). *Suppose that Assumptions SA-DGP, SA-SM, and SA-RP(i) hold. If $\frac{J^{\frac{\nu}{\nu-2}} \log J}{n} = o(1)$, then*

$$\sup_{x \in \mathcal{X}} \left| \hat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n [\hat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right| \lesssim_{\mathbb{P}} J^v \left(\frac{J \log J}{n} \right)^{1/2}.$$

Lemma SA-3.4 (Projection of Approximation Error). *Suppose that Assumptions SA-DGP, SA-SM, and SA-RP(i) hold. If $\frac{J^{\frac{\nu}{\nu-2}} \log J}{n} = o(1)$, then*

$$\begin{aligned} &\sup_{x \in \mathcal{X}} \left| \hat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n \left[\hat{\mathbf{b}}_{p,s}(x_i) \left(\eta_{i,1} \psi(y_i, \eta_i) - \eta^{(1)}(\hat{\mathbf{b}}_{p,s}(x_i)' \hat{\boldsymbol{\beta}}_0 + \mathbf{w}_i' \boldsymbol{\gamma}_0) \psi(y_i, \eta(\hat{\mathbf{b}}_{p,s}(x_i)' \hat{\boldsymbol{\beta}}_0 + \mathbf{w}_i' \boldsymbol{\gamma}_0)) \right) \right] \right| \\ &\lesssim_{\mathbb{P}} J^{-p-1+v} + J^{\frac{2v-p-1}{2}} \left(\frac{J \log J}{n} \right)^{1/2} + \frac{J^{1+v} \log J}{n}. \end{aligned}$$

Lemma SA-3.5 (Uniform Consistency). *Suppose that Assumptions SA-DGP, SA-SM, SA-HLE(i), and SA-RP(i) hold. If $\frac{J(J+d_{\gamma}) \log J}{n^{1-1/\nu}} = o(1)$, then*

$$\|\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0\|_{\infty} = o_{\mathbb{P}}(J^{-1/2}) \quad \text{and} \quad \sup_{x \in \mathcal{X}} |\hat{\mu}(x) - \mu_0(x)| = o_{\mathbb{P}}(1).$$

Remark SA-3.1 (Side rate conditions). When $\nu \rightarrow \infty$, the rate restriction in Lemma SA-3.5 tends to $J(J+d_\gamma) \log J/n = o(1)$, which reduces to $J^2 \log J/n = o(1)$ when $d_\gamma = o(J)$. We conjecture this rate restriction is stronger than needed. For piecewise polynomials (i.e., $s = 0$) and fixed d_γ , the weaker condition $\frac{J^{\frac{\nu}{\nu-1}} (\log J)^{\frac{\nu}{\nu-1}}}{n} = o(1)$ suffices to establish uniform consistency, and is redundant in our main theorems in view of the condition $\frac{J^{\frac{\nu}{\nu-2}} (\log n)^{\frac{\nu}{\nu-2}}}{n} = o(1)$ imposed below. $\lrcorner$

Our result holds without imposing any smoothness restrictions on the estimation space. Specifically, the estimation procedure (SA-2.3) searches for solutions in $\mathbb{R}^{K_{p,s}}$, leading to an estimation space $\{\widehat{\mathbf{b}}_{p,s}(x)' \boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{K_{p,s}}\}$. In contrast, many studies of series (or sieve) methods restrict the functions in the estimation space to satisfy certain smoothness conditions, e.g., Lipschitz continuity, to derive the uniform consistency. See, for example, Chernozhukov, Imbens and Newey (2007) and references therein.

Remark SA-3.2 (Improvements over literature). Most of the results in this subsection are new to the literature, even in the case of non-random partitioning and without covariate-adjustments, because they take advantage of the specific binscatter structure (i.e., locally bounded series basis). The closest antecedent in the literature is Belloni, Chernozhukov, Chetverikov and Fernandez-Val (2019), while it focuses on series-based quantile regression only. Furthermore, relative to prior work, our results allow for random partitioning schemes, formally taking into account both the potential randomness of the partition and the semi-linear regression estimation structure. Importantly, we highlight the key conditions on the possibly random partition (Assumptions SA-RP(i) and SA-RP(ii)) used to derive various properties of the Gram matrix, asymptotic variance and other quantities. $\lrcorner$

SA-3.2 Bahadur Representation

Define the dimension-adjusted entropy factor and Bahadur remainder

$$\ell_{\gamma,n} := \log n + \sqrt{d_\gamma \log n}, \quad \mathbf{r}_{\text{BR},n} := \left(\frac{J \log n}{n} \right)^{3/4} \ell_{\gamma,n} + J^{-\frac{p+1}{2}} \sqrt{\frac{J}{n}} \ell_{\gamma,n} + \frac{J \ell_{\gamma,n}^2}{n} + J^{-p-1} + \sqrt{d_\gamma} \mathbf{r}_\gamma.$$

The factor $\sqrt{d_\gamma} \mathbf{r}_\gamma$ controls the fitted-index error under coordinatewise bounded controls, while $\ell_{\gamma,n}$ accounts for the empirical-process complexity of estimating d_γ coefficients. For fixed d_γ , $\ell_{\gamma,n} \asymp$

$\log n$, and $J\ell_{\gamma,n}^2/n$ is absorbed by the first term under the imposed rate conditions. The second rate condition in the theorem below is an additional heavy-tail restriction used to keep this concise remainder.

Theorem SA-3.1 (Bahadur Representation). *Suppose that Assumptions SA-DGP, SA-SM, SA-HLE(i), and SA-RP(i) hold, $d_\gamma = o(J)$,*

$$\frac{J^{\frac{\nu}{\nu-2}} \log n}{n} + \frac{J \log n \ell_{\gamma,n}^{4/3}}{n} + \frac{J^{\frac{2\nu}{\nu-1}} (\log n)^{\frac{\nu}{\nu-1}}}{n} + \frac{\log n}{J} = o(1), \quad \frac{J\ell_{\gamma,n}^2}{n^{1-1/\nu}} \lesssim \sqrt{\frac{J \log n}{n}} + J^{-p-1}.$$

Then,

(i) $\hat{\mu}^{(v)}(x)$ satisfies that

$$\sup_{x \in \mathcal{X}} \left| \hat{\mu}^{(v)}(x) - \mu_0^{(v)}(x) + \hat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right| \lesssim_{\mathbb{P}} J^v \mathbf{r}_{\text{BR},n}.$$

(ii) $\hat{\vartheta}(x, \hat{\mathbf{w}})$ satisfies that

$$\begin{aligned} & \sup_{x \in \mathcal{X}} \left| \hat{\vartheta}(x, \hat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w}) + \eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \hat{\mathbf{b}}_{p,s}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right| \\ & \lesssim_{\mathbb{P}} \mathbf{r}_{\text{BR},n} + \|\hat{\mathbf{w}} - \mathbf{w}\|. \end{aligned}$$

(iii) $\hat{\zeta}(x, \hat{\mathbf{w}})$ satisfies that

$$\begin{aligned} & \sup_{x \in \mathcal{X}} \left| \hat{\zeta}(x, \hat{\mathbf{w}}) - \zeta_0(x, \mathbf{w}) + \eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \hat{\mathbf{b}}_{p,s}^{(1)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right| \\ & \lesssim_{\mathbb{P}} \left(\frac{J \log n}{n} \right)^{1/2} + J \mathbf{r}_{\text{BR},n} + \|\hat{\mathbf{w}} - \mathbf{w}\| \left(1 + J \left(\frac{J \log n}{n} \right)^{1/2} \right). \end{aligned}$$

The following corollary is an immediate result of Lemma SA-3.3 and Theorem SA-3.1, and hence its proof is omitted.

Corollary SA-3.1 (Uniform Convergence). *Suppose that the conditions of Theorem SA-3.1 hold and $\frac{J \log n \ell_{\gamma,n}^4}{n} + \frac{\ell_{\gamma,n}^2}{J \log n} \lesssim 1$. Then*

$$\sup_{x \in \mathcal{X}} |\hat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)| \lesssim_{\mathbb{P}} J^v \left(\left(\frac{J \log n}{n} \right)^{1/2} + J^{-p-1} \right).$$

If, in addition, $\|\widehat{\mathbf{w}} - \mathbf{w}\| \lesssim_{\mathbb{P}} \left(\frac{J \log n}{n}\right)^{1/2} + J^{-p-1}$, then

$$\begin{aligned} \sup_{x \in \mathcal{X}} |\widehat{\vartheta}(x, \widehat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w})| &\lesssim_{\mathbb{P}} \left(\frac{J \log n}{n}\right)^{1/2} + J^{-p-1} \quad \text{and} \\ \sup_{x \in \mathcal{X}} |\widehat{\zeta}(x, \widehat{\mathbf{w}}) - \zeta_0(x, \mathbf{w})| &\lesssim_{\mathbb{P}} J \left(\left(\frac{J \log n}{n}\right)^{1/2} + J^{-p-1}\right). \end{aligned}$$

The next theorem shows that the proposed variance estimator is consistent.

Theorem SA-3.2 (Variance Estimate). *Suppose that the conditions of Theorem SA-3.1 and Assumption SA-HLE(ii) hold. If $\frac{J^{\frac{\nu}{\nu-2}}(\log n)^{\frac{\nu}{\nu-2}}}{n} + \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} + \frac{J \log n \ell_{\gamma,n}^4}{n} + \frac{\ell_{\gamma,n}^2}{J \log n} = o(1)$, then*

$$\begin{aligned} \|\widehat{\boldsymbol{\Sigma}} - \bar{\boldsymbol{\Sigma}}\| &\lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2}, \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\mu^{(v)}}(x) - \bar{\Omega}_{\mu^{(v)}}(x) \right| &\lesssim_{\mathbb{P}} J^{1+2v} \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} \right), \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\vartheta}(x) - \bar{\Omega}_{\vartheta}(x) \right| &\lesssim_{\mathbb{P}} J \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \right), \quad \text{and} \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\zeta}(x) - \bar{\Omega}_{\zeta}(x) \right| &\lesssim_{\mathbb{P}} J^3 \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \right). \end{aligned}$$

If Assumption SA-RP(ii) also holds, then

$$\begin{aligned} \|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}_0\| &\lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \mathfrak{r}_{\text{RP}}, \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\mu^{(v)}}(x) - \Omega_{\mu^{(v)}}(x) \right| &\lesssim_{\mathbb{P}} J^{1+2v} \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \mathfrak{r}_{\text{RP}} \right), \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\vartheta}(x) - \Omega_{\vartheta}(x) \right| &\lesssim_{\mathbb{P}} J \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \mathfrak{r}_{\text{RP}} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \right), \quad \text{and} \\ \sup_{x \in \mathcal{X}} \left| \widehat{\Omega}_{\zeta}(x) - \Omega_{\zeta}(x) \right| &\lesssim_{\mathbb{P}} J^3 \left(J^{-p-1} + \left(\frac{J \log n}{n^{1-\frac{2}{\nu}}}\right)^{1/2} + \mathfrak{r}_{\text{RP}} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \right). \end{aligned}$$

Remark SA-3.3 (Improvements over literature). Theorem SA-3.1 and Corollary SA-3.1 construct the Bahadur representation and uniform convergence of nonlinear binscatter-based M-estimators, which improve upon prior results in the literature in at least two aspects. First, our results allow for random partitioning schemes, and the key condition imposed on the partition is Assumption SA-RP(i), i.e., the conditional independence between the partition and the outcome and the quasi-uniformity of the partition. The “convergence” of the random partition (Assumption SA-RP(ii)) is not required, which implies that our results can accommodate more complex partitioning schemes

other than evenly-spaced or empirical-quantile-spaced partitions.

Second, our results are established under weaker rate restrictions. When d_γ is fixed, we require $J^{8/3}/n = o(1)$ up to $\log n$ terms when $\nu \geq 4$, thus accommodating IMSE-optimal piecewise constant binscatter estimators. In fact, for piecewise polynomials ($s = 0$), we can show that the Bahadur representation still holds under $J/n = o(1)$ up to $\log n$ terms when a subexponential moment restriction holds for the “score” $\psi(y_i, \eta_i)$, which is analogous to the result for kernel-based estimators in the literature (see, e.g., [Kong et al., 2010](#)). For series estimators, similar results were established for particular choices of loss functions under more stringent conditions in the literature. For example, [Belloni, Chernozhukov, Chetverikov and Fernandez-Val \(2019\)](#) considers series-based quantile regression, and Theorem 2 and Corollary 2 therein can be used to establish a Bahadur representation and uniform convergence of the resulting estimators under $J^4/n^{1-\varepsilon} = o(1)$ for some $\varepsilon > 0$.

When d_γ may grow, the comparison is less direct. We focus on the binscatter component and use a profiling-type argument, which requires the estimation error of $\hat{\gamma}$ to be negligible relative to that of the nonparametric component and therefore imposes additional restrictions on d_γ . By contrast, no separate negligibility restriction is needed in [Belloni et al. \(2019\)](#), where $(\hat{\mathbf{b}}_{p,s}(x)', \mathbf{w}')'$ is treated as a single generic increasing-dimensional regressor vector. See Remark [SA-3.11](#) for details. Now, assuming that the required fast convergence of $\hat{\gamma}$ holds, when $\nu = 4$, the rate restrictions imposed directly in our Bahadur representation reduce to $J^{8/3}/n = o(1)$ and $Jd_\gamma^2/\sqrt{n} \lesssim 1$, ignoring logarithmic factors. More generally, these restrictions are $J^{2\nu/(\nu-1)}/n = o(1)$ and $Jd_\gamma^2/n^{1-2/\nu} \lesssim 1$, and approach $J^2/n = o(1)$ and $Jd_\gamma^2/n \lesssim 1$ as $\nu \rightarrow \infty$. They still improve upon those obtained in [Belloni et al. \(2019\)](#) by treating $(\hat{\mathbf{b}}_{p,s}(x)', \mathbf{w}')'$ as a generic increasing-dimensional regressor vector.

Note that the results in [Belloni et al. \(2019\)](#) are slightly stronger than our Theorem [SA-3.1](#) in the sense that the expansion holds uniformly over both the evaluation point $x \in \mathcal{X}$ and the desired quantiles $u \in \mathcal{U}$ for a compact set of quantile indices $\mathcal{U} \subset (0, 1)$. Our results regarding Bahadur representation can be extended to achieve the same level of uniformity. In general, the parameter of interest ([SA-2.1](#)) and the estimator ([SA-2.2](#)) are defined for each particular choice of the loss function within a function class $\mathcal{F}$. For the class of check functions used in quantile regression or other function classes with low complexity, it can be shown that the Bahadur representation still holds uniformly over the evaluation point $x \in \mathcal{X}$ and the loss function $\rho \in \mathcal{F}$ under rate restrictions

similar to those in Theorem SA-3.1, thereby providing an improvement over the literature. $\square$

SA-3.3 Pointwise Inference

Starting from this section, we consider statistical inference on $\mu_0^{(v)}(x)$, $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$ based on the following Studentized t -statistics:

$$\begin{aligned} T_{\mu^{(v)},p}(x) &= \frac{\hat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}}, \\ T_{\vartheta,p}(x) &= \frac{\hat{\vartheta}(x, \hat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w})}{\sqrt{\hat{\Omega}_{\vartheta}(x)/n}} \quad \text{and} \\ T_{\zeta,p}(x) &= \frac{\hat{\zeta}(x, \hat{\mathbf{w}}) - \zeta_0(x, \mathbf{w})}{\sqrt{\hat{\Omega}_{\zeta}(x)/n}}. \end{aligned}$$

The next theorem shows the pointwise asymptotic normality of the binscatter estimators.

Theorem SA-3.3 (Pointwise Asymptotic Distribution). *Suppose that the conditions of Theorem SA-3.1 and Assumption SA-HLE(ii) hold, $\sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \mathbb{E}[|\psi(y_i, \eta_i)|^\nu \mid x_i = x, \mathbf{w}_i = \mathbf{w}] \lesssim 1$ for some $\nu \geq 3$, and $\frac{J^{\frac{\nu}{\nu-2}}(\log n)^{\frac{\nu}{\nu-2}}}{n} + \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} + \frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} + \frac{\ell_{\gamma,n}^2}{J^{p+1}} + nJ^{-2p-3} = o(1)$. Then the following conclusions hold:*

(i) For $\hat{\mu}^{(v)}(x)$,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(T_{\mu^{(v)},p}(x) \leq u) - \Phi(u) \right| = o(1), \quad \text{for each } x \in \mathcal{X}.$$

(ii) For $\hat{\vartheta}(x, \hat{\mathbf{w}})$, if, in addition, $\|\hat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(\sqrt{J/n})$, then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(T_{\vartheta,p}(x) \leq u) - \Phi(u) \right| = o(1) \quad \text{for each } x \in \mathcal{X}.$$

(iii) For $\hat{\zeta}(x, \hat{\mathbf{w}})$, if, in addition,

$$\|\hat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}\left(\sqrt{J^3/n} \wedge \sqrt{1/\log n}\right),$$

then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(T_{\zeta,p}(x) \leq u) - \Phi(u) \right| = o(1) \quad \text{for each } x \in \mathcal{X}.$$

Remark SA-3.4 (Improvements over literature). The result in this subsection is new to the literature, even in the case of non-random partitioning and without covariate adjustments, because it takes advantage of the specific binscatter structure (i.e., locally bounded series basis). The closest antecedent in the literature is [Belloni et al. \(2019\)](#), which focuses on series-based quantile regression only. Furthermore, relative to prior work, our results allow for more general partitioning schemes, formally take into account the potential randomness of the partition, and account for the semi-linear regression estimation structure. The key condition imposed on the partition for pointwise inference is Assumption [SA-RP\(i\)](#), and the “convergence” of the random partition is not required. $\perp$

SA-3.4 Integrated Mean Squared Error

In this section we give a Nagar-type approximate IMSE expansion for each of the three estimators $\hat{\mu}^{(v)}(x)$, $\hat{\vartheta}(x, \hat{\mathbf{w}})$ and $\hat{\zeta}(x, \hat{\mathbf{w}})$, with explicit characterization of the leading constants. Define

$$r_{0,v}^*(x) = \frac{\mu_0^{(p+1)}(x)}{(p+1-v)!} h_x^{p+1-v} \mathcal{E}_{p+1-v}\left(\frac{x - \tau_x^L}{h_x}\right).$$

Here $\mathcal{E}_m(\cdot)$ is the m th Bernoulli polynomial for each $m \in \mathbb{Z}_+$, τ_x^L is the start of the interval in the non-random partition Δ_0 containing x , and h_x denotes its length.

Theorem SA-3.4 (IMSE). *Suppose that the conditions of Theorem [SA-3.1](#) and Assumption [SA-RP\(ii\)](#) hold, and $\mu_0(\cdot)$ is $(p+2)$ -times continuously differentiable with $\sup_{n \geq 1} \sup_{x \in \mathcal{X}} |\mu_0^{(p+2)}(x)| \lesssim$*

1. If $s > 0$, assume in addition that $\max_{2 \leq j \leq J} |h_j - h_{j-1}| = o(J^{-1})$. Let $\omega(x)$ be a continuous weighting function over $\mathcal{X}$ bounded away from zero. Also, assume that $\frac{J^{\frac{\nu-2}{2}} \log n}{n} + \frac{J^{\frac{2\nu}{\nu-1}} (\log n)^{\frac{\nu}{\nu-1}}}{n} + \frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} + \frac{\ell_{\gamma,n}^2}{J} = o(1)$.

(i) For $\hat{\mu}^{(v)}(x)$,

$$\int_{\mathcal{X}} \left(\hat{\mu}^{(v)}(x) - \mu_0^{(v)}(x) \right)^2 \omega(x) dx = \text{AISE}_{\mu^{(v)}} + o_{\mathbb{P}}\left(\frac{J^{1+2v}}{n} + J^{-2(p+1-v)}\right)$$

where

$$\begin{aligned}\mathbb{E}[\text{AISE}_{\mu^{(v)}}|\mathbf{X}, \mathbf{W}, \widehat{\Delta}] &= \frac{J^{1+2v}}{n} \mathcal{V}_n(p, s, v) + J^{-2(p+1-v)} \mathcal{B}_n(p, s, v) + o_{\mathbb{P}}\left(\frac{J^{1+2v}}{n} + J^{-2(p+1-v)}\right), \\ \mathcal{V}_n(p, s, v) &:= J^{-(1+2v)} \text{trace} \left(\mathbf{Q}_0^{-1} \Sigma_0 \mathbf{Q}_0^{-1} \int_{\mathcal{X}} \mathbf{b}_{p,s}^{(v)}(x) \mathbf{b}_{p,s}^{(v)}(x)' \omega(x) dx \right) \asymp 1, \\ \mathcal{B}_n(p, s, v) &:= J^{2p+2-2v} \int_{\mathcal{X}} \left(r_{0,v}^*(x) - \mathbf{b}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}[\mathbf{b}_{p,s}(x_i) \mathfrak{K}(x_i, \mathbf{w}_i) r_{0,0}^*(x_i)] \right)^2 \omega(x) dx \lesssim 1.\end{aligned}$$

(ii) For $\widehat{\vartheta}(x, \widehat{\mathbf{w}})$, if $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(\sqrt{J/n} + J^{-p-1})$, then

$$\int_{\mathcal{X}} \left(\widehat{\vartheta}(x, \widehat{\mathbf{w}}) - \vartheta_0(x, \mathbf{w}) \right)^2 \omega(x) dx = \text{AISE}_{\vartheta} + o_{\mathbb{P}}\left(\frac{J}{n} + J^{-2(p+1)}\right)$$

where

$$\begin{aligned}\mathbb{E}[\text{AISE}_{\vartheta}|\mathbf{X}, \mathbf{W}, \widehat{\Delta}] &= \frac{J}{n} \mathcal{V}_n(p, s) + J^{-2(p+1)} \mathcal{B}_n(p, s) + o_{\mathbb{P}}\left(\frac{J}{n} + J^{-2(p+1)}\right), \\ \mathcal{V}_n(p, s) &:= J^{-1} \text{trace} \left(\mathbf{Q}_0^{-1} \Sigma_0 \mathbf{Q}_0^{-1} \int_{\mathcal{X}} \eta_{0,1}(x, \mathbf{w})^2 \mathbf{b}_{p,s}(x) \mathbf{b}_{p,s}(x)' \omega(x) dx \right) \asymp 1, \\ \mathcal{B}_n(p, s) &:= J^{2p+2} \int_{\mathcal{X}} \left[\eta_{0,1}(x, \mathbf{w}) \left(r_{0,0}^*(x) - \mathbf{b}_{p,s}(x)' \mathbf{Q}_0^{-1} \mathbb{E}[\mathbf{b}_{p,s}(x_i) \mathfrak{K}(x_i, \mathbf{w}_i) r_{0,0}^*(x_i)] \right) \right]^2 \omega(x) dx \lesssim 1.\end{aligned}$$

(iii) For $\widehat{\zeta}(x, \widehat{\mathbf{w}})$, if

$$\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}\left(\frac{\sqrt{J^3/n + J^{-2p}}}{1 + \sqrt{J^3 \log n/n}}\right),$$

then

$$\int_{\mathcal{X}} \left(\widehat{\zeta}(x, \widehat{\mathbf{w}}) - \zeta_0(x, \mathbf{w}) \right)^2 \omega(x) dx = \text{AISE}_{\zeta} + o_{\mathbb{P}}\left(\frac{J^3}{n} + J^{-2p}\right)$$

where

$$\begin{aligned}\mathbb{E}[\text{AISE}_{\zeta}|\mathbf{X}, \mathbf{W}, \widehat{\Delta}] &= \frac{J^3}{n} \mathcal{V}_n(p, s) + J^{-2p} \mathcal{B}_n(p, s) + o_{\mathbb{P}}\left(\frac{J^3}{n} + J^{-2p}\right), \\ \mathcal{V}_n(p, s) &:= J^{-3} \text{trace} \left(\mathbf{Q}_0^{-1} \Sigma_0 \mathbf{Q}_0^{-1} \int_{\mathcal{X}} \eta_{0,1}(x, \mathbf{w})^2 \mathbf{b}_{p,s}^{(1)}(x) \mathbf{b}_{p,s}^{(1)}(x)' \omega(x) dx \right) \asymp 1, \\ \mathcal{B}_n(p, s) &:= J^{2p} \int_{\mathcal{X}} \left[\eta_{0,1}(x, \mathbf{w}) \left(r_{0,1}^*(x) - \mathbf{b}_{p,s}^{(1)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}[\mathbf{b}_{p,s}(x_i) \mathfrak{K}(x_i, \mathbf{w}_i) r_{0,0}^*(x_i)] \right) \right]^2 \omega(x) dx \lesssim 1.\end{aligned}$$

Remark SA-3.5 (Smoothness for the IMSE Expansion). The bound on $\mu_0^{(p+1)}$ in Assumption SA-SM(v) suffices to give an approximation error rate. However, for IMSE expansion, uniform equicontinuity of $\mu_0^{(p+1)}$ over n is needed to guarantee the remainder is of smaller order; the ad-

ditional assumption that $\mu_0^{(p+2)}$ is uniformly bounded in Theorem SA-3.4 is a simpler sufficient condition. For fixed μ_0 , continuity of $\mu_0^{(p+1)}$ suffices. $\lrcorner$

In general, $\mathcal{B}_n(p, s, v) \gtrsim 1$ (see Remark SA-3.7 in Cattaneo et al. (2024a)), and thus the above theorem implies that the (approximate) IMSE-optimal number of bins satisfies that $J_{\text{AIMSE}} \asymp n^{\frac{1}{2p+3}}$. Relying on the IMSE expansion in Theorem SA-3.4, one may design a data-driven procedure to select the IMSE-optimal number of bins for nonlinear binscatter-based M-estimators.

Remark SA-3.6 (Leading Bias under Quantile-Spaced Partitions). Suppose that Δ_0 is the population quantile-spaced partition, so that $\tau_j = F_X^{-1}(j/J)$. Assumption SA-DGP(ii) implies that $\max_{2 \leq j \leq J} |h_j - h_{j-1}| = O(J^{-2})$, so the additional condition in Theorem SA-3.4 is satisfied. By Assumption SA-DGP(ii), uniformly over $x \in \mathcal{X}$,

$$h_x = \frac{1}{Jf_X(x)}\{1 + o(1)\}.$$

Consequently,

$$r_{0,v}^*(x) = \frac{J^{-p-1+v}\mu_0^{(p+1)}(x)}{(p+1-v)!f_X(x)^{p+1-v}}\mathcal{E}_{p+1-v}\left(\frac{x - \tau_x^L}{h_x}\right) + o(J^{-p-1+v})$$

uniformly over $x \in \mathcal{X}$. Thus, the familiar density-based leading bias is recovered as a special case, while the definition above also applies to other limiting partitions covered by Theorem SA-3.4. The variance expression in Theorem SA-3.4 already accommodates quantile-spaced partitions through the corresponding population basis and matrices. $\lrcorner$

Remark SA-3.7 (Improvements over literature). The results in this subsection are new to the literature, even in the case of non-random partitioning and without covariate-adjustments, for both nonlinear series estimators and binscatter (piecewise polynomials and splines) nonlinear series estimators in particular. Furthermore, our results allow for random partitioning schemes, formally take into account the potential randomness of the partition, and account for the semi-linear regression estimation structure. We highlight the key conditions imposed on the partition (Assumption SA-RP) for the approximate IMSE expansion. The “convergence” of the random partition (Assumption SA-RP(ii)) is needed to derive the non-random variance and bias constants $\mathcal{V}_n(p, s)$ and $\mathcal{B}_n(p, s)$.

When $s > 0$, Theorem [SA-3.4](#) also requires adjacent bin widths in Δ_0 to be asymptotically equal. $\perp$

SA-3.5 Uniform Inference

Let $(a_n : n \geq 1)$ be a sequence of positive non-vanishing constants. We will first show that the (feasible) Studentized t -statistic processes $T_{\mu^{(v)},p}(\cdot)$, $T_{\vartheta,p}(\cdot)$ and $T_{\zeta,p}(\cdot)$ can be approximated by Gaussian processes in a proper sense at certain rate.

Theorem SA-3.5 (Strong Approximation). *Suppose that the conditions of Theorem [SA-3.1](#) and Assumption [SA-HLE\(ii\)](#) hold,*

$$\begin{aligned} & \frac{J(\log n)^2}{n^{1-\frac{2}{\nu}}} + \left(\frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} \right)^{1/2} + nJ^{-2p-3} + \frac{\ell_{\gamma,n}^2}{J^{p+1}} + nJ^{-1}d_{\gamma}\mathfrak{r}_{\gamma}^2 = o(a_n^{-2}), \\ & \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} = o(1). \end{aligned}$$

Then the following conclusions hold:

- (i) *On a properly enriched probability space, there exists some $K_{p,s}$ -dimensional standard normal random vector $\mathbf{N}_{K_{p,s}}$ such that for any $\xi > 0$,*

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x) - \bar{Z}_{\mu^{(v)},p}(x)| > \xi a_n^{-1}\right) = o(1), \quad \bar{Z}_{\mu^{(v)},p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)' \widehat{\mathbf{T}}_s' \bar{\mathbf{Q}}^{-1} \bar{\Sigma}^{1/2}}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \mathbf{N}_{K_{p,s}}.$$

If Assumption [SA-RP\(ii\)](#) also holds with $\mathfrak{r}_{\text{RP}} = o(a_n^{-1}(\log n)^{-1/2})$, then

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x) - Z_{\mu^{(v)},p}(x)| > \xi a_n^{-1}\right) = o(1), \quad Z_{\mu^{(v)},p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)' \mathbf{T}_s' \mathbf{Q}_0^{-1} \Sigma_0^{1/2}}{\sqrt{\Omega_{\mu^{(v)}}(x)}} \mathbf{N}_{K_{p,s}}.$$

- (ii) *If $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(a_n^{-1}\sqrt{J/n})$, then on a properly enriched probability space there exists some $K_{p,s}$ -dimensional standard normal random vector $\mathbf{N}_{K_{p,s}}$ such that for any $\xi > 0$,*

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\vartheta,p}(x) - \bar{Z}_{\vartheta,p}(x)| > \xi a_n^{-1}\right) = o(1), \quad \bar{Z}_{\vartheta,p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}(x)' \widehat{\mathbf{T}}_s' \eta_{0,1}(x, \mathbf{w}) \bar{\mathbf{Q}}^{-1} \bar{\Sigma}^{1/2}}{\sqrt{\Omega_{\vartheta}(x)}} \mathbf{N}_{K_{p,s}}.$$

If Assumption [SA-RP\(ii\)](#) also holds with $\mathfrak{r}_{\text{RP}} = o(a_n^{-1}(\log n)^{-1/2})$, then

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\vartheta,p}(x) - Z_{\vartheta,p}(x)| > \xi a_n^{-1}\right) = o(1), \quad Z_{\vartheta,p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}(x)' \mathbf{T}'_s \eta_{0,1}(x, \mathbf{w}) \mathbf{Q}_0^{-1}}{\sqrt{\Omega_{\vartheta}(x)}} \boldsymbol{\Sigma}_0^{1/2} \mathbf{N}_{K_{p,s}}.$$

(iii) If

$$\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}\left(\frac{a_n^{-1}}{\sqrt{n/J^3} + \sqrt{\log n}}\right),$$

then on a properly enriched probability space there exists some $K_{p,s}$ -dimensional standard normal random vector $\mathbf{N}_{K_{p,s}}$ such that for any $\xi > 0$,

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\zeta,p}(x) - \bar{Z}_{\zeta,p}(x)| > \xi a_n^{-1}\right) = o(1), \quad \bar{Z}_{\zeta,p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}^{(1)}(x)' \widehat{\mathbf{T}}'_s \eta_{0,1}(x, \mathbf{w}) \bar{\mathbf{Q}}^{-1}}{\sqrt{\bar{\Omega}_{\zeta}(x)}} \bar{\boldsymbol{\Sigma}}^{1/2} \mathbf{N}_{K_{p,s}}.$$

If Assumption [SA-RP\(ii\)](#) also holds with $\mathfrak{r}_{\text{RP}} = o(a_n^{-1}(\log n)^{-1/2})$, then

$$\mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\zeta,p}(x) - Z_{\zeta,p}(x)| > \xi a_n^{-1}\right) = o(1), \quad Z_{\zeta,p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}^{(1)}(x)' \mathbf{T}'_s \eta_{0,1}(x, \mathbf{w}) \mathbf{Q}_0^{-1}}{\sqrt{\Omega_{\zeta}(x)}} \boldsymbol{\Sigma}_0^{1/2} \mathbf{N}_{K_{p,s}}.$$

The approximating processes $\bar{Z}_{\mu^{(v)},p}(\cdot)$, $\bar{Z}_{\vartheta,p}(\cdot)$ and $\bar{Z}_{\zeta,p}(\cdot)$ are Gaussian processes conditional on $\mathbf{X}$, $\mathbf{W}$ and $\widehat{\Delta}$, and $Z_{\mu^{(v)},p}(\cdot)$, $Z_{\vartheta,p}(\cdot)$ and $Z_{\zeta,p}(\cdot)$ are Gaussian processes conditional on $\widehat{\Delta}$ by construction. In practice, one can replace all unknowns in $\bar{Z}_{\mu^{(v)},p}(\cdot)$, $\bar{Z}_{\vartheta,p}(\cdot)$ and $\bar{Z}_{\zeta,p}(\cdot)$ (or $Z_{\mu^{(v)},p}(\cdot)$, $Z_{\vartheta,p}(\cdot)$ and $Z_{\zeta,p}(\cdot)$) by their sample analogues, and then construct the following feasible (conditional) Gaussian processes:

$$\begin{aligned} \widehat{Z}_{\mu^{(v)},p}(x) &= \frac{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)' \widehat{\mathbf{T}}'_s \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)}} \mathbf{N}_{K_{p,s}}^* = \frac{\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)}} \mathbf{N}_{K_{p,s}}^*, \\ \widehat{Z}_{\vartheta,p}(x) &= \frac{\widehat{\mathbf{b}}_{p,0}(x)' \widehat{\mathbf{T}}'_s \widehat{\eta}_{0,1}(x, \widehat{\mathbf{w}}) \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\vartheta}(x)}} \mathbf{N}_{K_{p,s}}^* = \frac{\widehat{\mathbf{b}}_{p,s}(x)' \widehat{\eta}_{0,1}(x, \widehat{\mathbf{w}}) \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\vartheta}(x)}} \mathbf{N}_{K_{p,s}}^*, \\ \widehat{Z}_{\zeta,p}(x) &= \frac{\widehat{\mathbf{b}}_{p,0}^{(1)}(x)' \widehat{\mathbf{T}}'_s \widehat{\eta}_{0,1}(x, \widehat{\mathbf{w}}) \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\zeta}(x)}} \mathbf{N}_{K_{p,s}}^* = \frac{\widehat{\mathbf{b}}_{p,s}^{(1)}(x)' \widehat{\eta}_{0,1}(x, \widehat{\mathbf{w}}) \widehat{\mathbf{Q}}^{-1} \widehat{\boldsymbol{\Sigma}}^{1/2}}{\sqrt{\widehat{\Omega}_{\zeta}(x)}} \mathbf{N}_{K_{p,s}}^*, \end{aligned}$$

where $\mathbf{N}_{K_{p,s}}^*$ denotes a $K_{p,s}$ -dimensional standard normal vector independent of the data $\mathbf{D}$ and the partition $\widehat{\Delta}$.

For ease of presentation, we will always require a fast convergence rate of $\widehat{\mathbf{w}}$ hereafter: $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(a_n^{-1}\sqrt{J/n})$. Nevertheless, note that as shown in Theorem SA-3.5, such a rate restriction on $\widehat{\mathbf{w}}$ can be different for inference of $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$ and are unnecessary for inference of $\mu_0^{(v)}(x)$. Also, in the following theorem, the infeasible processes $\bar{Z}$ are understood as conditionally equal-in-distribution copies driven by the same external Gaussian vector $\mathbf{N}_{K_{p,s}}^*$ as their feasible counterparts, on a sufficiently rich probability space.

Theorem SA-3.6 (Plug-in Approximation). *Suppose that the conditions of Theorem SA-3.1 and Assumption SA-HLE(ii) hold,*

$$\begin{aligned} \frac{J(\log n)^2}{n^{1-\frac{2}{\nu}}} + \left(\frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} \right)^{1/2} + nJ^{-2p-3} + \frac{\ell_{\gamma,n}^2}{J^{p+1}} + nJ^{-1}d_{\gamma}\mathfrak{r}_{\gamma}^2 &= o(a_n^{-2}), \\ \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} &= o(1), \quad \text{and} \quad \|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(a_n^{-1}\sqrt{J/n}). \end{aligned}$$

Then on a properly enriched probability space, there exists a $K_{p,s}$ -dimensional standard normal random vector $\mathbf{N}_{K_{p,s}}^*$ independent of $\mathbf{D}$ and $\widehat{\Delta}$ such that for any $\xi > 0$,

- (i) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\mu^{(v)},p}(x) - \bar{Z}_{\mu^{(v)},p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1),$
- (ii) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\vartheta,p}(x) - \bar{Z}_{\vartheta,p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1),$
- (iii) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\zeta,p}(x) - \bar{Z}_{\zeta,p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1).$

If Assumption SA-RP(ii) also holds with $\mathfrak{r}_{\text{RP}} = o(a_n^{-1}(\log n)^{-1/2})$, then

- (iv) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\mu^{(v)},p}(x) - Z_{\mu^{(v)},p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1),$
- (v) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\vartheta,p}(x) - Z_{\vartheta,p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1),$
- (vi) $\mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\zeta,p}(x) - Z_{\zeta,p}(x)| > \xi a_n^{-1} \middle| \mathbf{D}, \widehat{\Delta}\right) = o_{\mathbb{P}}(1).$

Remark SA-3.8 (Improvements over literature). Theorems SA-3.5 and SA-3.6 provide empirical researchers with powerful tools for uniform inference based on binscatter methods. Importantly, we allow for random partitioning schemes, formally take into account the potential randomness of the partition, and construct a novel strong approximation of nonlinear binscatter-based M-estimators under mild rate restrictions. For fixed d_{γ} , $a_n = \sqrt{\log n}$ and $\nu \geq 4$, we require $J^{\frac{8}{3}}/n = o(1)$, up

to $\log n$ terms. In the literature, similar results were only available in some special cases under stringent rate restrictions. For instance, [Belloni et al. \(2019\)](#) considers strong approximations of more general series-based quantile regression estimators. For the binscatter basis considered in this paper, their Theorem 11 can be applied to construct strong approximation of the t -statistic process based on pivotal coupling that achieves the approximation rate $a_n^{-1} = n^{-\varepsilon'}$ under $J^4/n^{1-\varepsilon} = o(1)$ for some constants $\varepsilon, \varepsilon' > 0$, whereas their Theorem 12 can be used to construct strong approximation based on Gaussian processes under $J^5/n^{1-\varepsilon} = o(1)$. It should be noted that their notion of strong approximation is stronger than ours in the sense that it holds uniformly over both the evaluation point $x \in \mathcal{X}$ and the desired quantile $u \in \mathcal{U}$ for a compact set of quantile indices $\mathcal{U} \subset (0, 1)$. On the other hand, our methods allow for other loss functions (e.g., Huber regression), a large class of random partitions, and semi-linear covariate adjustment, leading to new results that were previously unavailable in the literature. $\square$

Theorems [SA-3.5](#) and [SA-3.6](#) offer a way to approximate the distribution of the *whole* t -statistic process based on $\hat{\mu}^{(v)}(\cdot)$, $\hat{\vartheta}(\cdot, \hat{\mathbf{w}})$ or $\hat{\zeta}(\cdot, \hat{\mathbf{w}})$. A direct application of these results is the distributional approximation to the suprema of these t -statistic processes. In subsequent results we take $a_n = \sqrt{\log J}$.

Theorem SA-3.7 (Supremum Approximation). *Suppose that the conditions of Theorem [SA-3.1](#) and Assumptions [SA-HLE\(ii\)](#) and [SA-RP\(ii\)](#) hold,*

$$\begin{aligned} & \frac{J(\log n)^2}{n^{1-\frac{2}{\nu}}} + \left(\frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} \right)^{1/2} + nJ^{-2p-3} + \frac{\ell_{\gamma,n}^2}{J^{p+1}} + nJ^{-1}d_\gamma \mathbf{r}_\gamma^2 = o((\log J)^{-1}), \\ & \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} = o(1), \quad \|\hat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}\left(\sqrt{\frac{J}{n \log J}}\right), \quad \text{and} \quad \mathbf{r}_{\text{RP}} = o\left(\frac{1}{\sqrt{\log n \log J}}\right). \end{aligned}$$

Then,

$$\begin{aligned} & \sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| \leq u\right) - \mathbb{P}\left(\sup_{x \in \mathcal{X}} |\hat{Z}_{\mu^{(v)},p}(x)| \leq u \mid \mathbf{D}, \hat{\Delta}\right) \right| = o_{\mathbb{P}}(1), \\ & \sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\vartheta,p}(x)| \leq u\right) - \mathbb{P}\left(\sup_{x \in \mathcal{X}} |\hat{Z}_{\vartheta,p}(x)| \leq u \mid \mathbf{D}, \hat{\Delta}\right) \right| = o_{\mathbb{P}}(1), \quad \text{and} \\ & \sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x \in \mathcal{X}} |T_{\zeta,p}(x)| \leq u\right) - \mathbb{P}\left(\sup_{x \in \mathcal{X}} |\hat{Z}_{\zeta,p}(x)| \leq u \mid \mathbf{D}, \hat{\Delta}\right) \right| = o_{\mathbb{P}}(1). \end{aligned}$$

SA-3.6 Confidence Bands

Let

$$\begin{aligned}\widehat{I}_{\mu^{(v)},p}(x) &= \left[\widehat{\mu}^{(v)}(x) \pm \mathfrak{c}_{\mu^{(v)}} \sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)/n} \right], \\ \widehat{I}_{\vartheta,p}(x, \mathbf{w}) &= \left[\widehat{\vartheta}(x, \widehat{\mathbf{w}}) \pm \mathfrak{c}_{\vartheta} \sqrt{\widehat{\Omega}_{\vartheta}(x)/n} \right] \quad \text{and} \\ \widehat{I}_{\zeta,p}(x, \mathbf{w}) &= \left[\widehat{\zeta}(x, \widehat{\mathbf{w}}) \pm \mathfrak{c}_{\zeta} \sqrt{\widehat{\Omega}_{\zeta}(x)/n} \right]\end{aligned}$$

be confidence bands for $\mu_0^{(v)}(\cdot)$, $\vartheta_0(\cdot, \mathbf{w})$ and $\zeta_0(\cdot, \mathbf{w})$ respectively, where $\mathfrak{c}_{\mu^{(v)}}$, $\mathfrak{c}_{\vartheta}$ and $\mathfrak{c}_{\zeta}$ are corresponding critical values to be specified. Recall that $\mathbf{w}$ here is taken as a fixed evaluation point for the control variables, and these bands are constructed based on a certain choice of J and the p th-order binscatter basis. Using the previous results, we have the following theorem.

Theorem SA-3.8. *Suppose that the conditions of Theorem SA-3.1 and Assumption SA-HLE(ii) hold,*

$$\begin{aligned}\frac{J(\log n)^2}{n^{1-\frac{2}{\nu}}} + \left(\frac{J(\log n)^3 \ell_{\gamma,n}^4}{n} \right)^{1/2} + nJ^{-2p-3} + \frac{\ell_{\gamma,n}^2}{J^{p+1}} + nJ^{-1}d_{\gamma}\mathfrak{r}_{\gamma}^2 &= o((\log J)^{-1}), \\ \frac{J^{\frac{2\nu}{\nu-1}}(\log n)^{\frac{\nu}{\nu-1}}}{n} &= o(1), \quad \text{and} \quad \|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}\left(\sqrt{\frac{J}{n \log J}}\right).\end{aligned}$$

(i) *If $\mathfrak{c}_{\mu^{(v)}} = \inf \left\{ c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\widehat{Z}_{\mu^{(v)},p}(x)| \leq c \mid \mathbf{D}, \widehat{\Delta}] \geq 1 - \alpha \right\}$, then*

$$\mathbb{P}\left[\mu_0^{(v)}(x) \in \widehat{I}_{\mu^{(v)},p}(x), \text{ for all } x \in \mathcal{X}\right] = 1 - \alpha + o(1).$$

(ii) *If $\mathfrak{c}_{\vartheta} = \inf \left\{ c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\widehat{Z}_{\vartheta,p}(x)| \leq c \mid \mathbf{D}, \widehat{\Delta}] \geq 1 - \alpha \right\}$, then*

$$\mathbb{P}\left[\vartheta_0(x, \mathbf{w}) \in \widehat{I}_{\vartheta,p}(x, \mathbf{w}), \text{ for all } x \in \mathcal{X}\right] = 1 - \alpha + o(1).$$

(iii) *If $\mathfrak{c}_{\zeta} = \inf \left\{ c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\widehat{Z}_{\zeta,p}(x)| \leq c \mid \mathbf{D}, \widehat{\Delta}] \geq 1 - \alpha \right\}$, then*

$$\mathbb{P}\left[\zeta_0(x, \mathbf{w}) \in \widehat{I}_{\zeta,p}(x, \mathbf{w}), \text{ for all } x \in \mathcal{X}\right] = 1 - \alpha + o(1).$$

Remark SA-3.9. The above results construct valid uniform confidence bands for nonlinear

binscatter-based M-estimators under mild rate restrictions. Specifically, when d_γ is fixed and $\nu \geq 4$, we require $J^{\frac{8}{3}}/n = o(1)$, up to $\log n$ terms. In contrast, Belloni et al. (2019) considers more general series-based quantile regression estimators, and Theorem 15 therein can be used to construct confidence bands for binscatter estimators via various resampling methods under $J^4/n^{1-\varepsilon} = o(1)$ for some $\varepsilon > 0$. Furthermore, our results allow for random partitioning schemes, formally accounting for their randomness and generic structure. The key condition imposed on the partition for the validity of confidence bands is Assumption SA-RP(i), but the “convergence” of the random partition (Assumption SA-RP(ii)) is not necessary. $\perp$

SA-3.7 Parametric Specification Tests

We can test parametric specifications of and shape restrictions on $\mu_0^{(v)}(x)$, $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$. Before we present the results, some clarifications are needed. We will use $\bar{\gamma} \in \mathbb{R}^{d_\gamma}$ to denote the (non-random) benchmark coefficient corresponding to $\hat{\gamma}$. $\bar{\gamma}$ may vary with n , although this dependence is suppressed. Under the coordinatewise support bound on $\mathbf{w}$, the contribution of estimating the benchmark index is bounded by

$$\sqrt{d_\gamma} \|\tilde{\gamma} - \bar{\gamma}\| + \|\hat{\mathbf{w}} - \mathbf{w}\| \|\bar{\gamma}\|,$$

so the displayed function-level convergence conditions below can, for example, be verified by requiring this quantity, together with the contribution from estimating $\bar{\theta}$, to have the corresponding stated order.

We first consider parametric specification tests. We introduce the following tests:

$$\begin{aligned} \dot{H}_0^{\mu^{(v)}} : \quad & \sup_{x \in \mathcal{X}} \left| \mu_0^{(v)}(x) - m^{(v)}(x; \boldsymbol{\theta}) \right| = 0, \quad \text{for some } \boldsymbol{\theta}, \quad \text{vs.} \\ \dot{H}_A^{\mu^{(v)}} : \quad & \sup_{x \in \mathcal{X}} \left| \mu_0^{(v)}(x) - m^{(v)}(x; \boldsymbol{\theta}) \right| > 0, \quad \text{for all } \boldsymbol{\theta}. \end{aligned}$$

where $m(x; \boldsymbol{\theta})$ is some known function depending on some finite dimensional parameter $\boldsymbol{\theta}$. This testing problem can be viewed as a two-sided test where the equality between two functions holds *uniformly* over $x \in \mathcal{X}$. In this case, we introduce $\tilde{\theta}$ and $\tilde{\gamma}$ as consistent estimators of $\boldsymbol{\theta}$ and γ_0

under $\dot{H}_0^{\mu^{(v)}}$. Then we rely on the following test statistic:

$$\dot{T}_{\mu^{(v)},p}(x) := \frac{\widehat{\mu}^{(v)}(x) - m^{(v)}(x; \tilde{\theta})}{\sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)},p}(x)| > \mathbf{c}_{\mu^{(v)}}$ for some critical value $\mathbf{c}_{\mu^{(v)}}$.

Similarly, to test the specification of $\vartheta_0(x, \mathbf{w})$, we introduce

$$\begin{aligned} \dot{H}_0^\vartheta : \quad & \sup_{x \in \mathcal{X}} \left| \vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) \right| = 0, \quad \text{for some } \boldsymbol{\theta}, \quad vs. \\ \dot{H}_A^\vartheta : \quad & \sup_{x \in \mathcal{X}} \left| \vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) \right| > 0, \quad \text{for all } \boldsymbol{\theta}. \end{aligned}$$

where $M(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) = \eta(m(x; \boldsymbol{\theta}) + \mathbf{w}'\gamma_0)$. We rely on the following test statistic:

$$\dot{T}_{\vartheta,p}(x) := \frac{\widehat{\vartheta}(x, \widehat{\mathbf{w}}) - M(x, \widehat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\gamma})}{\sqrt{\widehat{\Omega}_\vartheta(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} |\dot{T}_{\vartheta,p}(x)| > \mathbf{c}_\vartheta$ for some critical value $\mathbf{c}_\vartheta$.

To test the specification of $\zeta_0(x, \mathbf{w})$, we introduce

$$\begin{aligned} \dot{H}_0^\zeta : \quad & \sup_{x \in \mathcal{X}} \left| \zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) \right| = 0, \quad \text{for some } \boldsymbol{\theta}, \quad vs. \\ \dot{H}_A^\zeta : \quad & \sup_{x \in \mathcal{X}} \left| \zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) \right| > 0, \quad \text{for all } \boldsymbol{\theta}. \end{aligned}$$

where $M^{(1)}(x, \mathbf{w}; \boldsymbol{\theta}, \gamma_0) := \eta^{(1)}(m(x; \boldsymbol{\theta}) + \mathbf{w}'\gamma_0)m^{(1)}(x; \boldsymbol{\theta})$. We rely on the following test statistic:

$$\dot{T}_{\zeta,p}(x) := \frac{\widehat{\zeta}(x, \widehat{\mathbf{w}}) - M^{(1)}(x, \widehat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\gamma})}{\sqrt{\widehat{\Omega}_\zeta(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} |\dot{T}_{\zeta,p}(x)| > \mathbf{c}_\zeta$ for some critical value $\mathbf{c}_\zeta$.

Theorem SA-3.9 (Specification Tests). *Suppose that the conditions in Theorem [SA-3.8](#) hold.*

(i) Let $\mathbf{c}_{\mu^{(v)}} = \inf\{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\widehat{Z}_{\mu^{(v)},p}(x)| \leq c | \mathbf{D}, \widehat{\Delta}] \geq 1 - \alpha\}$.

Under $\dot{H}_0^{\mu^{(v)}}$, if $\sup_{x \in \mathcal{X}} |\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})| = o_{\mathbb{P}}\left(\sqrt{\frac{J^{1+2v}}{n \log J}}\right)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)}, p}(x)| > \mathfrak{c}_{\mu^{(v)}}\right] = \alpha.$$

Under $\dot{H}_A^{\mu^{(v)}}$, if there exist some fixed $\bar{\boldsymbol{\theta}}$ such that $\sup_{x \in \mathcal{X}} |m^{(v)}(x; \tilde{\boldsymbol{\theta}}) - m^{(v)}(x; \bar{\boldsymbol{\theta}})| = o_{\mathbb{P}}(1)$,

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} |\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})| > 0,$$

and $J^v \left(\frac{J \log J}{n}\right)^{1/2} = o(1)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)}, p}(x)| > \mathfrak{c}_{\mu^{(v)}}\right] = 1.$$

(ii) Let $\mathfrak{c}_{\vartheta} = \inf\{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\hat{Z}_{\vartheta, p}(x)| \leq c | \mathbf{D}, \hat{\Delta}] \geq 1 - \alpha\}$.

Under $\dot{H}_0^{\vartheta}$, if $\sup_{x \in \mathcal{X}} |\vartheta_0(x, \mathbf{w}) - M(x, \hat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\gamma}})| = o_{\mathbb{P}}\left(\sqrt{\frac{J}{n \log J}}\right)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\vartheta, p}(x)| > \mathfrak{c}_{\vartheta}\right] = \alpha.$$

Under $\dot{H}_A^{\vartheta}$, if there exist some fixed $\bar{\boldsymbol{\theta}}$ and a (non-random) benchmark $\bar{\boldsymbol{\gamma}} \in \mathbb{R}^{d_{\gamma}}$ such that $\sup_n \|\bar{\boldsymbol{\gamma}}\| < \infty$ and the benchmark index $m(x; \bar{\boldsymbol{\theta}}) + \mathbf{w}' \bar{\boldsymbol{\gamma}}$ is uniformly contained in a fixed compact subset of the interior of the domain of η , $\sup_{x \in \mathcal{X}} |M(x, \hat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\gamma}}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| = o_{\mathbb{P}}(1)$,

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} |\vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| > 0,$$

and $\left(\frac{J \log J}{n}\right)^{1/2} = o(1)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\vartheta, p}(x)| > \mathfrak{c}_{\vartheta}\right] = 1.$$

(iii) Let $\mathfrak{c}_{\zeta} = \inf\{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} |\hat{Z}_{\zeta, p}(x)| \leq c | \mathbf{D}, \hat{\Delta}] \geq 1 - \alpha\}$.

Under $\dot{H}_0^\zeta$, if $\sup_{x \in \mathcal{X}} |\zeta_0(x, \mathbf{w}) - M^{(1)}(x, \hat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\gamma}})| = o_{\mathbb{P}}\left(\sqrt{\frac{J^3}{n \log J}}\right)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\zeta,p}(x)| > \mathfrak{c}_\zeta\right] = \alpha.$$

Under $\dot{H}_A^\zeta$, if there exist some fixed $\bar{\boldsymbol{\theta}}$ and a (non-random) benchmark $\bar{\boldsymbol{\gamma}} \in \mathbb{R}^{d_\gamma}$ such that $\sup_n \|\bar{\boldsymbol{\gamma}}\| < \infty$ and the benchmark index $m(x; \bar{\boldsymbol{\theta}}) + \mathbf{w}'\bar{\boldsymbol{\gamma}}$ is uniformly contained in a fixed compact subset of the interior of the domain of η , $\sup_{x \in \mathcal{X}} |M^{(1)}(x, \hat{\mathbf{w}}; \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\gamma}}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| = o_{\mathbb{P}}(1)$,

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} |\zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| > 0,$$

and $J\left(\frac{J \log J}{n}\right)^{1/2} = o(1)$, then

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\dot{T}_{\zeta,p}(x)| > \mathfrak{c}_\zeta\right] = 1.$$

SA-3.8 Shape Restriction Tests

The third application of our results is to test certain shape restrictions on $\mu_0^{(v)}(x)$, $\vartheta_0(x, \mathbf{w})$ and $\zeta_0(x, \mathbf{w})$. To be specific, consider the following problem:

$$\begin{aligned} \ddot{H}_0^{\mu^{(v)}} : \sup_{x \in \mathcal{X}} (\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})) &\leq 0 \text{ for certain } \bar{\boldsymbol{\theta}} \quad \text{v.s.} \\ \ddot{H}_A^{\mu^{(v)}} : \sup_{x \in \mathcal{X}} (\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})) &> 0 \text{ for } \bar{\boldsymbol{\theta}}. \end{aligned}$$

This testing problem can be viewed as a one-sided test where the inequality holds *uniformly* over $x \in \mathcal{X}$. Importantly, it should be noted that under both $\ddot{H}_0^{\mu^{(v)}}$ and $\ddot{H}_A^{\mu^{(v)}}$, we fix $\bar{\boldsymbol{\theta}}$. In such a case, we introduce $\tilde{\boldsymbol{\theta}}$ as a consistent estimator of $\bar{\boldsymbol{\theta}}$ under both $\ddot{H}_0^{\mu^{(v)}}$ and $\ddot{H}_A^{\mu^{(v)}}$. Then we will rely on the following test statistic:

$$\ddot{T}_{\mu^{(v)},p}(x) := \frac{\hat{\mu}^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)},p}(x) > \mathfrak{c}_{\mu^{(v)}}$ for some critical value $\mathfrak{c}_{\mu^{(v)}}$.

For the following transformed targets, we fix $\bar{\boldsymbol{\theta}}$ and use the same benchmark coefficient $\bar{\boldsymbol{\gamma}}$ under the null and alternative, and introduce $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\gamma}}$ as consistent estimators of $\bar{\boldsymbol{\theta}}$ and $\bar{\boldsymbol{\gamma}}$. Similarly,

define the test for the shape of $\vartheta_0(x, \mathbf{w})$:

$$\ddot{\mathbf{H}}_0^\vartheta : \sup_{x \in \mathcal{X}} (\vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\gamma})) \leq 0 \text{ for certain } \bar{\boldsymbol{\theta}} \text{ and } \bar{\gamma} \quad \text{v.s.}$$

$$\ddot{\mathbf{H}}_A^\vartheta : \sup_{x \in \mathcal{X}} (\vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\gamma})) > 0 \text{ for } \bar{\boldsymbol{\theta}} \text{ and } \bar{\gamma}.$$

We will rely on the following test statistic:

$$\ddot{T}_{\vartheta,p}(x) := \frac{\widehat{\vartheta}(x, \widehat{\mathbf{w}}) - M(x, \widehat{\mathbf{w}}; \widetilde{\boldsymbol{\theta}}, \widetilde{\gamma})}{\sqrt{\widehat{\Omega}_\vartheta(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} \ddot{T}_{\vartheta,p}(x) > \mathbf{c}_\vartheta$ for some critical value $\mathbf{c}_\vartheta$.

Also, define the test for the shape of $\zeta_0(x, \mathbf{w})$:

$$\ddot{\mathbf{H}}_0^\zeta : \sup_{x \in \mathcal{X}} (\zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\gamma})) \leq 0 \text{ for certain } \bar{\boldsymbol{\theta}} \text{ and } \bar{\gamma} \quad \text{v.s.}$$

$$\ddot{\mathbf{H}}_A^\zeta : \sup_{x \in \mathcal{X}} (\zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\gamma})) > 0 \text{ for } \bar{\boldsymbol{\theta}} \text{ and } \bar{\gamma}.$$

We will rely on the following test statistic:

$$\ddot{T}_{\zeta,p}(x) := \frac{\widehat{\zeta}(x, \widehat{\mathbf{w}}) - M^{(1)}(x, \widehat{\mathbf{w}}; \widetilde{\boldsymbol{\theta}}, \widetilde{\gamma})}{\sqrt{\widehat{\Omega}_\zeta(x)/n}}.$$

The null hypothesis is rejected if $\sup_{x \in \mathcal{X}} \ddot{T}_{\zeta,p}(x) > \mathbf{c}_\zeta$ for some critical value $\mathbf{c}_\zeta$.

The following theorem characterizes the size and power of such tests.

Theorem SA-3.10 (Shape Restriction Tests). *Suppose that the conditions in Theorem SA-3.8 hold.*

$$(i) \text{ Assume } \sup_{x \in \mathcal{X}} |m^{(v)}(x; \widetilde{\boldsymbol{\theta}}) - m^{(v)}(x; \bar{\boldsymbol{\theta}})| = o_{\mathbb{P}}\left(\sqrt{\frac{J^{1+2v}}{n \log J}}\right).$$

$$\text{Let } \mathbf{c}_{\mu^{(v)}} = \inf\{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} \widehat{Z}_{\mu^{(v)},p}(x) \leq c | \mathbf{D}, \widehat{\Delta}] \geq 1 - \alpha\}.$$

$$\text{Under } \ddot{\mathbf{H}}_0^{\mu^{(v)}},$$

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)},p}(x) > \mathbf{c}_{\mu^{(v)}}\right] \leq \alpha.$$

Under $\ddot{H}_A^{\mu^{(v)}}$, if

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} \{\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})\} > 0,$$

$$\text{and } J^v \left(\frac{J \log J}{n} \right)^{1/2} = o(1),$$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)}, p}(x) > \mathfrak{c}_{\mu^{(v)}} \right] = 1.$$

(ii) Assume that the (non-random) benchmark $\bar{\boldsymbol{\gamma}} \in \mathbb{R}^{d_\gamma}$ satisfies $\sup_n \|\bar{\boldsymbol{\gamma}}\| < \infty$ and that the benchmark index $m(x; \bar{\boldsymbol{\theta}}) + \mathbf{w}'\bar{\boldsymbol{\gamma}}$ is uniformly contained in a fixed compact subset of the interior of the domain of η . Assume $\sup_{x \in \mathcal{X}} |M(x, \hat{\mathbf{w}}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| = o_{\mathbb{P}} \left(\sqrt{\frac{J}{n \log J}} \right)$. Let $\mathfrak{c}_\vartheta = \inf \{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} \hat{Z}_{\vartheta, p}(x) \leq c | \mathbf{D}, \hat{\Delta}] \geq 1 - \alpha\}$.

Under $\ddot{H}_0^\vartheta$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{x \in \mathcal{X}} \ddot{T}_{\vartheta, p}(x) > \mathfrak{c}_\vartheta \right] \leq \alpha.$$

Under $\ddot{H}_A^\vartheta$, if

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} \{\vartheta_0(x, \mathbf{w}) - M(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})\} > 0,$$

$$\text{and } \left(\frac{J \log J}{n} \right)^{1/2} = o(1),$$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{x \in \mathcal{X}} \ddot{T}_{\vartheta, p}(x) > \mathfrak{c}_\vartheta \right] = 1.$$

(iii) Assume that the (non-random) benchmark $\bar{\boldsymbol{\gamma}} \in \mathbb{R}^{d_\gamma}$ satisfies $\sup_n \|\bar{\boldsymbol{\gamma}}\| < \infty$ and that the benchmark index $m(x; \bar{\boldsymbol{\theta}}) + \mathbf{w}'\bar{\boldsymbol{\gamma}}$ is uniformly contained in a fixed compact subset of the interior of the domain of η . Assume $\sup_{x \in \mathcal{X}} |M^{(1)}(x, \hat{\mathbf{w}}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})| = o_{\mathbb{P}} \left(\sqrt{\frac{J^3}{n \log J}} \right)$. Let $\mathfrak{c}_\zeta = \inf \{c \in \mathbb{R}_+ : \mathbb{P}[\sup_{x \in \mathcal{X}} \hat{Z}_{\zeta, p}(x) \leq c | \mathbf{D}, \hat{\Delta}] \geq 1 - \alpha\}$.

Under $\ddot{H}_0^\zeta$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{x \in \mathcal{X}} \ddot{T}_{\zeta, p}(x) > \mathfrak{c}_\zeta \right] \leq \alpha.$$

Under $\ddot{H}_A^\zeta$, if

$$\liminf_{n \rightarrow \infty} \sup_{x \in \mathcal{X}} \{\zeta_0(x, \mathbf{w}) - M^{(1)}(x, \mathbf{w}; \bar{\boldsymbol{\theta}}, \bar{\boldsymbol{\gamma}})\} > 0,$$

and $J\left(\frac{J \log J}{n}\right)^{1/2} = o(1)$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left[\sup_{x \in \mathcal{X}} \ddot{T}_{\zeta,p}(x) > \mathfrak{c}_\zeta\right] = 1.$$

Remark SA-3.10 (Improvements over literature). The results in Sections SA-3.6–SA-3.8 are new to the literature, even in the case of non-random partitioning and without covariate-adjustments, because they take advantage of the specific binscatter structure (i.e., locally bounded series basis). Furthermore, relative to prior work, our results allow for a large class of random partitioning schemes, formally take into account the potential randomness of the partition, account for the generalized semi-linear structure, and consider an array of possibly nonlinear estimation and inference problems. In particular, the approach taken in Theorems SA-3.5 and SA-3.7 to establish strong approximation and related distributional approximations for nonlinear binscatter statistics may be of independent interest. The key condition imposed on the partition for uniform inference (confidence bands and hypothesis testing) is Assumption SA-RP(i), while “convergence” of the random partition (Assumption SA-RP(ii)) is not required. $\lrcorner$

SA-3.9 Convergence Rate of Covariate Coefficients

In this section, we establish the convergence rate of $\hat{\gamma}$ under primitive conditions. Set $\hat{\mathbf{b}}_i = \hat{\mathbf{b}}_{p,s}(x_i)$, $\varpi_i = \mu_0(x_i) + \mathbf{w}_i' \gamma_0$, and

$$\begin{aligned} \mathcal{X}_i &= \mathcal{X}(x_i, \mathbf{w}_i) = \Psi_1(x_i, \mathbf{w}_i; \eta_i) \eta_{i,1}^2, & \bar{\mathbf{Q}} &= \mathbb{E}_n[\mathcal{X}_i \hat{\mathbf{b}}_i \hat{\mathbf{b}}_i'], \\ \boldsymbol{\pi}_w(x_i) &= \mathbb{E}_n[\mathbf{w}_i \hat{\mathbf{b}}_i' \mathcal{X}_i] \bar{\mathbf{Q}}^{-1} \hat{\mathbf{b}}_i, & \tilde{\mathbf{w}}_i &= \mathbf{w}_i - \boldsymbol{\pi}_w(x_i), \\ \hat{\mathbf{G}}_w &= \mathbb{E}_n[\mathcal{X}_i \tilde{\mathbf{w}}_i \tilde{\mathbf{w}}_i'], & \boldsymbol{\pi}_{w,0}(x) &= \frac{\mathbb{E}[\mathcal{X}_i \mathbf{w}_i \mid x_i = x]}{\mathbb{E}[\mathcal{X}_i \mid x_i = x]}, \\ \boldsymbol{\pi}_{w,J}(x) &= \mathbb{E}_{\hat{\Delta}}[\mathcal{X}_i \mathbf{w}_i \hat{\mathbf{b}}_i'] \mathbb{E}_{\hat{\Delta}}[\mathcal{X}_i \hat{\mathbf{b}}_i \hat{\mathbf{b}}_i']^{-1} \hat{\mathbf{b}}_{p,s}(x). \end{aligned}$$

Define the directional projection error by

$$\mathfrak{a}_{\pi,J} = \sup_{\|\boldsymbol{\alpha}\|=1} \left\{ \mathbb{E}_{\hat{\Delta}} \left[\mathbb{E}[\mathcal{X}_i \mid x_i] \{ \boldsymbol{\alpha}' (\boldsymbol{\pi}_{w,0}(x_i) - \boldsymbol{\pi}_{w,J}(x_i)) \}^2 \right] \right\}^{1/2}, \quad (\text{SA-3.1})$$

and set

$$\begin{aligned}\boldsymbol{\theta}_0 &= \widehat{\boldsymbol{\beta}}_0 + \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_i \mathbf{w}_i' \boldsymbol{\varkappa}_i] \boldsymbol{\gamma}_0, & u_i(\boldsymbol{\theta}) &= \widehat{\mathbf{b}}_i'(\boldsymbol{\theta} - \boldsymbol{\theta}_0), \\ \mathbf{r}_{\theta,n} &= \sqrt{J/n} + J^{-p-1}, \\ \mathbf{r}_{\gamma,n} &= \frac{\sqrt{\bar{\lambda}_{w,n} d_\gamma/n} + J^{-p-1} \mathbf{a}_{\pi,J} + \sqrt{\bar{\lambda}_{w,n} \mathbf{r}_{\theta,n}^2} + \sqrt{\bar{\lambda}_{w,n} J \mathbf{r}_{\theta,n} \log n/n} + J \sqrt{d_\gamma \log n/n}}{\underline{\lambda}_{w,n}}.\end{aligned}$$

Lemma SA-3.6 (Covariate Coefficients Rate). *Suppose that $\widehat{\boldsymbol{\gamma}}$ is obtained from the joint estimator in (SA-2.2). Suppose that Assumptions SA-DGP, SA-SM, and SA-RP(i) hold uniformly over the triangular array with $\mathbf{w}_i \in \mathbb{R}^{d_\gamma}$ and $\mathcal{W} \subset \mathbb{R}^{d_\gamma}$. Suppose also that $d_\gamma = o(J)$. Assume the following additional conditions:*

(i) *There exist deterministic sequences $0 < \underline{\lambda}_{w,n} \leq \bar{\lambda}_{w,n} < \infty$ such that, w.p.a. 1,*

$$\underline{\lambda}_{w,n} \leq \lambda_{\min}(\widehat{\mathbf{G}}_w) \leq \lambda_{\max}(\widehat{\mathbf{G}}_w) \leq \bar{\lambda}_{w,n}.$$

(ii) *The sample projection satisfies*

$$\left\| \mathbb{E}_n[\boldsymbol{\varkappa}_i \widetilde{\mathbf{w}}_i \widehat{r}_{0,0}(x_i)] - \mathbb{E}_{\widehat{\Delta}}[\boldsymbol{\varkappa}_i \{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\} \widehat{r}_{0,0}(x_i)] \right\| \lesssim_{\mathbb{P}} J^{-p-1} \sqrt{\bar{\lambda}_{w,n} d_\gamma/n}.$$

(iii) *For every fixed $C < \infty$, uniformly over $\|\boldsymbol{\alpha}\| = 1$ and $\boldsymbol{\theta}$ such that $\mathbb{E}_n[u_i(\boldsymbol{\theta})^2] \leq C \mathbf{r}_{\theta,n}^2$ and $\max_i |u_i(\boldsymbol{\theta})| \leq C \sqrt{J} \mathbf{r}_{\theta,n}$,*

$$\mathbb{E}_n[(\boldsymbol{\alpha}' \widetilde{\mathbf{w}}_i)^2 u_i(\boldsymbol{\theta})^2] \lesssim_{\mathbb{P}} \bar{\lambda}_{w,n} \mathbb{E}_n[u_i(\boldsymbol{\theta})^2].$$

Moreover,

$$\mathbf{r}_{\theta,n} \left(\sqrt{J} + \sqrt{\frac{d_\gamma}{\underline{\lambda}_{w,n}}} \right) = o_{\mathbb{P}}(1), \quad \frac{\bar{\lambda}_{w,n}}{\underline{\lambda}_{w,n}} \left\{ \sqrt{J} \mathbf{r}_{\theta,n} + \sqrt{d_\gamma} \mathbf{r}_{\gamma,n} \right\} = o_{\mathbb{P}}(1).$$

(iv) The following rate restrictions hold:

$$\begin{aligned} \frac{J}{n^{1-1/\nu}} \left(\sqrt{J} + \sqrt{\frac{d_\gamma}{\lambda_{w,n}}} \right) &= o_{\mathbb{P}}(\mathbf{r}_{\theta,n}), & \frac{d_\gamma^{3/2}}{n^{1-1/\nu}} &= o_{\mathbb{P}}(\lambda_{w,n} \mathbf{r}_{\gamma,n}), \\ \frac{J^{3/2} \mathbf{r}_{\theta,n}}{n} \sqrt{d_\gamma} \left\{ n^{1/\nu} + \sqrt{J} \mathbf{r}_{\theta,n} + \sqrt{d_\gamma} \mathbf{r}_{\gamma,n} \right\} &= o_{\mathbb{P}}(\lambda_{w,n} \mathbf{r}_{\gamma,n}). \end{aligned}$$

Then $\|\hat{\gamma} - \gamma_0\| = O_{\mathbb{P}}(\mathbf{r}_{\gamma,n})$. If, in addition, there exists a deterministic sequence $\mathbf{r}_\gamma$ such that

$$\mathbf{r}_{\gamma,n} = O_{\mathbb{P}}(\mathbf{r}_\gamma), \quad \sqrt{d_\gamma} \mathbf{r}_\gamma = o\left(\sqrt{J/n} + J^{-p-1}\right),$$

then the rate condition on $\hat{\gamma}$ in Assumption [SA-HLE\(i\)](#) holds.

Remark SA-3.11 (Fast convergence of covariate coefficients). The last conclusion of Lemma [SA-3.6](#) verifies the fast-rate condition in Assumption [SA-HLE\(i\)](#) whenever $\mathbf{r}_{\gamma,n} = O_{\mathbb{P}}(\mathbf{r}_\gamma)$ and $\sqrt{d_\gamma} \mathbf{r}_\gamma = o(\sqrt{J/n} + J^{-p-1})$. As a simple benchmark, suppose that $\nu = 4$, the Gram matrix for the residualized controls is well conditioned, $\mathbf{r}_{\theta,n} \asymp \sqrt{J/n}$, and the projection and quadratic terms in $\mathbf{r}_{\gamma,n}$ are no larger than $\sqrt{d_\gamma/n}$. Then

$$\mathbf{r}_{\gamma,n} \asymp \sqrt{d_\gamma/n} + (J/n)^{3/4} \sqrt{\log n} + J \sqrt{d_\gamma} \log n/n.$$

Comparing $\sqrt{d_\gamma} \mathbf{r}_{\gamma,n}$ with $\sqrt{J/n}$, the first term in the display requires $d_\gamma = o(\sqrt{J})$, while each of the other two terms requires $d_\gamma^2 J (\log n)^2 / n = o(1)$. Because our analysis allows $\mathbf{w}$ to be generic, sharper conditions may be available when $\mathbf{w}$ has additional structure, such as sparsity or local support. $\square$

Remark SA-3.12 (Verification of Conditions in Lemma [SA-3.6](#)). We can verify conditions imposed in Lemma [SA-3.6](#) under more primitive conditions.

Condition (i). Suppose that there exist deterministic sequences $0 < c_n \leq C_n < \infty$ such that the following primitive covariance condition holds uniformly over $x \in \mathcal{X}$:

$$c_n \mathbf{I}_{d_\gamma} \leq \mathbb{E} \left[(\mathbf{w}_i - \mathbb{E}[\mathbf{w}_i | x_i])(\mathbf{w}_i - \mathbb{E}[\mathbf{w}_i | x_i])' \mid x_i = x \right] \leq C_n \mathbf{I}_{d_\gamma}. \quad (\text{SA-3.2})$$

Under [\(SA-3.2\)](#), since κ_i is bounded and bounded away from zero and $\pi_{w,0}(x)$ is the weighted

conditional mean, the conditional second-moment matrix of $\mathcal{X}_i^{1/2}\{\mathbf{w}_i - \boldsymbol{\pi}_{w,0}(x_i)\}$ has eigenvalues bounded below and above by constant multiples of c_n and C_n , respectively, uniformly over x . Suppose, in addition, that $\mathbf{a}_{\pi,J}^2 \lesssim C_n$. Coordinatewise boundedness of $\mathbf{w}_i$, local support of the binscatter basis, and the bounded row-sum norm of the inverse weighted Gram matrix imply the required projection stability. If the uniform within-bin matrix-concentration error is $o_{\mathbb{P}}(c_n)$, then condition (i) holds with $\underline{\lambda}_{w,n} \asymp c_n$ and $\bar{\lambda}_{w,n} \asymp C_n$. Under coordinatewise boundedness, a sufficient concentration restriction is $C_n d_\gamma J \log(J d_\gamma) / (n c_n^2) = o(1)$. When $c_n \asymp C_n \asymp 1$, this reduces to $d_\gamma J \log(J d_\gamma) / n = o(1)$.

Condition (ii). The difference in condition (ii) equals

$$(\mathbb{E}_n - \mathbb{E}_{\hat{\Delta}})[\mathcal{X}_i\{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\}\hat{r}_{0,0}(x_i)] - \mathbb{E}_n[\mathcal{X}_i\{\boldsymbol{\pi}_w(x_i) - \boldsymbol{\pi}_{w,J}(x_i)\}\hat{r}_{0,0}(x_i)].$$

The projection normal equations imply

$$\begin{aligned} \mathbb{E}_n[\mathcal{X}_i\{\boldsymbol{\pi}_w(x_i) - \boldsymbol{\pi}_{w,J}(x_i)\}\hat{r}_{0,0}(x_i)] &= (\mathbb{E}_n - \mathbb{E}_{\hat{\Delta}})[\mathcal{X}_i\{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\}\hat{\mathbf{b}}_i']\bar{\mathbf{Q}}^{-1} \\ &\quad \times \mathbb{E}_n[\mathcal{X}_i\hat{\mathbf{b}}_i'\hat{r}_{0,0}(x_i)]. \end{aligned}$$

Decomposing $\bar{\mathbf{Q}}^{-1}$ around $\{\mathbb{E}_{\hat{\Delta}}[\mathcal{X}_i\hat{\mathbf{b}}_i\hat{\mathbf{b}}_i']\}^{-1}$ and applying the uniform approximation bound, vector concentration, and projection stability bounds the resulting leading terms by $J^{-p-1}\sqrt{\bar{\lambda}_{w,n}d_\gamma/n}$.

The remaining term is

$$(\mathbb{E}_n - \mathbb{E}_{\hat{\Delta}})[\mathcal{X}_i\{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\}\hat{\mathbf{b}}_i'] \times \left(\bar{\mathbf{Q}}^{-1} - \{\mathbb{E}_{\hat{\Delta}}[\mathcal{X}_i\hat{\mathbf{b}}_i\hat{\mathbf{b}}_i']\}^{-1} \right) \mathbb{E}_n[\mathcal{X}_i\hat{\mathbf{b}}_i'\hat{r}_{0,0}(x_i)],$$

which is bounded by $J^{-p-1}J\sqrt{d_\gamma/n}$ and is of smaller order provided $J^2/(nC_n) = o(1)$. This condition reduces to $J^2/n = o(1)$ in the well-conditioned case. For a data-dependent partition, the concentration is applied uniformly over the admissible partition class.

Condition (iii). Under (SA-3.2), conditioning on x_i bounds the conditional second moment of the weighted conditional-mean residual $\mathbf{w}_i - \boldsymbol{\pi}_{w,0}(x_i)$ uniformly by C_n . Smoothness and projection stability control the approximation and sample projection errors $\boldsymbol{\pi}_{w,0}(x_i) - \boldsymbol{\pi}_{w,J}(x_i)$ and $\boldsymbol{\pi}_{w,J}(x_i) - \boldsymbol{\pi}_w(x_i)$, respectively. Iterated expectations and the same uniform concentration arguments therefore

give, uniformly over $\|\boldsymbol{\alpha}\| = 1$ and the stated neighborhood of $\boldsymbol{\theta}$,

$$\mathbb{E}_n[(\boldsymbol{\alpha}'\tilde{\mathbf{w}}_i)^2 u_i(\boldsymbol{\theta})^2] \lesssim_{\mathbb{P}} C_n \mathbb{E}_n[u_i(\boldsymbol{\theta})^2],$$

which is the first bound in condition (iii). Moreover,

$$\mathbf{r}_{\gamma,n} \asymp \frac{\sqrt{C_n d_\gamma/n} + J^{-p-1} \mathbf{a}_{\pi,J} + \sqrt{C_n} \mathbf{r}_{\theta,n}^2 + \sqrt{C_n J \mathbf{r}_{\theta,n} \log n/n} + J \sqrt{d_\gamma} \log n/n}{c_n}.$$

The last two terms account for crossing the finitely many discontinuities of a piecewise-Lipschitz score and may be omitted when $\psi^\dagger(\cdot)$ is globally Lipschitz. Thus, for general c_n and C_n , the remaining restrictions in condition (iii) require

$$\mathbf{r}_{\theta,n} \left(\sqrt{J} + \sqrt{d_\gamma/c_n} \right) = o_{\mathbb{P}}(1), \quad \frac{C_n}{c_n} \left\{ \sqrt{J} \mathbf{r}_{\theta,n} + \sqrt{d_\gamma} \mathbf{r}_{\gamma,n} \right\} = o_{\mathbb{P}}(1).$$

In the well-conditioned case $c_n \asymp C_n \asymp 1$, compact support and projection stability also give $\mathbf{a}_{\pi,J} \lesssim_{\mathbb{P}} \sqrt{d_\gamma}$. Therefore,

$$\sqrt{J} \mathbf{r}_{\theta,n} = o(1), \quad \sqrt{d_\gamma} \mathbf{r}_{\gamma,n} \lesssim_{\mathbb{P}} \frac{d_\gamma}{\sqrt{n}} + d_\gamma J^{-p-1} + \sqrt{d_\gamma} \mathbf{r}_{\theta,n}^2 + \sqrt{d_\gamma J \mathbf{r}_{\theta,n} \log n/n} + J d_\gamma \log n/n = o(1),$$

where the last relations follow from $J^2 \log n/n = o(1)$ and $d_\gamma = o(J)$. These relations imply both rate restrictions in condition (iii) in this well-conditioned case.

Condition (iv). For general c_n and C_n , condition (iv) retains the corresponding factors displayed in the lemma. In the well-conditioned case $c_n \asymp C_n \asymp 1$, it reduces to

$$\frac{J^{3/2}}{n^{1-1/\nu}} = o(\mathbf{r}_{\theta,n}), \quad \frac{d_\gamma^{3/2}}{n^{1-1/\nu}} = o(\mathbf{r}_{\gamma,n}), \quad \frac{J^{3/2} \sqrt{d_\gamma} \mathbf{r}_{\theta,n}}{n^{1-1/\nu}} = o(\mathbf{r}_{\gamma,n}).$$

For finite ν , all three relations follow, for example, if $J^{2\nu/(\nu-2)}/n = o(1)$, which also controls the corresponding d_γ terms because $d_\gamma = o(J)$. If the score is uniformly bounded, condition (iv) follows directly from $J^2/n = o(1)$ and $d_\gamma = o(J)$.

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Remark SA-3.13 (Projection approximation). The directional approximation error $\mathbf{a}_{\pi,J}$ is needed

only to bound the population component of $\mathbb{E}_n[\varkappa_i \widehat{\mathbf{w}}_i \widehat{r}_{0,0}(x_i)]$. A primitive sufficient condition is, for some $s_\pi > 0$ and a deterministic sequence $C_{\pi,n} < \infty$,

$$\max_{1 \leq k \leq d_\gamma} \sup_{x \in \mathcal{X}} |\pi_{w,0,k}(x) - \pi_{w,J,k}(x)| \lesssim C_{\pi,n} J^{-s_\pi}. \quad (\text{SA-3.3})$$

Condition (SA-3.3) and boundedness of $\mathbb{E}[\varkappa_i | x_i]$ give $\mathbf{a}_{\pi,J} \lesssim \sqrt{d_\gamma} C_{\pi,n} J^{-s_\pi}$, so the corresponding contribution to $\mathbf{r}_{\gamma,n}$ is $\sqrt{d_\gamma} C_{\pi,n} J^{-p-1-s_\pi} / c_n$. In particular, the condition $\mathbf{a}_{\pi,J}^2 \lesssim C_n$ in the preceding remark follows if $d_\gamma C_{\pi,n}^2 J^{-2s_\pi} \lesssim C_n$. The displayed L_∞ approximation follows from coordinatewise smoothness of $\pi_{w,0}(x)$ with bound $C_{\pi,n}$, sparsity of the binscatter basis, and stability of the weighted projection.

In general, it is not enough to approximate $\mathbb{E}[\mathbf{w}_i | x_i = x]$, because

$$\pi_{w,0}(x) - \mathbb{E}[\mathbf{w}_i | x_i = x] = \frac{\mathbb{E}[\varkappa_i \{\mathbf{w}_i - \mathbb{E}[\mathbf{w}_i | x_i]\} | x_i = x]}{\mathbb{E}[\varkappa_i | x_i = x]}.$$

An L_∞ approximation condition on the ordinary conditional mean is sufficient if the numerator is zero, as when $\varkappa_i$ depends only on x_i . $\perp$

SA-4 Implementation Details

SA-4.1 Standard Error Computation

In Section SA-3, we have given the variance formulas $\widehat{\Omega}_{\mu^{(v)}}(x)$, $\widehat{\Omega}_\vartheta(x)$ and $\widehat{\Omega}_\zeta(x)$ that can be used to obtain the standard errors of $\widehat{\mu}^{(v)}(x)$, $\widehat{\vartheta}(x, \widehat{\mathbf{w}})$ and $\widehat{\zeta}(x, \widehat{\mathbf{w}})$. Recall that the formula for the estimator $\widehat{\Sigma}$ of Σ_0 is

$$\widehat{\Sigma} = \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \psi(y_i, \widehat{\eta}_i)^2 \eta^{(1)}(\widehat{\mu}(x_i) + \mathbf{w}_i' \widehat{\gamma})^2 \right].$$

It only relies on known or estimable quantities such as the derivative of the loss function $\psi(\cdot)$, the derivative of the inverse link function $\eta^{(1)}(\cdot)$, the residual $\widehat{\epsilon}_i$ and the binscatter estimates $\widehat{\mu}(\cdot)$ and $\widehat{\gamma}$. Thus, $\widehat{\Sigma}$ and other types of heteroskedasticity-robust “meat” matrix estimators can be easily constructed using the data. Then, it remains to obtain an estimator $\widehat{\mathbf{Q}}$ of $\bar{\mathbf{Q}}$ (or $\mathbf{Q}_0$), which in general relies on an estimator $\widehat{\Psi}_1(\cdot)$ of $\Psi_1(\cdot)$ and can be constructed on a case-by-case basis. In the following we discuss several examples.

Example 1 (Least Squares Regression). For least squares regression, the loss function $\rho(y; \eta) = \frac{1}{2}(y - \eta)^2$ and the (inverse) link function $\eta(\theta) = \theta$. Therefore, $\psi(y_i, \eta_i) = -\epsilon_i$ and $\eta_{i,1} = 1$. Thus, the formula for $\hat{\mathbf{Q}}$ given in Section SA-3 reduces to $\mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i)\hat{\mathbf{b}}_{p,s}(x_i)']$, which is immediately feasible in practice.

Example 2 (Logistic Regression). For logistic regression, the loss function is given by the corresponding likelihood function, i.e., $-\rho(y; \eta) = y \log \eta + (1 - y) \log(1 - \eta)$, and the inverse link is given by the logistic function $\eta(\theta) = \frac{e^\theta}{1 + e^\theta}$. Accordingly, an estimator of $\mathbf{Q}_0$ is given by

$$\hat{\mathbf{Q}} = \mathbb{E}_n \left[\hat{\mathbf{b}}_{p,s}(x_i) \hat{\mathbf{b}}_{p,s}(x_i)' \hat{\eta}_i (1 - \hat{\eta}_i) \right], \quad \hat{\eta}_i = \eta(\hat{\mu}(x_i) + \mathbf{w}_i' \hat{\gamma}).$$

Example 3 (Quantile Regression). For quantile regression, $\rho(y; \eta) = (q - \mathbb{1}(y < \eta))(y - \eta)$ for some $q \in (0, 1)$ and $\eta(\theta) = \theta$. Accordingly, $\psi(y_i, \eta_i) = \mathbb{1}(\epsilon_i < 0) - q$, and one needs to estimate

$$\mathbf{Q}_0 = \mathbb{E} \left[\mathbf{b}_{p,s}(x_i) \mathbf{b}_{p,s}(x_i)' f_{Y|XW}(\mu_0(x_i) + \mathbf{w}_i' \gamma_0 | x_i, \mathbf{w}_i) \right].$$

The key is to estimate the conditional density $f_{Y|XW}(\cdot | x_i, \mathbf{w}_i)$ evaluated at the conditional quantile of interest $(\mu_0(x_i) + \mathbf{w}_i' \gamma_0)$, whose reciprocal is termed “sparsity function” in the literature. Many different methods have been proposed. For example, the sparsity function is simply the derivative of the conditional quantile function with respect to the quantile, which can be estimated by using the difference quotient of the estimated conditional quantile function. Alternatively, $\mathbf{Q}_0$ can be viewed as a matrix-weighted density function, and one can construct a corresponding estimator based on kernel density estimation ideas. In addition, one can use bootstrapping methods to estimate the variance, avoiding the technical difficulty of estimating the sparsity function. See Section 3.4 and Section 3.9 of [Koenker \(2005\)](#) for more discussion of variance estimation for quantile regression.

SA-4.2 Number of Bins Selector

We discuss the implementation details for data-driven selection of the number of bins, based on the approximate integrated mean squared error expansion in Theorem SA-3.4.

We offer two procedures for estimating the bias and variance constants, and once these estimates

$(\widehat{\mathcal{B}}_n(p, s, v)$ and $\widehat{\mathcal{V}}_n(p, s, v)$) are available, the estimated optimal J is

$$\widehat{J}_{\text{IMSE}} = \left\lceil \left(\frac{2(p-v+1)\widehat{\mathcal{B}}_n(p, s, v)}{(1+2v)\widehat{\mathcal{V}}_n(p, s, v)} \right)^{\frac{1}{2p+3}} n^{\frac{1}{2p+3}} \right\rceil.$$

We always let $\omega(x) = f_X(x)$ as weighting function for concreteness.

SA-4.2.1 Rule-of-thumb Selector

For quantile-spaced partitions, a rule-of-thumb choice of J can be obtained based on Corollary SA-3.2 in [Cattaneo et al. \(2024a\)](#), which gives an explicit characterization of the variance and bias constants for least squares binscatter using piecewise polynomials ($s = 0$). For other partitions covered by Theorem [SA-3.4](#), the partition-specific constants can instead be estimated by the direct plug-in method below.

Specifically, the variance constant $\mathcal{V}(p, 0, v)$ is estimated by

$$\widehat{\mathcal{V}}(p, 0, v) = \text{trace} \left\{ \left(\int_0^1 \boldsymbol{\varphi}(z) \boldsymbol{\varphi}(z)' \, dz \right)^{-1} \int_0^1 \boldsymbol{\varphi}^{(v)}(z) \boldsymbol{\varphi}^{(v)}(z)' \, dz \right\} \times \frac{1}{n} \sum_{i=1}^n \widehat{\sigma}^2(x_i, \mathbf{w}_i) \widehat{f}_X(x_i)^{2v}$$

where $\boldsymbol{\varphi}(z) = (1, z, \dots, z^p)'$, $\widehat{\sigma}^2(x_i, \mathbf{w}_i)$ is some estimate of the conditional variance $\mathbb{V}[y_i | x_i, \mathbf{w}_i]$ and $\widehat{f}_X(x_i)$ is some estimate of the density $f_X(x_i)$. On the other hand, the bias constant $\mathcal{B}(p, 0, v)$ is estimated by

$$\widehat{\mathcal{B}}(p, 0, v) = \frac{\int_0^1 [\mathcal{B}_{p+1-v}(z)]^2 \, dz}{((p+1-v)!)^2} \times \frac{1}{n} \sum_{i=1}^n \frac{[\widehat{\mu}^{(p+1)}(x_i)]^2}{\widehat{f}_X(x_i)^{2p+2-2v}}.$$

where $\mathcal{B}_p(z) = (-1)^p \sum_{k=0}^p \binom{p}{k} \binom{p+k}{k} (-z)^k / \binom{2p}{p}$ for each $p \in \mathbb{Z}_+$ and $\widehat{\mu}^{(p+1)}(x_i)$ is some preliminary estimate of $\mu_0^{(p+1)}(x_i)$. The details about getting the estimates $\widehat{\sigma}^2(x_i, \mathbf{w}_i)$, $\widehat{f}_X(x_i)$ and $\widehat{\mu}^{(p+1)}(x_i)$ can be found in Section SA-4.1 in [Cattaneo et al. \(2024a\)](#).

This procedure still yields a choice of J with the correct rate, though the constant approximations are inconsistent for general loss.

SA-4.2.2 Direct-plug-in Selector

The direct-plug-in selector is implemented based on nonlinear binscatter estimators, which applies to any user-specified p , s and v . It requires a preliminary choice of partitioning denoted by J_{pre} .

More generally, suppose that a preliminary choice J_{pre} is given, and then a binscatter basis

$\hat{\mathbf{b}}_{p,s}(x)$ (of order p) can be constructed immediately on the preliminary partition. Implementing a nonlinear binscatter estimation using this basis and partitioning, we can obtain the variance constant estimate using the variance matrix estimators discussed in Section SA-4.1.

Regarding the bias constant, the key unknown in the expression of the leading approximation error $r_{0,v}^*(x)$ in Theorem SA-3.4 is $\mu_0^{(p+1)}(x)$, which can be estimated by implementing a nonlinear binscatter estimation of order $p + 1$. The local width h_{x_i} can be replaced directly by $\hat{h}_{x_i}$, the length of the interval in the preliminary partition containing x_i . Thus, this construction does not require replacing h_{x_i} by the quantile-specific approximation $1/(Jf_X(x_i))$. All other quantities in the expression of $\mathcal{B}(p, s, v)$ can be replaced by their sample analogues. Then, a bias constant estimate is available. Under the conditions of Theorem SA-3.4, and provided that $J_{\text{pre}} \rightarrow \infty$ and $J_{\text{pre}}^{2p+3}/n = o(1)$, this construction employs consistent constant approximations for any nonlinear binscatter with any choice of p , s and v .

SA-4.3 Fixed J and choice of polynomial order

Our theory treats J as diverging with the sample size. This reflects the standard approach wherein a researcher selects p and s in advance (often $s = p = 0$ or $s = p = 3$) and then selects J given the data, and further, is interested in recovery of the true nonparametric function $\mu_0(x)$. The partition must get finer to remove the nonparametric smoothing bias in estimating the function $\mu_0(x)$ (and along with it, $\vartheta_0(x, \mathbf{w})$ or $\zeta_0(x, \mathbf{w})$). Correct recovery (either for estimation or visualization) of these functions is the primary use of binscatter. However, researchers sometimes prefer to pre-specify a fixed $J = \mathbf{J}$, and we also discuss implementation and interpretation of binscatter in this case. In this subsection, when considering fixed $J = \mathbf{J}$, we also treat d_γ as fixed, as this maintains focus on the relationship of y_i and x_i . If $J = \mathbf{J}$ is fixed and d_γ diverges this is high-dimensional linear model and is no longer in the spirit of binscatter, both in terms of applications and econometric theory.

A researcher may want to use a fixed $J = \mathbf{J}$, often small and/or round, instead of the optimal choice, to make the estimate more visually appealing or because the results can be directly interpreted. In this case, instead of recovering the functions $\mu_0^{(v)}(x)$, $\vartheta_0(x, \mathbf{w})$, and $\zeta_0(x, \mathbf{w})$, the binscatter is interpreted as estimating their coarsened versions: the distribution of y_i conditional on x_i lying in a (fixed) bin, rather than at a single point. For some $\mathbf{J}$, this remains interpretable and can be viewed as a finite-dimensional estimation problem. For example, in our application

we can take $J = 10$ and study the distribution of uninsured rate for each decile of income. The confidence bands then become pointwise confidence intervals that are simultaneously valid. This could be used to examine inequality in health care access by asking if median uninsured rates are statistically different between the top and bottom decile. All our inference results still apply to this case, with the caveat that if control variables $\mathbf{w}_i$ are added, the estimation error in $\hat{\gamma}$ cannot be ignored.

A fixed J is also interpretable, and applicable, if x_i is discrete. Then each mass point is given its own bin and the results apply to the conditional distribution of y_i given $x_i = x$. Again, our theoretical results apply directly to this case and one obtains simultaneous inference over the set of points. [Cattaneo et al. \(2024a\)](#) give further discussion and examples.

As a practical compromise between the visual appeal and interpretation of a small, fixed J and the demand for consistent estimation, we propose a novel, albeit ad-hoc, adjustment to the estimator aimed at addressing the smoothing bias left by fixing J by adjusting the choice of polynomial order p . The standard approach fixes p in advance and selects J based on the data, but we can invert this and search for the value of p for which the fixed J would be IMSE-optimal. That is, we solve for

$$\hat{p}_{\text{IMSE}}(J, s, v) = \arg \min_{p \in \mathcal{P}} \left| \hat{J}_{\text{IMSE}}(p, s, v) - J \right|, \quad (\text{SA-4.1})$$

where in principle the set $\mathcal{P}$ contains all integers $p \geq \max\{s, v\}$ (and $p \geq 1$ for ζ_0), but in practice $\mathcal{P} = \{p_{\min}, p_{\min} + 1, \dots, p_{\max} - 1, p_{\max}\}$, for some integers $\max\{s, v\} \leq p_{\min} \leq p_{\max}$ (and $p_{\min} \geq 1$ for ζ_0). The (in)flexibility of fixed $J = J$ is offset by changing the polynomial approximation. This may have some practical appeal, but our theoretical results in the next section continue in the standard case of fixed p and diverging J .

To implement the data-driven choice $\hat{p}_{\text{IMSE}}(J, s, v)$, users need to specify the desired (often small) number of bins J , the smoothness order s , the derivative order v of interest, and a (finite) set $\mathcal{P}$ of acceptable polynomial orders. The size of $\mathcal{P}$ is usually small since in practice $p = 3$ or 4 often suffices to yield a small IMSE-optimal number of bins. Then, for each value of p in $\mathcal{P}$, we can implement the rule-of-thumb or direct plug-in procedure as described in [Section SA-4.2](#) to obtain $\hat{J}_{\text{IMSE}}(p, s, v)$. The “optimal” choice $\hat{p}_{\text{IMSE}}(J, s, v)$ is the value of p with the resulting $\hat{J}_{\text{IMSE}}(p, s, v)$

closest to J.

SA-5 Proofs

We begin with a subsection collecting some technical lemmas used in the proofs of our main results. We then collect all the proofs of the results presented in this supplemental appendix, which are in several cases more general than those discussed in the main text. Some of our technical results may be of broader independent interest in the nonlinear series estimation literature.

SA-5.1 Technical Lemmas

We first give several simple facts about $\hat{\Delta}$ in the following lemma, which are immediate from Assumption SA-RP(ii).

Lemma SA-5.1 (Quasi-Uniformity). *Suppose that Assumption SA-RP(ii) holds. Then, (i) $J^{-1} \lesssim \min_{1 \leq j \leq J} h_j \leq \max_{1 \leq j \leq J} h_j \lesssim J^{-1}$, (ii) $\max_{0 \leq j \leq J} |\hat{\tau}_j - \tau_j| \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}$, and (iii) $\hat{\Delta} \in \Pi_{3c_{\text{QU}}} \text{ w.p.a. } 1$.*

Proof. By Assumption SA-RP(ii), $\text{len}(\mathcal{X}) = \sum_{j=1}^J h_j \geq J \min_{1 \leq j \leq J} h_j \geq c_{\text{QU}}^{-1} J \max_{1 \leq j \leq J} h_j$ where $\text{len}(\mathcal{X})$ denotes the length of $\mathcal{X}$ (which is a fixed number). On the other hand, $\text{len}(\mathcal{X}) \leq J \max_{1 \leq j \leq J} h_j \leq c_{\text{QU}} J \min_{1 \leq j \leq J} h_j$. Therefore, $c_{\text{QU}}^{-1} J^{-1} \text{len}(\mathcal{X}) \leq \min_{1 \leq j \leq J} h_j \leq \max_{1 \leq j \leq J} h_j \leq c_{\text{QU}} J^{-1} \text{len}(\mathcal{X})$.

Next, by Assumption SA-RP(ii),

$$\max_{0 \leq j \leq J} |\hat{\tau}_j - \tau_j| \leq |\hat{\tau}_0 - \tau_0| + J \max_{1 \leq l \leq J} |\hat{h}_l - h_l| \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}.$$

In addition, $\max_{1 \leq j \leq J} |\hat{h}_j - h_j| \leq \frac{1}{2} c_{\text{QU}}^{-1} J^{-1} \text{len}(\mathcal{X}) \leq \frac{1}{2} \min_{1 \leq j \leq J} h_j \text{ w.p.a. } 1$, and thus

$$\frac{\max_{1 \leq j \leq J} \hat{h}_j}{\min_{1 \leq j \leq J} \hat{h}_j} = \frac{\max_{1 \leq j \leq J} h_j + \max_{1 \leq j \leq J} |\hat{h}_j - h_j|}{\min_{1 \leq j \leq J} h_j - \max_{1 \leq j \leq J} |\hat{h}_j - h_j|} \leq 3c_{\text{QU}}, \quad \text{w.p.a.1.}$$

Then, the proof is complete. $\square$

The next lemma then verifies Assumption SA-RP(ii) for the special case of quantile-spaced partitions. The proof is available in the supplemental appendix of Cattaneo et al. (2024a) (see Section SA-3.1 therein) and thus omitted here.

Lemma SA-5.2 (Quasi-Uniformity of Quantile-Spaced Partitions). *Suppose that Assumptions SA-DGP(i) and SA-DGP(ii) hold and $\hat{\Delta}$ is the quantile-based partition defined in Section SA-2, i.e., $\hat{\tau}_0 = \inf \mathcal{X}$, $\hat{\tau}_J = \sup \mathcal{X}$, and $\hat{\tau}_j = \hat{F}_X^{-1}(j/J)$ for $j = 1, \dots, J-1$. If $\frac{J \log J}{n} = o(1)$ and $\frac{\log n}{J} = o(1)$,*

then Assumption [SA-RP\(ii\)](#) holds with $\tau_0 = \inf \mathcal{X}$, $\tau_J = \sup \mathcal{X}$, $\tau_j = F_X^{-1}(j/J)$ for $j = 1, \dots, J-1$, and $\mathfrak{r}_{\text{RP}} = \left(\frac{J \log J}{n} \right)^{1/2}$.

The next three lemmas [SA-5.3–SA-5.5](#) concern the properties of binscatter basis functions. Their proofs are the same as those for quantile-based partitions that are available in the supplemental appendix of [Cattaneo et al. \(2024a\)](#) (see Section SA-3.1 therein) and are omitted here to conserve space.

Lemma SA-5.3 (Transformation Matrix). *Suppose that Assumption [SA-RP\(i\)](#) holds. Then $\widehat{\mathbf{b}}_{p,s}(x) = \widehat{\mathbf{T}}_s \widehat{\mathbf{b}}_{p,0}(x)$ with $\|\widehat{\mathbf{T}}_s\|_\infty \lesssim_{\mathbb{P}} 1$ and $\|\widehat{\mathbf{T}}_s\| \lesssim_{\mathbb{P}} 1$. If, in addition, Assumption [SA-RP\(ii\)](#) holds, then $\|\widehat{\mathbf{T}}_s - \mathbf{T}_s\|_\infty \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}$ and $\|\widehat{\mathbf{T}}_s - \mathbf{T}_s\| \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}$.*

Lemma SA-5.4 (Local Basis). *Suppose that Assumption [SA-RP\(i\)](#) holds. Then $\sup_{x \in \mathcal{X}} \|\widehat{\mathbf{b}}_{p,s}^{(v)}(x)\|_0 \leq (p+1)^2$ and $\sup_{x \in \mathcal{X}} \|\widehat{\mathbf{b}}_{p,s}^{(v)}(x)\| \lesssim_{\mathbb{P}} J^{\frac{1}{2}+v}$.*

The following lemma provides a particular way to define $\beta_0(\Delta)$ and $\widehat{\beta}_0$ so that the required approximation rate is achieved. We define

$$\beta_0^{\text{LS}}(\Delta) := \arg \min_{\beta \in \mathbb{R}^{K_{p,s}}} \mathbb{E}[(\mu_0(x_i) - \mathbf{b}_{p,s}(x_i; \Delta)' \beta)^2], \quad \widehat{\beta}_0^{\text{LS}} = \beta_0^{\text{LS}}(\widehat{\Delta}).$$

Lemma SA-5.5 (Approximation Error). *Suppose that Assumptions [SA-DGP\(i\)\(ii\)](#), [SA-SM\(v\)](#) and [SA-RP\(i\)](#) hold. Then*

$$\sup_{\Delta \in \Pi_C} \sup_{x \in \mathcal{X}} |\mathbf{b}_{p,s}^{(v)}(x; \Delta)' \beta_0^{\text{LS}}(\Delta) - \mu_0^{(v)}(x)| \lesssim J^{-p-1+v} \quad \text{and} \quad \sup_{x \in \mathcal{X}} |\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\beta}_0^{\text{LS}} - \mu_0^{(v)}(x)| \lesssim_{\mathbb{P}} J^{-p-1+v}.$$

Next, the following maximal inequality is useful in our analysis and was established as Lemma SA24 in [Cattaneo et al. \(2024b\)](#). Its proof is available therein and thus omitted here.

Lemma SA-5.6 (Maximal Inequality). *Let $Z_1, \dots, Z_n$ be independent but not necessarily identically distributed random variables taking values in a measurable space $(\mathcal{S}; \mathcal{S})$. Denote the joint distribution of $Z_1, \dots, Z_n$ by $\mathbb{P}$ and the marginal distribution of Z_i by $\mathbb{P}_i$, and let $\bar{\mathbb{P}} = \frac{1}{n} \sum_{i=1}^n \mathbb{P}_i$. Let $\mathcal{F}$ be a class of Borel measurable functions from $\mathcal{S}$ to $\mathbb{R}$ which is pointwise measurable. Let $\bar{F}$ be a measurable envelope function for $\mathcal{F}$. Suppose that $\|\bar{F}\|_{L_2(\bar{\mathbb{P}})} < \infty$. Let $\bar{\sigma} > 0$ satisfy*

$\sup_{f \in \mathcal{F}} \|f\|_{L_2(\bar{\mathbb{P}})} \leq \bar{\sigma} \leq \|\bar{F}\|_{L_2(\bar{\mathbb{P}})}$ and define $\bar{\bar{F}} = \max_{1 \leq i \leq n} \bar{F}(Z_i)$. Then, with $\delta = \bar{\sigma} / \|\bar{F}\|_{L_2(\bar{\mathbb{P}})}$,

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(f(Z_i) - \mathbb{E}[f(Z_i)] \right) \right| \right] \lesssim \|\bar{F}\|_{L_2(\bar{\mathbb{P}})} J(\delta, \mathcal{F}, \bar{F}) + \frac{\|\bar{\bar{F}}\|_{L_2(\bar{\mathbb{P}})} J(\delta, \mathcal{F}, \bar{F})^2}{\delta^2 \sqrt{n}},$$

where

$$J(\delta, \mathcal{F}, \bar{F}) = \int_0^\delta \sqrt{1 + \sup_{\mathbb{Q}} \log N(\mathcal{F}, L_2(\mathbb{Q}), \varepsilon \|\bar{F}\|_{L_2(\mathbb{Q})})} d\varepsilon.$$

SA-5.2 Proof of Lemma SA-3.1

Proof. We write $\Psi_{i,1} := \Psi_1(x_i, \mathbf{w}_i; \eta_i)$.

(i) We first prove a convergence result for $\bar{\mathbf{Q}}$. In view of Lemma SA-5.3, it suffices to show the convergence for $s = 0$. Let $\mathcal{A}_n$ denote the event on which $\hat{\Delta} \in \Pi_C$. By Assumption SA-RP(i), $\mathbb{P}(\mathcal{A}_n^c) = o(1)$. On $\mathcal{A}_n$,

$$\begin{aligned} & \left\| \mathbb{E}_n[\hat{\mathbf{b}}_{p,0}(x_i) \hat{\mathbf{b}}_{p,0}(x_i)' \Psi_{i,1} \eta_{i,1}^2] - \mathbb{E}_{\hat{\Delta}}[\hat{\mathbf{b}}_{p,0}(x_i) \hat{\mathbf{b}}_{p,0}(x_i)' \Psi_{i,1} \eta_{i,1}^2] \right\| \\ & \leq \sup_{\Delta \in \Pi_C} \left\| \mathbb{E}_n[\mathbf{b}_{p,0}(x_i; \Delta) \mathbf{b}_{p,0}(x_i; \Delta)' \Psi_{i,1} \eta_{i,1}^2] - \mathbb{E}[\mathbf{b}_{p,0}(x_i; \Delta) \mathbf{b}_{p,0}(x_i; \Delta)' \Psi_{i,1} \eta_{i,1}^2] \right\|_\infty. \end{aligned}$$

Let a_{kl} be a generic (k, l) th entry of the matrix inside the norm, i.e.,

$$|a_{kl}| = \left| \mathbb{E}_n[b_{p,0,k}(x_i; \Delta) b_{p,0,l}(x_i; \Delta) \Psi_{i,1} \eta_{i,1}^2] - \mathbb{E}[b_{p,0,k}(x_i; \Delta) b_{p,0,l}(x_i; \Delta) \Psi_{i,1} \eta_{i,1}^2] \right|.$$

Clearly, if $b_{p,0,k}(\cdot; \Delta)$ and $b_{p,0,l}(\cdot; \Delta)$ are basis functions with different supports, a_{kl} is zero. Now define the following function class

$$\mathcal{G} = \left\{ (x_1, \mathbf{w}_1) \mapsto b_{p,0,k}(x_1; \Delta) b_{p,0,l}(x_1; \Delta) \Psi_1(x_1, \mathbf{w}_1; \eta_1) \eta_{1,1}^2 : 1 \leq k, l \leq J(p+1), \Delta \in \Pi_C \right\}.$$

We have $\sup_{g \in \mathcal{G}} |g|_\infty \lesssim J$ and $\sup_{g \in \mathcal{G}} \mathbb{V}[g] \leq \sup_{g \in \mathcal{G}} \mathbb{E}[g^2] \lesssim J$, by Assumption SA-SM. Also, by Proposition 3.6.12 of Giné and Nickl (2016), the collection $\mathcal{G}$ is of VC type with a bounded index. Then, by Lemma SA-5.6,

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n g(x_i) - \mathbb{E}[g(x_i)] \right| \lesssim_{\mathbb{P}} \sqrt{J \log J / n},$$

which implies $\|\mathbb{E}_n[\widehat{\mathbf{b}}_{p,0}(x_i)\widehat{\mathbf{b}}_{p,0}(x_i)'\Psi_{i,1}\eta_{i,1}^2] - \mathbb{E}_{\widehat{\Delta}}[\widehat{\mathbf{b}}_{p,0}(x_i)\widehat{\mathbf{b}}_{p,0}(x_i)'\Psi_{i,1}\eta_{i,1}^2]\| \lesssim_{\mathbb{P}} \sqrt{J \log J/n}$.

Then, the lower bound on the minimum eigenvalue of $\bar{\mathbf{Q}}$ follows by Theorem 4.42 of [Schumaker \(2007\)](#) and Assumption [SA-RP\(i\)](#). The upper bound immediately follows by Assumption [SA-RP\(i\)](#) and Lemmas [SA-5.3](#) and [SA-5.4](#).

Given the above fact, it follows that $\|\bar{\mathbf{Q}}^{-1}\| \lesssim_{\mathbb{P}} 1$. Notice that $\bar{\mathbf{Q}}$ is a banded matrix with a finite band width. Then, the bounds on the elements of $\bar{\mathbf{Q}}^{-1}$ and $\|\bar{\mathbf{Q}}^{-1}\|_{\infty}$ hold by Theorem 2.2 of [Demko \(1977\)](#).

(ii) By Assumption [SA-DGP](#) and [SA-SM](#), $\Psi_{i,1}\eta_{i,1}^2$ is bounded and bounded away from zero uniformly over $1 \leq i \leq n$. Then, $\mathbb{E}[\mathbf{b}_{p,s}(x_i)\mathbf{b}_{p,s}(x_i)'] \lesssim \mathbf{Q}_0 \lesssim \mathbb{E}[\mathbf{b}_{p,s}(x_i)\mathbf{b}_{p,s}(x_i)']$. The desired bounds on the minimum and maximum eigenvalues of $\mathbf{Q}_0$ follow from Lemma SA-3.5 of [Cattaneo et al. \(2024a\)](#).

Next, we show the convergence of $\bar{\mathbf{Q}}$ to $\mathbf{Q}_0$. Let α_{kl} be a generic (k, l) th entry of

$$\mathbb{E}_{\widehat{\Delta}}[\widehat{\mathbf{b}}_{p,0}(x_i)\widehat{\mathbf{b}}_{p,0}(x_i)'\Psi_{i,1}\eta_{i,1}^2]/J - \mathbb{E}[\mathbf{b}_{p,0}(x_i)\mathbf{b}_{p,0}(x_i)'\Psi_{i,1}\eta_{i,1}^2]/J.$$

By definition, it is either equal to zero or

$$\begin{aligned} \alpha_{kl} &= \int_{\widehat{\mathcal{B}}_j} \left(\frac{x - \hat{\tau}_{j-1}}{\hat{h}_j} \right)^{\ell} \varphi(x) f_X(x) dx - \int_{\mathcal{B}_j} \left(\frac{x - \tau_{j-1}}{h_j} \right)^{\ell} \varphi(x) f_X(x) dx \\ &= \hat{h}_j \int_0^1 z^{\ell} \varphi(z\hat{h}_j + \hat{\tau}_{j-1}) f_X(z\hat{h}_j + \hat{\tau}_{j-1}) dz - h_j \int_0^1 z^{\ell} \varphi(zh_j + \tau_{j-1}) f_X(zh_j + \tau_{j-1}) dz \\ &= (\hat{h}_j - h_j) \int_0^1 z^{\ell} \varphi(z\hat{h}_j + \hat{\tau}_{j-1}) f_X(z\hat{h}_j + \hat{\tau}_{j-1}) dz \\ &\quad + h_j \int_0^1 z^{\ell} \left(\varphi(z\hat{h}_j + \hat{\tau}_{j-1}) f_X(z\hat{h}_j + \hat{\tau}_{j-1}) - \varphi(zh_j + \tau_{j-1}) f_X(zh_j + \tau_{j-1}) \right) dz \end{aligned}$$

for some $1 \leq j \leq J$ and $0 \leq \ell \leq 2p$ and $\varphi(x) = \mathbb{E}[\boldsymbol{\varkappa}(x, \mathbf{w}_i) \mid x_i = x]$. Because φf_X is bounded and Lipschitz, Lemma [SA-5.1](#) and Assumption [SA-RP\(ii\)](#) imply $\max_{k,l} |\alpha_{kl}| \lesssim_{\mathbb{P}} J^{-1} \mathfrak{r}_{\text{RP}}$. Since the matrices have uniformly bounded bandwidth, rescaling by J therefore gives

$$\|\mathbb{E}_{\widehat{\Delta}}[\widehat{\mathbf{b}}_{p,0}(x_i)\widehat{\mathbf{b}}_{p,0}(x_i)'\Psi_{i,1}\eta_{i,1}^2] - \mathbf{Q}_0\| \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}.$$

Since $\bar{\mathbf{Q}}$ and $\mathbf{Q}_0$ are banded matrices with finite band widths. Then, the bound $\|\bar{\mathbf{Q}}^{-1} - \mathbf{Q}_0^{-1}\|_{\infty}$

hold by Theorem 2.2 of [Demko \(1977\)](#). This completes the proof. $\square$

SA-5.3 Proof of Lemma SA-3.2

Proof. Since $\mathbb{E}[\psi(y_i, \eta_i)^2 | x_i = x, \mathbf{w}_i = \mathbf{w}]$ and $(\eta^{(1)}(\mu_0(x) + \mathbf{w}'\gamma_0))^2$ is bounded and bounded away from zero uniformly over $x \in \mathcal{X}$ and $\mathbf{w} \in \mathcal{W}$, $\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i)\widehat{\mathbf{b}}_{p,s}(x_i)'] \lesssim \bar{\Sigma} \lesssim \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i)\widehat{\mathbf{b}}_{p,s}(x_i)']$. By the same argument in the proof of Lemma SA-3.1 (we can simply drop the additional term $\Psi_{i,1}\eta_{i,1}^2$ in $\bar{\mathbf{Q}}$), the eigenvalues of $\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i)\widehat{\mathbf{b}}_{p,s}(x_i)']$ and thus $\bar{\Sigma}$ are bounded and bounded away from zero. Then, the desired results follow from Lemma SA-3.1 and the fact that $\inf_{x \in \mathcal{X}} \|\widehat{\mathbf{b}}_{p,s}^{(v)}(x)\| \gtrsim J^{1/2+v}$ w.p.a. 1 (it was shown in the proof of Lemma SA-3.6 of [Cattaneo et al. \(2024a\)](#)). Under Assumption SA-RP(ii), Lemmas SA-5.3 and SA-5.4 also give $\sup_{x \in \mathcal{X}} \|(\widehat{\mathbf{T}}_s - \mathbf{T}_s)\widehat{\mathbf{b}}_{p,0}^{(v)}(x)\| \lesssim_{\mathbb{P}} \tau_{\text{RP}} J^{1/2+v} = o_{\mathbb{P}}(J^{1/2+v})$, so $\mathbf{T}_s\widehat{\mathbf{b}}_{p,0}^{(v)}(x)$ obeys the same uniform upper and lower bounds as $\widehat{\mathbf{b}}_{p,s}^{(v)}(x)$. This proves the claims for the population variances under their revised definitions. $\square$

SA-5.4 Proof of Lemma SA-3.3

Proof. By Lemmas SA-5.3, SA-5.4 and SA-3.1, $\sup_{x \in \mathcal{X}} \|\widehat{\mathbf{b}}_{p,s}^{(v)}(x)\|_1 \lesssim_{\mathbb{P}} J^{1/2+v}$, $\|\bar{\mathbf{Q}}^{-1}\|_{\infty} \lesssim_{\mathbb{P}} 1$ and $\|\widehat{\mathbf{T}}_s\|_{\infty} \lesssim_{\mathbb{P}} 1$. Recall that by Assumption SA-SM, $\psi(y_i, \eta_i) = \psi^\dagger(y_i - \eta_i)\psi^\dagger(\eta_i) = \psi^\dagger(\epsilon_i)\psi^\dagger(\eta_i)$. Define the following function class

$$\mathcal{G} = \left\{ (x_1, \mathbf{w}_1, \epsilon_1) \mapsto b_{p,0,l}(x_1; \Delta) \eta^{(1)}(\mu_0(x_1) + \mathbf{w}_1'\gamma_0) \psi^\dagger(\epsilon_1) \psi^\dagger(\eta_1) : 1 \leq l \leq J(p+1), \Delta \in \Pi_C \right\}.$$

Then, $\sup_{g \in \mathcal{G}} |g| \lesssim \sqrt{J} |\psi^\dagger(\epsilon_1)|$, and hence take an envelop $\bar{G} = C\sqrt{J} |\psi^\dagger(\epsilon_1)|$ for some C large enough. Moreover, $\sup_{g \in \mathcal{G}} \mathbb{V}[g] \lesssim 1$ and $\mathcal{G}$ is of VC type with a bounded index. By Proposition 6.1 of [Belloni et al. \(2015\)](#),

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n g(x_i, \epsilon_i) \right| \lesssim_{\mathbb{P}} \sqrt{\frac{\log J}{n}} + \frac{J^{\frac{\nu}{2(\nu-2)}} \log J}{n} \lesssim \sqrt{\frac{\log J}{n}},$$

and the desired result follows. $\square$

SA-5.5 Proof of Lemma SA-3.4

Proof. Let $\hat{z}_i = \hat{\mathbf{b}}_{p,s}(x_i)' \hat{\beta}_0 + \mathbf{w}_i' \gamma_0$ and $\mathbf{r}(x_i, \mathbf{w}_i, y_i) := \mathbf{r}(x_i, \mathbf{w}_i, y_i; \hat{\Delta}) := \eta_{i,1} \psi(y_i, \eta_i) - \eta^{(1)}(\hat{z}_i) \psi(y_i, \eta(\hat{z}_i))$
 $= A_1(x_i, \mathbf{w}_i, y_i) + A_2(x_i, \mathbf{w}_i, y_i)$ where

$$\begin{aligned} A_1(x_i, \mathbf{w}_i, y_i) &:= A_1(x_i, \mathbf{w}_i, y_i; \hat{\Delta}) := [\eta_{i,1} \psi^\dagger(\eta_i) - \eta^{(1)}(\hat{z}_i) \psi^\dagger(\eta(\hat{z}_i))] \psi^\dagger(y_i - \eta_i) \quad \text{and} \\ A_2(x_i, \mathbf{w}_i, y_i) &:= A_2(x_i, \mathbf{w}_i, y_i; \hat{\Delta}) := \eta^{(1)}(\hat{z}_i) \psi^\dagger(\eta(\hat{z}_i)) [\psi^\dagger(y_i - \eta_i) - \psi^\dagger(y_i - \eta(\hat{z}_i))]. \end{aligned}$$

First, by Assumption SA-SM and Lemma SA-5.5, $\sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} |\eta_{i,1} \psi^\dagger(\eta_i) - \eta^{(1)}(\hat{z}_i) \psi^\dagger(\eta(\hat{z}_i))| \lesssim J^{-p-1}$ w.p.a. 1. Also, for every $1 \leq l \leq K_{p,s}$ and $\Delta \in \Pi_C$,

$$\begin{aligned} &b_{p,s,l}(x; \Delta) \left(\eta_{i,1} \psi^\dagger(\eta_i) - \eta^{(1)}(\mathbf{b}_{p,s}(x; \Delta)' \beta_0(\Delta) + \mathbf{w}' \gamma_0) \psi^\dagger(\eta(\mathbf{b}_{p,s}(x; \Delta)' \beta_0(\Delta) + \mathbf{w}' \gamma_0)) \right) \\ &= b_{p,s,l}(x; \Delta) \eta_{i,1} \psi^\dagger(\eta_i) - \\ &b_{p,s,l}(x; \Delta) \eta^{(1)} \left(\sum_{k=\underline{k}_l}^{k_l+p} b_{p,s,k}(x; \Delta) \beta_{0,k}(\Delta) + \mathbf{w}' \gamma_0 \right) \psi^\dagger \left(\eta \left(\sum_{k=\underline{k}_l}^{k_l+p} b_{p,s,k}(x; \Delta) \beta_{0,k}(\Delta) + \mathbf{w}' \gamma_0 \right) \right) \end{aligned}$$

for some integer $\underline{k}_l \in [1, K_{p,s}]$ where $\beta_{0,k}(\Delta)$ denotes the k th element in $\beta_0(\Delta)$. Then, the function class $\mathcal{G} = \{(x, \mathbf{w}, y) \mapsto b_{p,s,l}(x; \Delta) A_1(x, \mathbf{w}, y; \Delta) : 1 \leq l \leq K_{p,s}, \Delta \in \Pi_C\}$ is of VC type with a bounded index. By the same argument given in the proof of Lemma SA-3.3,

$$\|\mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) A_1(x_i, \mathbf{w}_i, y_i)]\|_\infty \lesssim_{\mathbb{P}} J^{-p-1} \left(\frac{\log J}{n} \right)^{1/2}.$$

Next, let $\mathcal{F}_{XW\Delta}$ be the σ -field generated by $\{(x_i, \mathbf{w}_i)\}_{i=1}^n$ and $\hat{\Delta}$. Note that

$$\begin{aligned} \mathbb{E}_n[\hat{\mathbf{b}}_{p,s}(x_i) A_2(x_i, \mathbf{w}_i, y_i)] &= \mathbb{E}_n[\mathbb{E}[\hat{\mathbf{b}}_{p,s}(x_i) A_2(x_i, \mathbf{w}_i, y_i) | \mathcal{F}_{XW\Delta}]] + \\ &\mathbb{E}_n \left[\hat{\mathbf{b}}_{p,s}(x_i) A_2(x_i, \mathbf{w}_i, y_i) - \mathbb{E}[\hat{\mathbf{b}}_{p,s}(x_i) A_2(x_i, \mathbf{w}_i, y_i) | \mathcal{F}_{XW\Delta}] \right]. \end{aligned}$$

By Assumption SA-SM(iii) and Lemma SA-5.5,

$$\begin{aligned} &\max_{1 \leq i \leq n} |\mathbb{E}[A_2(x_i, \mathbf{w}_i, y_i) | \mathcal{F}_{XW\Delta}]| \\ &= \max_{1 \leq i \leq n} | -\eta^{(1)}(\hat{\mathbf{b}}_{p,s}(x_i)' \hat{\beta}_0 + \mathbf{w}_i' \gamma_0) \Psi(x_i, \mathbf{w}_i; \eta(\hat{z}_i)) | \lesssim_{\mathbb{P}} J^{-p-1}. \end{aligned}$$

Then, $\|\mathbb{E}_n[\mathbb{E}[\widehat{\mathbf{b}}_{p,s}(x_i)A_2(x_i, \mathbf{w}_i, y_i)|\mathcal{F}_{XW\Delta}]]\|_\infty \lesssim_{\mathbb{P}} J^{-p-1-1/2}$ by the same argument in the proof of Lemma SA-3.1. On the other hand, define the following function class

$$\mathcal{G} := \left\{ (x, \mathbf{w}, y) \mapsto b_{p,s,l}(x; \Delta)A_2(x, \mathbf{w}, y; \Delta) : 1 \leq l \leq K_{p,s}, \Delta \in \Pi_C \right\}.$$

By Assumption SA-SM, $\sup_{g \in \mathcal{G}} \|g\|_\infty \lesssim J^{1/2}$, and $\sup_{g \in \mathcal{G}} \mathbb{V}[g(x_i, \mathbf{w}_i, y_i)] \lesssim J^{-p-1}$. By a similar argument given before, this function class is of VC type with a bounded index. Then, by Lemma SA-5.6,

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n (g(x_i, \mathbf{w}_i, y_i) - \mathbb{E}[g(x_i, \mathbf{w}_i, y_i)]) \right| \lesssim_{\mathbb{P}} J^{-\frac{p+1}{2}} \sqrt{\frac{\log J}{n}} + \frac{J^{1/2} \log J}{n}.$$

Collecting these results, we conclude that

$$\sup_{x \in \mathcal{X}} \left| \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \mathbf{r}(x_i, \mathbf{w}_i, y_i)] \right| \lesssim_{\mathbb{P}} J^{-p-1+v} + J^{\frac{2v-p-1}{2}} \left(\frac{J \log J}{n} \right)^{1/2} + \frac{J^{1+v} \log J}{n}.$$

The proof is complete. $\square$

SA-5.6 Proof of Lemma SA-3.5

Proof. By convexity of $\rho(y; \eta(\cdot))$, we only need to consider $\boldsymbol{\beta} = \widehat{\boldsymbol{\beta}}_0 + \varepsilon \boldsymbol{\alpha} / \sqrt{J}$ for any sufficiently small fixed $\varepsilon > 0$ and $\boldsymbol{\alpha} \in \mathbb{R}^{K_{p,s}}$ such that $\|\boldsymbol{\alpha}\| = 1$. For notational simplicity, let $\widehat{\mathbf{b}}_i := \widehat{\mathbf{b}}_{p,s}(x_i)$. For this choice of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma} \in \mathbb{R}^{d_\gamma}$,

$$\begin{aligned} \delta_i(\boldsymbol{\beta}, \boldsymbol{\gamma}) &= \rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\beta} + \mathbf{w}_i' \boldsymbol{\gamma})) - \rho(y_i; \eta(\widehat{\mathbf{b}}_i' \widehat{\boldsymbol{\beta}}_0 + \mathbf{w}_i' \boldsymbol{\gamma})) \\ &= \int_0^{\varepsilon \widehat{\mathbf{b}}_i' \boldsymbol{\alpha} / \sqrt{J}} \psi\left(y_i, \eta(\widehat{\mathbf{b}}_i' \widehat{\boldsymbol{\beta}}_0 + \mathbf{w}_i' \boldsymbol{\gamma} + t)\right) \eta^{(1)}(\widehat{\mathbf{b}}_i' \widehat{\boldsymbol{\beta}}_0 + \mathbf{w}_i' \boldsymbol{\gamma} + t) dt. \end{aligned}$$

Let $\mathcal{F}_{XW\Delta}$ be the σ -field generated by $\{(x_i, \mathbf{w}_i)\}_{i=1}^n$ and $\widehat{\Delta}$. We have

$$\mathbb{E}_n[\delta_i(\boldsymbol{\beta}, \widehat{\boldsymbol{\gamma}})] = \frac{1}{\sqrt{n}} \mathbb{G}_n[\delta_i(\boldsymbol{\beta}, \widehat{\boldsymbol{\gamma}})] + \mathbb{E}_n\left[\mathbb{E}[\delta_i(\boldsymbol{\beta}, \widehat{\boldsymbol{\gamma}})|\mathcal{F}_{XW\Delta}]\right],$$

where $\mathbb{G}_n[\cdot]$ denotes $\sqrt{n}(\mathbb{E}_n[\cdot] - \mathbb{E}[\cdot|\mathcal{F}_{XW\Delta}])$, and $\mathbb{E}[\delta_i(\boldsymbol{\beta}, \hat{\boldsymbol{\gamma}})|\mathcal{F}_{XW\Delta}] := \mathbb{E}[\delta_i(\boldsymbol{\beta}, \boldsymbol{\gamma})|\mathcal{F}_{XW\Delta}]|_{\boldsymbol{\gamma}=\hat{\boldsymbol{\gamma}}}$, i.e., the conditional expectation with $\hat{\boldsymbol{\gamma}}$ viewed as fixed. By Assumption SA-SM,

$$\begin{aligned}\mathbb{E}[\delta_i(\boldsymbol{\beta}, \hat{\boldsymbol{\gamma}})|\mathcal{F}_{XW\Delta}] &= \int_0^{\varepsilon \hat{\mathbf{b}}'_i \boldsymbol{\alpha} / \sqrt{J}} \Psi(x_i, \mathbf{w}_i; \eta(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \hat{\boldsymbol{\gamma}} + t)) \eta^{(1)}(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \hat{\boldsymbol{\gamma}} + t) dt \\ &= \int_0^{\varepsilon \hat{\mathbf{b}}'_i \boldsymbol{\alpha} / \sqrt{J}} \Psi_1(x_i, \mathbf{w}_i; \xi_{i,t}) (\eta(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \hat{\boldsymbol{\gamma}} + t) - \eta_i) \eta^{(1)}(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \hat{\boldsymbol{\gamma}} + t) dt,\end{aligned}$$

where $\xi_{i,t}$ is between $\eta(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \hat{\boldsymbol{\gamma}} + t)$ and $\eta(\mu_0(x_i) + \mathbf{w}'_i \boldsymbol{\gamma}_0)$ and we use the fact that $\Psi(x, \mathbf{w}_i; \eta_i) = 0$. By Lemma SA-5.5 and the imposed conditions that $\eta(\cdot)$ is strictly monotonic, $\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0 \lesssim_{\mathbb{P}} \mathbf{r}_{\boldsymbol{\gamma}}$, $\sup_i |\mathbf{w}'_i (\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0)| \lesssim_{\mathbb{P}} \sqrt{d_{\boldsymbol{\gamma}}} \mathbf{r}_{\boldsymbol{\gamma}} = o_{\mathbb{P}}(\sqrt{J/n} + J^{-p-1})$ and $\sqrt{J}(J^{-p-1} + \sqrt{d_{\boldsymbol{\gamma}}} \mathbf{r}_{\boldsymbol{\gamma}}) = o(1)$, we have $\mathbb{E}_n[\mathbb{E}[\delta_i(\boldsymbol{\beta}, \hat{\boldsymbol{\gamma}})|\mathcal{F}_{XW\Delta}]] \gtrsim_{\mathbb{P}} \varepsilon^2 \boldsymbol{\alpha}' \mathbb{E}_n[\hat{\mathbf{b}}_i \hat{\mathbf{b}}'_i] \boldsymbol{\alpha} / J \gtrsim_{\mathbb{P}} J^{-1} \varepsilon^2$.

On the other hand, let $\mathcal{H} := \{\boldsymbol{\gamma} : \|\boldsymbol{\gamma} - \boldsymbol{\gamma}_0\| \leq C \mathbf{r}_{\boldsymbol{\gamma}}\}$ and define the following function class

$$\mathcal{G} := \left\{ (x_i, \mathbf{w}_i, y_i) \mapsto \delta_i(\boldsymbol{\beta}, \boldsymbol{\gamma}) : \boldsymbol{\alpha} \in \mathcal{S}^{K_{p,s}}, \boldsymbol{\gamma} \in \mathcal{H} \right\}.$$

Note that

$$\begin{aligned}\delta_i(\boldsymbol{\beta}, \boldsymbol{\gamma}) &= \int_0^{\varepsilon \hat{\mathbf{b}}'_i \boldsymbol{\alpha} / \sqrt{J}} \left(\psi(y_i, \eta(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \boldsymbol{\gamma} + t)) - \psi(y_i, \eta_i) \right) \eta^{(1)}(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \boldsymbol{\gamma} + t) dt + \\ &\quad \int_0^{\varepsilon \hat{\mathbf{b}}'_i \boldsymbol{\alpha} / \sqrt{J}} \psi(y_i, \eta_i) \eta^{(1)}(\hat{\mathbf{b}}'_i \hat{\boldsymbol{\beta}}_0 + \mathbf{w}'_i \boldsymbol{\gamma} + t) dt.\end{aligned}$$

By Assumption SA-SM, we have $\sup_{g \in \mathcal{G}} |g| \lesssim \varepsilon(1 + |\psi(y_i, \eta_i)|)$, $\|\max_{1 \leq i \leq n} |\psi(y_i, \eta_i)|\|_{L_2(\mathbb{P})} \lesssim n^{1/\nu}$, $\sup_{g \in \mathcal{G}} \mathbb{E}_n[\mathbb{E}[g^2|\mathcal{F}_{XW\Delta}]] \lesssim_{\mathbb{P}} J^{-1} \varepsilon^2$, and local support and the piecewise-Lipschitz score condition give the covering bound

$$\sup_{\mathbb{Q}} \log N(\mathcal{G}, L_2(\mathbb{Q}), t \|\varepsilon(1 + |\psi|)\|_{\mathbb{Q},2}) \lesssim (K_{p,s} + d_{\boldsymbol{\gamma}}) \log(C/t).$$

Therefore, by Lemma SA-5.6 and the rate restriction,

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{\sqrt{n}} \mathbb{G}_n[\delta_i(\boldsymbol{\beta}, \boldsymbol{\gamma})] \right| \lesssim_{\mathbb{P}} J^{-1} \left(\frac{J(J + d_{\boldsymbol{\gamma}}) \log J}{n} \right)^{1/2} \varepsilon + J^{-1} \frac{J(J + d_{\boldsymbol{\gamma}}) \log J}{n^{1-\frac{1}{\nu}}} \varepsilon = o(\varepsilon/J).$$

Thus, for any fixed (sufficiently small) $\varepsilon > 0$, $\mathbb{E}_n[\delta_i(\boldsymbol{\beta}, \hat{\boldsymbol{\gamma}})] > 0$ when n is sufficiently large. Thus,

$\|\widehat{\beta} - \widehat{\beta}_0\| = o_{\mathbb{P}}(J^{-1/2})$, implying $\|\widehat{\beta} - \widehat{\beta}_0\|_{\infty} = o_{\mathbb{P}}(J^{-1/2})$ immediately. $\square$

SA-5.7 Proof of Theorem SA-3.1

Proof. The proof is long. We divide it into several steps.

Step 0: We first prepare some notation and useful facts. To simplify the presentation, in this proof we drop the scaling factor $\sqrt{J}$ in the basis by defining

$$\check{\mathbf{b}}_i := \widehat{\mathbf{b}}_{p,s}(x_i)/\sqrt{J} = (\widehat{b}_{p,s,1}(x_i), \dots, \widehat{b}_{p,s,K_{p,s}}(x_i))'/\sqrt{J} \quad \text{and} \quad \check{\beta} := \sqrt{J}\widehat{\beta}, \quad \check{\beta}_0 = \sqrt{J}\widehat{\beta}_0.$$

The Hessian in this rescaled parameterization is

$$\check{\mathbf{Q}} := \frac{\bar{\mathbf{Q}}}{J}, \quad \check{\mathbf{Q}}^{-1} = J\bar{\mathbf{Q}}^{-1}.$$

Throughout the proof, $C, c, C_1, c_1, C_2, c_2, \dots$ denote (strictly positive) absolute constants, $\mathcal{F}_{XW\Delta}$ denotes the σ -field generated by $\{(x_i, \mathbf{w}_i)\}_{i=1}^n$ and $\widehat{\Delta}$, and $\text{supp}(g(\cdot))$ denotes the support of a generic function $g(\cdot)$. Moreover, define

$$\begin{aligned} \mathcal{V} &= \{(v_1, \dots, v_{K_{p,s}})' : \exists k \in \{1, \dots, K_{p,s}\}, |v_\ell| \leq \varrho^{|k-\ell|} \varepsilon_n \text{ for } |\ell - k| \leq M_n \text{ and } v_\ell = 0 \text{ otherwise}\}, \\ \mathcal{H}_l &= \{\mathbf{v} \in \mathbb{R}^{K_{p,s}} : \|\mathbf{v}\|_{\infty} \leq r_{l,n}\} \text{ for } l = 1, 2, \quad \text{and} \quad \mathcal{H}_3 = \{\mathbf{v} \in \mathbb{R}^{d_\gamma} : \|\mathbf{v}\| \leq r_{3,n}\}, \end{aligned}$$

where $\varrho \in (0, 1)$ is the constant given in Lemma SA-3.1, $r_{1,n} = C_1[(J \log n/n)^{1/2} + J^{-p-1}]$, $r_{2,n} = \mathfrak{z} \mathfrak{r}_{2,n}$ for $\mathfrak{z} > 0$, $\varepsilon_n = \mathfrak{z}' \mathfrak{r}_{2,n}$ for $\mathfrak{z}' > 0$, $\mathfrak{r}_{2,n} = \mathfrak{r}_{\text{BR},n}$, $r_{3,n} = C \mathfrak{r}_\gamma$, and $M_n = c_1 \log n$. In the last step of the proof, we will consider $\mathfrak{z} = 2^\ell$, $\ell = L, L+1, \dots, \bar{L}$ where $\bar{L}$ is the smallest number such that $2^{\bar{L}} \mathfrak{r}_{2,n} \geq c$ for some sufficiently small constant $c > 0$, and ε_n is a quantity that we can choose. By Assumption SA-HLE(i), $\widehat{\gamma} - \gamma_0 \in \mathcal{H}_3$ with probability approaching one for C large enough, and by Lemma SA-3.5, $\|\check{\beta} - \check{\beta}_0\| \leq c$ with probability approaching one.

For any $\beta_1 \in \mathcal{H}_1, \beta_2 \in \mathcal{H}_2, \mathbf{v} \in \mathcal{V}$ and $\gamma := \gamma_0 + \gamma_1$ with $\gamma_1 \in \mathcal{H}_3$, define

$$\begin{aligned} \delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) &= \rho\left(y_i; \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma)\right) - \rho\left(y_i; \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma)\right) \\ &\quad - \left[\eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma) - \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma) \right] \\ &\quad \times \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \\ &= \int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \left[\psi\left(y_i, \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t)\right) - \psi\left(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)\right) \right] \\ &\quad \times \eta^{(1)}\left(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t\right) dt. \end{aligned}$$

Note that $\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \neq 0$ only if $\check{\mathbf{b}}'_i \mathbf{v} \neq 0$. For each $\mathbf{v} \in \mathcal{V}$, let $\mathcal{J}_{\mathbf{v}} = \{j : v_j \neq 0\}$. By construction, the cardinality of $\mathcal{J}_{\mathbf{v}}$ is bounded by $2M_n + 1$. We have $\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \neq 0$ only if $\check{b}_j(x_i) \neq 0$ for some $j \in \mathcal{J}_{\mathbf{v}}$, which happens only when $x_i \in \text{supp}(\check{b}_j(\cdot))$ for some $j \in \mathcal{J}_{\mathbf{v}}$. Let $\mathcal{I}_{\mathbf{v}} = \cup_{j \in \mathcal{J}_{\mathbf{v}}} \text{supp}(\check{b}_j(\cdot))$. Since the basis functions are locally supported, $\mathcal{I}_{\mathbf{v}}$ includes at most $c_2 M_n$ (connected) intervals for all $\mathbf{v} \in \mathcal{V}$. Moreover, at most $c_3 M_n$ basis functions in $\check{\mathbf{b}}(\cdot)$ have supports overlapping with $\mathcal{I}_{\mathbf{v}}$. Denote the set of indices for such basis functions by $\bar{\mathcal{J}}_{\mathbf{v}}$. Let $\check{\beta}_{0,j}$, $\beta_{1,j}$ and $\beta_{2,j}$ be the j th entries of $\check{\beta}_0$, β_1 , and β_2 respectively, and v_j be the j th entry of $\mathbf{v}$. Based on the above observations, we have $\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \equiv \delta_i(\beta_{1,\bar{\mathcal{J}}_{\mathbf{v}}}, \beta_{2,\bar{\mathcal{J}}_{\mathbf{v}}}, \mathbf{v}, \gamma)$ where

$$\begin{aligned} \delta_i(\beta_{1,\bar{\mathcal{J}}_{\mathbf{v}}}, \beta_{2,\bar{\mathcal{J}}_{\mathbf{v}}}, \mathbf{v}, \gamma) &:= \int_{-\sum_{j \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,j} v_j}^0 \left[\psi\left(y_i, \eta\left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l}(\check{\beta}_{0,l} + \beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma + t\right)\right) \right. \\ &\quad \left. - \psi\left(y_i, \eta\left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} \check{\beta}_{0,l} + \mathbf{w}'_i \gamma_0\right)\right) \right] \times \eta^{(1)}\left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l}(\check{\beta}_{0,l} + \beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma + t\right) dt \mathbf{1}_{i,\mathbf{v}}, \end{aligned}$$

$\mathbf{1}_{i,\mathbf{v}} = \mathbf{1}(x_i \in \mathcal{I}_{\mathbf{v}})$, and $\beta_{1,\bar{\mathcal{J}}_{\mathbf{v}}}$ and $\beta_{2,\bar{\mathcal{J}}_{\mathbf{v}}}$ respectively denote the subvectors of β_1 and β_2 whose indices belong to $\bar{\mathcal{J}}_{\mathbf{v}}$. Accordingly, define the following function class

$$\begin{aligned} \mathcal{G} &= \left\{ (x_i, \mathbf{w}_i, y_i) \mapsto \delta_i(\tilde{\beta}_1, \tilde{\beta}_2, \mathbf{v}, \gamma) : \mathbf{v} \in \mathcal{V}, \tilde{\beta}_1 \in \mathbb{R}^{c_3 M_n}, \tilde{\beta}_2 \in \mathbb{R}^{c_3 M_n}, \right. \\ &\quad \left. \|\tilde{\beta}_1\|_{\infty} \leq r_{1,n}, \|\tilde{\beta}_2\|_{\infty} \leq r_{2,n}, \gamma - \gamma_0 \in \mathcal{H}_3 \right\}. \end{aligned}$$

Step 1: We bound $\sup_{g \in \mathcal{G}} |\mathbb{E}_n[g(x_i, \mathbf{w}_i, y_i)] - \mathbb{E}[g(x_i, \mathbf{w}_i, y_i) | \mathcal{F}_{XW\Delta}]|$ in this step. Let $a_i(t) :=$

$\eta(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} \check{\beta}_{0,l} + \mathbf{w}'_i \gamma_0 + t)$. Define

$$\underline{a}_i = \min \left\{ a_i(0), a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 \right), a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 + \sum_{j \in \mathcal{J}_{\mathbf{v}}} \check{b}_{i,j} v_j \right) \right\} \text{ and}$$

$$\bar{a}_i = \max \left\{ a_i(0), a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 \right), a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 + \sum_{j \in \mathcal{J}_{\mathbf{v}}} \check{b}_{i,j} v_j \right) \right\}.$$

Consider the following two cases.

First, suppose that $(y_i - \bar{a}_i, y_i - \underline{a}_i)$ does not contain any discontinuity points. By Assumption [SA-SM](#), for all t in the interval of integration $[-\sum_{j \in \mathcal{J}_{\mathbf{v}}} \check{b}_{i,j} v_j, 0]$ (or $[0, -\sum_{j \in \mathcal{J}_{\mathbf{v}}} \check{b}_{i,j} v_j]$),

$$\left| \psi \left(y_i, a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 + t \right) \right) - \psi(y_i, a_i(0)) \right| \lesssim (1 + |\psi(y_i, \eta_i)|)(r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n}).$$

Second, if $(y_i - \bar{a}_i, y_i - \underline{a}_i)$ contains at least one discontinuity point, say j . For any t in the interval of integration, by Assumption [SA-SM](#),

$$\begin{aligned} & \left| \psi \left(y_i, a_i \left(\sum_{l \in \bar{\mathcal{J}}_{\mathbf{v}}} \check{b}_{i,l} (\beta_{1,l} + \beta_{2,l}) + \mathbf{w}'_i \gamma_1 + t \right) \right) - \psi(y_i, a_i(0)) \right| \\ & \lesssim 1 + \sqrt{d_\gamma} r_{3,n} + (1 + |\psi(y_i, \eta_i)|)(r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n}). \end{aligned}$$

for any $(x_i, \mathbf{w}_i, y_i)$, and in this case $y_i \in (j + \underline{a}_i, j + \bar{a}_i)$. By Assumption [SA-SM](#),

$$|\bar{a}_i - \underline{a}_i| \lesssim (r_{1,n} + r_{2,n} + \sqrt{d_\gamma} r_{3,n} + \varepsilon_n)(|\eta_{i,1}| + r_{1,n} + r_{2,n} + \sqrt{d_\gamma} r_{3,n} + \varepsilon_n).$$

By construction, for each $\mathbf{v} \in \mathcal{V}$, there exists some $k_{\mathbf{v}}$ such that $|v_\ell| \leq \varrho^{|\ell - k_{\mathbf{v}}|} \varepsilon_n$ for $|\ell - k_{\mathbf{v}}| \leq M_n$. Therefore, we can further write $\mathbb{1}_{i,\mathbf{v}} = \sum_{j: \hat{\mathcal{B}}_j \subset \mathcal{I}_{\mathbf{v}}} \mathbb{1}_{i,\mathbf{v},j}$ where each $\mathbb{1}_{i,\mathbf{v},j}$ is an indicator of the subinterval involved in $\mathcal{I}_{\mathbf{v}}$, and the above facts imply that for any $x_i \in \hat{\mathcal{B}}_l$ for some $\hat{\mathcal{B}}_l \subset \mathcal{I}_{\mathbf{v}}$,

$$\begin{aligned} \mathbb{V}[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) | \mathcal{F}_{XW\Delta}] & \lesssim \varrho^{2|(p-s+1)l - k_{\mathbf{v}}|} \varepsilon_n^2 \\ & \times (r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n})(|\eta_{i,1}| + r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n}). \end{aligned}$$

In addition, since $\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \neq 0$ only if $x_i \in \mathcal{I}_{\mathbf{v}}$, for all $g \in \mathcal{G}$ (each corresponds to a particular

$\mathbf{v})$,

$$\mathbb{E}_n[\mathbb{V}[g(x_i, \mathbf{w}_i, y_i)|\mathcal{F}_{XW\Delta}]] \lesssim \varepsilon_n^2(r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n}) \sum_{l: \widehat{\mathcal{B}}_l \subset \mathcal{I}_\mathbf{v}} \mathbb{E}_n[\mathbb{1}_{i,\mathbf{v},l}] \varrho^{2|(p-s+1)l-k_\mathbf{v}|}.$$

This inequality holds for any event in $\mathcal{F}_{XW\Delta}$. Define an event $\mathcal{A}_1$ on which $\sup_{1 \leq j \leq J} \mathbb{E}_n[\mathbb{1}_{i,j}] \leq C_2 J^{-1}$ for some large enough $C_2 > 0$ where $\mathbb{1}_{i,j} = \mathbb{1}(x_i \in \widehat{\mathcal{B}}_j)$. By the argument in Lemma SA-3.1, $\mathbb{P}(\mathcal{A}_1^c) \rightarrow 0$. On $\mathcal{A}_1$,

$$\bar{\sigma}^2 := \sup_{g \in \mathcal{G}} \mathbb{E}_n[\mathbb{V}[g(x_i, \mathbf{w}_i, y_i)|\mathcal{F}_{XW\Delta}]] \lesssim \varepsilon_n^2 J^{-1} (r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n}).$$

On the other hand,

$$\begin{aligned} \bar{G} := \sup_{g \in \mathcal{G}} |g(x_i, \mathbf{w}_i, y_i)| &\lesssim \varepsilon_n \{1 + \sqrt{d_\gamma}r_{3,n} + |\psi(y_i, \eta_i)|(r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n})\} \\ &\quad \times (|\eta_{i,1}| + r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n}). \end{aligned}$$

Also, for any $g, \tilde{g} \in \mathcal{G}$, denote the corresponding parameters defining g and $\tilde{g}$ by $(\beta_1, \beta_2, \mathbf{v}, \gamma)$ and $(\tilde{\beta}_1, \tilde{\beta}_2, \tilde{\mathbf{v}}, \tilde{\gamma})$. We have

$$\begin{aligned} |\tilde{g}(x_i, \mathbf{w}_i, y_i) - g(x_i, \mathbf{w}_i, y_i)| &= \left| \int_0^{\Lambda_1} \left[\psi(y_i, \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t)) \right. \right. \\ &\quad \left. \left. - \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right] \times \eta^{(1)}(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) dt \right. \\ &\quad \left. - \int_0^{\Lambda_2} \left[\psi(y_i, \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma + t)) \right. \right. \\ &\quad \left. \left. - \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right] \times \eta^{(1)}(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma + t) dt \right| \\ &\lesssim \{1 + \sqrt{d_\gamma}r_{3,n} + |\psi(y_i, \eta_i)|(r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n})\} \\ &\quad \times \left(|\eta_{i,1}| + r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma}r_{3,n} + |\Lambda_1| + |\Lambda_2| \right) \\ &\quad \times \left(\|\tilde{\beta}_1 - \beta_1\|_\infty + \|\tilde{\beta}_2 - \beta_2\|_\infty + \|\tilde{\mathbf{v}} - \mathbf{v}\|_\infty + \sqrt{d_\gamma} \|\tilde{\gamma} - \gamma\| \right), \end{aligned}$$

where $\Lambda_1 = \check{\mathbf{b}}'_i(\tilde{\beta}_1 + \tilde{\beta}_2 - \beta_1 - \beta_2) + \mathbf{w}'_i(\tilde{\gamma} - \gamma)$ and $\Lambda_2 = \Lambda_1 - \check{\mathbf{b}}'_i(\tilde{\mathbf{v}} - \mathbf{v})$. Based on these observations,

$$\begin{aligned} \|\bar{G}\|_{\mathbb{P},2} & \int_0^{\frac{\bar{\sigma}}{\|\bar{G}\|_{\mathbb{P},2}}} \sqrt{1 + \sup_{\mathbb{Q}} \log N(\mathcal{G}, L_2(\mathbb{Q}), t\|\bar{G}\|_{\mathbb{Q},2})} dt \\ & \lesssim \bar{\sigma} \left(\sqrt{\log J} + \sqrt{(d_\gamma + \log n) \log \frac{e\|\bar{G}\|_{\mathbb{P},2}}{\bar{\sigma}}} \right) \lesssim \bar{\sigma} \ell_{\gamma,n}. \end{aligned}$$

where the supremum is taken over all finite discrete probability measures $\mathbb{Q}$. Then, by Lemma SA-5.6,

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} \left| \mathbb{G}_n[g(x_i, \mathbf{w}_i, y_i)] \right| \middle| \mathcal{F}_{XW\Delta} \right] \lesssim \bar{\sigma} \ell_{\gamma,n} + \frac{\sqrt{\mathbb{E}[\bar{G}^2]} \ell_{\gamma,n}^2}{\sqrt{n}},$$

where $\bar{G} = \max_{1 \leq i \leq n} \bar{G}(x_i, \mathbf{w}_i, y_i)$. Note that $(\mathbb{E}[\bar{G}^2])^{1/2} \lesssim \varepsilon_n(1 + n^{1/\nu}(r_{1,n} + r_{2,n} + \sqrt{d_\gamma} r_{3,n} + \varepsilon_n))$.

Therefore, on $\mathcal{A}_1$ (whose probability approaches one),

$$\begin{aligned} & \sup_{\beta_1 \in \mathcal{H}_1, \beta_2 \in \mathcal{H}_2, \mathbf{v} \in \mathcal{V}, \gamma_1 \in \mathcal{H}_3} \left| \mathbb{E}_n \left[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \right] - \mathbb{E}_n \left[\mathbb{E}[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) | \mathcal{F}_{XW\Delta}] \right] \right| \\ & \lesssim \left(J^{-1} \varepsilon_n \sqrt{\mathfrak{L}_n} \sqrt{\frac{J}{n}} \ell_{\gamma,n} + \frac{\varepsilon_n(1 + n^{1/\nu} \mathfrak{L}_n) \ell_{\gamma,n}^2}{n} \right) \end{aligned}$$

for $\mathfrak{L}_n = r_{1,n} + r_{2,n} + \sqrt{d_\gamma} r_{3,n} + \varepsilon_n$.

Step 2: For $\tilde{\mathbf{Q}} := \mathbb{E}_n[\check{\mathbf{b}}_i \check{\mathbf{b}}'_i \Psi_1(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0))(\eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0))^2]$, by Assumption SA-SM and the same argument in the proof of Lemma SA-3.1,

$$\|\check{\mathbf{Q}} - \tilde{\mathbf{Q}}\|_\infty \vee \|\check{\mathbf{Q}} - \tilde{\mathbf{Q}}\| \lesssim J^{-p-2}.$$

Therefore,

$$\sup_{\beta_1 \in \mathcal{H}_1, \beta_2 \in \mathcal{H}_2, \mathbf{v} \in \mathcal{V}} |\mathbf{v}'(\tilde{\mathbf{Q}} - \check{\mathbf{Q}})(\beta_1 + \beta_2)| \lesssim J^{-p-2} \varepsilon_n(r_{1,n} + r_{2,n}).$$

In addition, by Lemmas SA-3.3 and SA-3.4, $\|\bar{\beta}\|_\infty \leq r_{1,n}$ with probability approaching one for C_1 large enough, where

$$\bar{\beta} := -\check{\mathbf{Q}}^{-1} \mathbb{E}_n \left[\check{\mathbf{b}}_i \eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \psi \left(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \right) \right].$$

Step 3: Let $\tilde{z}_i = \Psi_1(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0))(\eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0))^2$. By Taylor expansion, we have

$$\begin{aligned}
& \mathbb{E}_n \left[\mathbb{E}[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) | \mathcal{F}_{XW\Delta}] \right] \\
&= \mathbb{E}_n \left[\int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \left\{ \Psi(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t)) \right. \right. \\
&\quad \left. \left. - \Psi(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right\} \times \eta^{(1)}(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) dt \right] \\
&= \mathbb{E}_n \left[\int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \left\{ \Psi_1(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \left(\eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) (\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 + t) \right. \right. \right. \\
&\quad \left. \left. \left. + \frac{1}{2} \eta^{(2)}(\xi_{i,t}) (\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 + t)^2 \right) \right. \right. \\
&\quad \left. \left. + \frac{1}{2} \Psi_2(x_i, \mathbf{w}_i; \xi_{i,t}) \left(\eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) - \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \right)^2 \right\} \right. \\
&\quad \left. \times \left(\eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) + \eta^{(2)}(\xi_{i,t}) (\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 + t) \right) dt \right] \\
&= \mathbf{v}' \tilde{\mathbf{Q}}(\beta_1 + \beta_2) + \mathbf{v}' \mathbb{E}_n[\check{\mathbf{b}}_i \tilde{z}_i \mathbf{w}'_i] \gamma_1 - \frac{1}{2} \mathbf{v}' \tilde{\mathbf{Q}} \mathbf{v} + \text{I} + \text{II} + \text{III},
\end{aligned}$$

where $\xi_{i,t}$ and $\tilde{\xi}_{i,t}$ are between $\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0$ and $\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t$, $\tilde{\xi}_{i,t}$ is between $\eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)$ and $\eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t)$, $\sup_{1 \leq i \leq n} |\mathbf{w}'_i \gamma_1| \lesssim \sqrt{d_\gamma} r_{3,n}$, $\mathbb{E}_n[|\mathbf{v}' \check{\mathbf{b}}_i|] \lesssim \varepsilon_n/J$, and hence $|\mathbf{v}' \mathbb{E}_n[\check{\mathbf{b}}_i \tilde{z}_i \mathbf{w}'_i] \gamma_1| \lesssim \varepsilon_n \sqrt{d_\gamma} r_{3,n}/J$, $|\mathbf{v}' \tilde{\mathbf{Q}} \mathbf{v}| \lesssim \varepsilon_n^2/J$, and I, II, and III are defined and bounded as follows:

$$\begin{aligned}
\text{I} &= \mathbb{E}_n \left[\int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \Psi_1(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \right. \\
&\quad \left. \times \eta^{(2)}(\xi_{i,t}) (\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 + t)^2 dt \mathbf{1}_{i,v} \right] \lesssim \varepsilon_n J^{-1} (r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n})^2, \\
\text{II} &= \mathbb{E}_n \left[\int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \Psi_1(x_i, \mathbf{w}_i; \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \times \frac{1}{2} \eta^{(2)}(\xi_{i,t}) (\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 + t)^2 \right. \\
&\quad \left. \times \eta^{(1)}(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) dt \mathbf{1}_{i,v} \right] \lesssim \varepsilon_n J^{-1} (r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n})^2, \\
\text{III} &= \mathbb{E}_n \left[\int_{-\check{\mathbf{b}}'_i \mathbf{v}}^0 \frac{1}{2} \Psi_2(x_i, \mathbf{w}_i; \xi_{i,t}) \left(\eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) - \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \right)^2 \right. \\
&\quad \left. \times \eta^{(1)}(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma + t) dt \mathbf{1}_{i,v} \right] \lesssim \varepsilon_n J^{-1} (r_{1,n} + r_{2,n} + \varepsilon_n + \sqrt{d_\gamma} r_{3,n})^2.
\end{aligned}$$

These bounds hold uniformly for $\mathbf{v} \in \mathcal{V}$, $\beta_1 \in \mathcal{H}_1$, $\beta_2 \in \mathcal{H}_2$ and $\gamma_1 \in \mathcal{H}_3$ (that is, uniformly over the function class $\mathcal{G}$), and on an event $\mathcal{A}_1 \cap \mathcal{A}_2$ where $\mathcal{A}_2 = \{\lambda_{\max}(\tilde{\mathbf{Q}}) \leq c_4 J^{-1}\}$ for some large enough $c_4 > 0$. Note that $\mathbb{P}(\mathcal{A}_1 \cap \mathcal{A}_2) \rightarrow 1$ by Lemma SA-3.1.

Step 4: By Assumption SA-SM and the integral form of Taylor's expansion,

$$\begin{aligned}
\text{IV} &= \mathbb{E}_n \left[\left(\eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma) - \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma) \right) \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right] \\
&\quad - \mathbb{E}_n \left[\mathbf{v}' \check{\mathbf{b}}_i \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \eta^{(1)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \right] \\
&= \sum_j v_j \mathbb{E}_n \left[\check{b}_{i,j} \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \int_0^1 \int_0^{\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 - t \check{\mathbf{b}}'_i \mathbf{v}} \eta^{(2)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0 + s) \, ds \, dt \right] \\
&\lesssim J^{-1} \left\{ \sqrt{\frac{J(d_\gamma + \log J)}{n}} + \frac{J(d_\gamma + \log J)}{n^{1-1/\nu}} + J^{-p-1} \right\} (\varepsilon_n + r_{1,n} + r_{2,n} + \sqrt{d_\gamma} r_{3,n}) \varepsilon_n,
\end{aligned}$$

The last line holds uniformly over $\beta_1 \in \mathcal{H}_1$, $\beta_2 \in \mathcal{H}_2$, $\mathbf{v} \in \mathcal{V}$, and $\gamma_1 \in \mathcal{H}_3$ on $\mathcal{A}_1 \cap \mathcal{A}_3$, where $\mathbb{P}(\mathcal{A}_3) \rightarrow 1$. To verify it, fix j in the preceding sum and, recalling $\mathfrak{L}_n$ from Step 1, write

$$\begin{aligned}
R_{i,j} &:= \check{b}_{i,j} \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \int_0^1 \int_0^{\check{\mathbf{b}}'_i(\beta_1 + \beta_2) + \mathbf{w}'_i \gamma_1 - t \check{\mathbf{b}}'_i \mathbf{v}} \eta^{(2)}(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0 + s) \, ds \, dt, \\
|R_{i,j}| &\lesssim |\check{b}_{i,j} \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0))| \mathfrak{L}_n, \quad \mathbb{E}_n[\mathbb{1}(\check{b}_{i,j} \neq 0)] \lesssim J^{-1} \quad \text{on } \mathcal{A}_1.
\end{aligned}$$

On the support of $\check{b}_{i,j}$, local support implies that only a fixed number of coordinates of β_1 , β_2 , and $\mathbf{v}$ enter $R_{i,j}$. Its smoothness and the variation in γ_1 therefore give covering entropy bounded by $C d_\gamma \log(C/t)$ relative to the displayed envelope. Taking the union over $O(J)$ possible values of j adds $\log J$. Consequently, with each supremum below taken over $1 \leq j \leq K_{p,s}$ and the parameter sets above, Lemma SA-5.6 and the conditional mean bound give

$$\begin{aligned}
\sup_{j, \beta_1, \beta_2, \mathbf{v}, \gamma_1} |\mathbb{E}_n[R_{i,j}] - \mathbb{E}[R_{i,j} \mid \mathcal{F}_{XW\Delta}]| &\lesssim J^{-1} \left\{ \sqrt{\frac{J(d_\gamma + \log J)}{n}} + \frac{J(d_\gamma + \log J)}{n^{1-1/\nu}} \right\} \mathfrak{L}_n, \\
\sup_{j, \beta_1, \beta_2, \mathbf{v}, \gamma_1} |\mathbb{E}_n[\mathbb{E}[R_{i,j} \mid \mathcal{F}_{XW\Delta}]]| &\lesssim J^{-1} J^{-p-1} \mathfrak{L}_n, \\
|\text{IV}| &\leq \sum_j |v_j| \sup_{j, \beta_1, \beta_2, \mathbf{v}, \gamma_1} |\mathbb{E}_n[R_{i,j}]| \lesssim J^{-1} \left\{ \sqrt{\frac{J(d_\gamma + \log J)}{n}} + \frac{J(d_\gamma + \log J)}{n^{1-1/\nu}} + J^{-p-1} \right\} \mathfrak{L}_n \varepsilon_n.
\end{aligned}$$

Here the second line uses $\mathbb{E}[\psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \mid \mathcal{F}_{XW\Delta}] \lesssim J^{-p-1}$, while the last line uses the geometric bound $\sum_j |v_j| \lesssim \varepsilon_n$. This also makes clear why no factor proportional to M_n appears.

Step 5: Let

$$\bar{\mathbf{v}} = c_5 \varepsilon_n J^{-1} \text{sgn}(\beta_{2,k}) [\check{\mathbf{Q}}^{-1}]_{\cdot k}$$

for some k such that $|\beta_{2,k}| = \|\beta_2\|_\infty$ and some $c_5 > 0$, where $[\check{\mathbf{Q}}^{-1}]_{\cdot k}$ denotes the k th column of $\check{\mathbf{Q}}^{-1}$. Since $J^{-1}\check{\mathbf{Q}}^{-1} = \bar{\mathbf{Q}}^{-1}$,

$$\bar{\mathbf{v}}' \check{\mathbf{Q}} \beta_2 = c_5 \varepsilon_n J^{-1} |\beta_{2,k}|.$$

Take $\mathbf{v} = (v_1, \dots, v_{K_{p,s}})$ where $v_j = \bar{v}_j$ for $|j - k| \leq M_n$ and zero otherwise. Clearly, $\mathbf{v} \in \mathcal{V}$ on an event $\mathcal{A}_4$ with $\mathbb{P}(\mathcal{A}_4) \rightarrow 1$. On $\mathcal{A}_2 \cap \mathcal{A}_4$,

$$|(\mathbf{v} - \bar{\mathbf{v}})' \check{\mathbf{Q}} \beta_2| \lesssim \varepsilon_n J^{-1} r_{2,n} n^{-c_6}$$

for some large $c_6 > 0$ if we let c_1 be sufficiently large.

Step 6: Partition the parameter space into shells $\mathcal{O} = \cup_{\ell=-\infty}^{\bar{L}} \mathcal{O}_\ell$, where

$$\mathcal{O}_\ell = \left\{ \beta \in \mathbb{R}^{K_{p,s}} : 2^{\ell-1} \mathbf{r}_{2,n} \leq \|\beta - \check{\beta}_0 - \bar{\beta}\|_\infty \leq 2^\ell \mathbf{r}_{2,n} \right\},$$

and let $\bar{L}$ be the smallest integer such that $2^{\bar{L}} \mathbf{r}_{2,n} \geq c$. Also, define

$$\mathcal{H}_{2,\ell} = \left\{ \beta_2 \in \mathbb{R}^{K_{p,s}} : 2^{\ell-1} \mathbf{r}_{2,n} \leq \|\beta_2\|_\infty \leq 2^\ell \mathbf{r}_{2,n} \right\}.$$

By construction,

$$\check{\mathbf{Q}} \bar{\beta} = -\mathbb{E}_n \left[\check{\mathbf{b}}_i \eta^{(1)} (\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0) \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right].$$

For generic $(\beta_1, \beta_2, \mathbf{v}, \gamma)$ in the sets defined in Step 0, let

$$\begin{aligned} A_i(\beta_1, \beta_2, \mathbf{v}, \gamma) &= \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma) \\ &\quad - \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma), \\ s_i^0 &= \psi(y_i, \eta(\check{\mathbf{b}}'_i \check{\beta}_0 + \mathbf{w}'_i \gamma_0)). \end{aligned}$$

The definition of δ_i in Step 0 gives the exact identity

$$\begin{aligned} &\mathbb{E}_n \left[\rho(y_i; \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2) + \mathbf{w}'_i \gamma)) - \rho(y_i; \eta(\check{\mathbf{b}}'_i(\check{\beta}_0 + \beta_1 + \beta_2 - \mathbf{v}) + \mathbf{w}'_i \gamma)) \right] \\ &= \mathbb{E}_n \left[\mathbb{E}[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \mid \mathcal{F}_{XW\Delta}] \right] + \mathbb{E}_n [A_i(\beta_1, \beta_2, \mathbf{v}, \gamma) s_i^0] + \frac{1}{\sqrt{n}} \mathbb{G}_n[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma)], \end{aligned}$$

where, throughout this step, $\mathbb{G}_n[f_i] = \sqrt{n} \mathbb{E}_n[f_i - \mathbb{E}[f_i \mid \mathcal{F}_{XW\Delta}]]$. Conditional expectations in-

volving random parameter values are understood as the generic-parameter conditional expectations evaluated at those values; the bounds in Steps 1–4 are uniform over the corresponding parameter sets.

Let $\mathcal{A} = \cap_{j=1}^4 \mathcal{A}_j$. For $\beta_1 = \bar{\beta}$, $\beta_2 \in \mathcal{H}_{2,\ell}$, and $\gamma_1 = \hat{\gamma} - \gamma_0$, Steps 2–5 and the signed direction constructed in Step 5 give, the uniform lower bound

$$\begin{aligned} & \mathbb{E}_n \left[\mathbb{E} \left[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \mid \mathcal{F}_{XW\Delta} \right] \right] + \mathbb{E}_n \left[A_i(\beta_1, \beta_2, \mathbf{v}, \gamma) s_i^0 \right] \\ & \geq c_5 J^{-1} \varepsilon_n 2^{\ell-1} \mathbf{r}_{2,n} - C J^{-1} \varepsilon_n \left\{ \sqrt{d_\gamma} r_{3,n} + \varepsilon_n + \mathfrak{L}_{n,\ell}^2 + q_n \mathfrak{L}_{n,\ell} + J^{-p-1} \mathfrak{L}_{n,\ell} + 2^\ell \mathbf{r}_{2,n} n^{-c_6} \right\}, \end{aligned}$$

with

$$\begin{aligned} \mathfrak{L}_{n,\ell} &= r_{1,n} + 2^\ell \mathbf{r}_{2,n} + \sqrt{d_\gamma} r_{3,n} + \varepsilon_n, \\ q_n &= \left(\frac{J \log n}{n} \right)^{1/2} + \sqrt{\frac{J(d_\gamma + \log J)}{n}} + \frac{J(d_\gamma + \log J)}{n^{1-1/\nu}} + J^{-p-1}. \end{aligned}$$

To verify the required absorption explicitly, the definition of $\mathbf{r}_{2,n}$ and Young's inequality give

$$\sqrt{\frac{J}{n}} \ell_{\gamma,n} \sqrt{r_{1,n} + \sqrt{d_\gamma} r_{3,n}} \lesssim \left(\frac{J \log n}{n} \right)^{3/4} \ell_{\gamma,n} + J^{-\frac{p+1}{2}} \sqrt{\frac{J}{n}} \ell_{\gamma,n} + \frac{J \ell_{\gamma,n}^2}{n} + \sqrt{d_\gamma} r_{3,n}. \quad (\text{SA-5.1})$$

$$(r_{1,n} + \sqrt{d_\gamma} r_{3,n})^2 \lesssim \left(\frac{J \log n}{n} \right)^{3/4} \ell_{\gamma,n} + J^{-p-1} + \sqrt{d_\gamma} r_{3,n}. \quad (\text{SA-5.2})$$

The second rate condition in the theorem further gives

$$\frac{J \ell_{\gamma,n}^2}{n^{1-1/\nu}} (r_{1,n} + \sqrt{d_\gamma} r_{3,n}) \lesssim (r_{1,n} + \sqrt{d_\gamma} r_{3,n})^2, \quad (\text{SA-5.3})$$

while $J(d_\gamma + \log J)/n \lesssim J \ell_{\gamma,n}^2/n$ and $J(d_\gamma + \log J)/n^{1-1/\nu} \lesssim r_{1,n}$ show that the Step 4 base-radius terms are absorbed by (SA-5.1)–(SA-5.3). Set $\varepsilon_n = c_7 2^L \mathbf{r}_{2,n}$ for a sufficiently small fixed $c_7 > 0$. The outer-shell terms must instead be compared with the shell radius. Uniformly for $L \leq \ell \leq \bar{L}$,

$$\mathfrak{L}_{n,\ell}^2 \lesssim \mathbf{r}_{2,n} + C c 2^\ell \mathbf{r}_{2,n}, \quad q_n \mathfrak{L}_{n,\ell} \lesssim \mathbf{r}_{2,n} + o(1) 2^\ell \mathbf{r}_{2,n},$$

and the same second bound holds with q_n replaced by J^{-p-1} . Indeed, $2^\ell \mathbf{r}_{2,n} \lesssim c$, $\varepsilon_n \leq c_7 2^\ell \mathbf{r}_{2,n}$, the base-radius bounds in (SA-5.1) and (SA-5.2) are $O(\mathbf{r}_{2,n})$, and $q_n = o(1)$. Consequently, the terms independent of the shell radius are absorbed by taking L large, while the terms proportional to

$2^\ell \mathfrak{r}_{2,n}$ are absorbed by taking c and c_7 small. Hence, by first taking the localization constant c and c_7 sufficiently small and then L sufficiently large, the preceding display implies, uniformly on $\mathcal{A}$,

$$\mathbb{E}_n \left[\mathbb{E}[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma) \mid \mathcal{F}_{XW\Delta}] \right] + \mathbb{E}_n [A_i(\beta_1, \beta_2, \mathbf{v}, \gamma) s_i^0] \geq C_4 J^{-1} 2^\ell \mathfrak{r}_{2,n} \varepsilon_n$$

for some $C_4 > 0$. Exact optimality of $\check{\beta}$ yields a non-strict objective comparison. Therefore, by Lemma SA-3.5, the preceding display, and conditional Markov's inequality,

$$\begin{aligned} & \mathbb{P} \left(\|\check{\beta} - \check{\beta}_0 - \bar{\beta}\|_\infty \geq 2^L \mathfrak{r}_{2,n} \mid \mathcal{F}_{XW\Delta} \right) \\ & \leq \mathbb{P} \left(\bigcup_{\ell=L}^{\bar{L}} \left\{ \sup_{\beta_1 \in \mathcal{H}_1} \sup_{\beta_2 \in \mathcal{H}_{2,\ell}} \sup_{\gamma_1 \in \mathcal{H}_3} \sup_{\mathbf{v} \in \mathcal{V}} \frac{1}{\sqrt{n}} |\mathbb{G}_n[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma)]| \geq C_4 J^{-1} 2^\ell \mathfrak{r}_{2,n} \varepsilon_n \right\} \cap \mathcal{A} \mid \mathcal{F}_{XW\Delta} \right) + o_{\mathbb{P}}(1) \\ & \leq \sum_{\ell=L}^{\bar{L}} (C_6 J^{-1} 2^\ell \mathfrak{r}_{2,n} \varepsilon_n)^{-1} \mathbb{1}(\mathcal{A}_1) \mathbb{E} \left[\sup_{\beta_1 \in \mathcal{H}_1} \sup_{\beta_2 \in \mathcal{H}_{2,\ell}} \sup_{\gamma_1 \in \mathcal{H}_3} \sup_{\mathbf{v} \in \mathcal{V}} \frac{1}{\sqrt{n}} |\mathbb{G}_n[\delta_i(\beta_1, \beta_2, \mathbf{v}, \gamma)]| \mid \mathcal{F}_{XW\Delta} \right] + o_{\mathbb{P}}(1). \end{aligned}$$

Using Step 1, dividing its bound by the shell threshold in the preceding display, and summing over the shells gives

$$\begin{aligned} & \sum_{\ell=L}^{\bar{L}} \left\{ \frac{\sqrt{J/n} \ell_{\gamma,n} \sqrt{r_{1,n} + \sqrt{d_\gamma} r_{3,n} + 2^\ell \mathfrak{r}_{2,n} + \varepsilon_n}}{2^\ell \mathfrak{r}_{2,n}} + \frac{J \ell_{\gamma,n}^2}{n 2^\ell \mathfrak{r}_{2,n}} \right. \\ & \quad \left. + \frac{J \ell_{\gamma,n}^2 (r_{1,n} + \sqrt{d_\gamma} r_{3,n} + 2^\ell \mathfrak{r}_{2,n} + \varepsilon_n)}{n^{1-1/\nu} 2^\ell \mathfrak{r}_{2,n}} \right\} \\ & \lesssim \underbrace{2^{-L/2}}_{\text{first summand}} + \underbrace{2^{-L}}_{\text{second summand}} + \underbrace{\left(2^{-L} + \frac{J \ell_{\gamma,n}^2}{n^{1-1/\nu}} \bar{L} \right)}_{\text{third summand}}. \end{aligned}$$

The first group follows from (SA-5.1), the definition of $\mathfrak{r}_{2,n}$, and $\varepsilon_n = c_7 2^L \mathfrak{r}_{2,n}$. The second follows from $J \ell_{\gamma,n}^2 / n \leq \mathfrak{r}_{2,n}$. For the third group, (SA-5.3) followed by (SA-5.2) gives the 2^{-L} term, while the second rate condition in the theorem gives

$$\frac{J \ell_{\gamma,n}^2}{n^{1-1/\nu}} \bar{L} \lesssim r_{1,n} \log J = o(1),$$

because $\bar{L} = O(\log(1/\mathfrak{r}_{2,n})) = O(\log J)$. Thus the displayed shell sum can be made arbitrarily small

by first choosing L sufficiently large and then letting $n \rightarrow \infty$. Hence

$$\|\check{\beta} - \check{\beta}_0 - \bar{\beta}\|_\infty = O_{\mathbb{P}}(\mathfrak{r}_{2,n}).$$

Moreover,

$$\frac{\bar{\beta}}{\sqrt{J}} = -\bar{\mathbf{Q}}^{-1} \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \eta^{(1)}(\widehat{\mathbf{b}}_{p,s}(x_i)' \bar{\beta}_0 + \mathbf{w}'_i \gamma_0) \psi(y_i, \eta(\widehat{\mathbf{b}}_{p,s}(x_i)' \bar{\beta}_0 + \mathbf{w}'_i \gamma_0)) \right].$$

Therefore, Lemma SA-3.4 gives

$$\begin{aligned} & \sup_{x \in \mathcal{X}} \left| \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \left\{ \frac{\bar{\beta}}{\sqrt{J}} + \bar{\mathbf{Q}}^{-1} \mathbb{E}_n [\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right\} \right| \\ & \lesssim_{\mathbb{P}} J^v \left\{ J^{-p-1} + J^{-(p+1)/2} \left(\frac{J \log J}{n} \right)^{1/2} + \frac{J \log J}{n} \right\}. \end{aligned}$$

The last two terms are absorbed by the first two stochastic terms in $\mathfrak{r}_{2,n}$ under the theorem's rate conditions. Combining these displays with the approximation bound $\sup_x |\widehat{r}_{0,v}(x)| \lesssim_{\mathbb{P}} J^{-p-1+v}$ and converting back through $\check{\beta} = \sqrt{J} \bar{\beta}$ proves the leading term and the remainder stated in part (i).

Step 7: To show part (ii) and part (iii), by Taylor expansion and the result in part (i),

$$\begin{aligned} & \eta(\widehat{\mu}(x) + \widehat{\mathbf{w}}' \widehat{\gamma}) - \eta(\mu_0(x) + \mathbf{w}' \gamma_0) \\ &= \eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \left(\widehat{\mathbf{b}}_{p,s}(x)' \widehat{\beta} - \mu_0(x) \right) \\ & \quad + O_{\mathbb{P}} \left(\|\widehat{\mathbf{w}} - \mathbf{w}\| + \sqrt{d_\gamma} \mathfrak{r}_\gamma + \frac{J \log n}{n} + J^{-2p-2} + \mathfrak{r}_{\text{BR},n}^2 \right) \\ &= -\eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \widehat{\mathbf{b}}_{p,s}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n [\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \\ & \quad + O_{\mathbb{P}} \left(\mathfrak{r}_{\text{BR},n} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \right), \end{aligned}$$

and

$$\begin{aligned} & \eta^{(1)}(\widehat{\mu}(x) + \widehat{\mathbf{w}}' \widehat{\gamma}) \widehat{\mu}^{(1)}(x) - \eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \mu_0^{(1)}(x) \\ &= \eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \left(\widehat{\mu}^{(1)}(x) - \mu_0^{(1)}(x) \right) \\ & \quad + O_{\mathbb{P}} \left(\left(\frac{J \log n}{n} \right)^{1/2} + J^{-p-1} + \|\widehat{\mathbf{w}} - \mathbf{w}\| + \mathfrak{r}_{\text{BR},n} \right) O_{\mathbb{P}} \left(1 + J \left(\left(\frac{J \log n}{n} \right)^{1/2} + J^{-p-1} + \mathfrak{r}_{\text{BR},n} \right) \right) \\ &= -\eta^{(1)}(\mu_0(x) + \mathbf{w}' \gamma_0) \widehat{\mathbf{b}}_{p,s}^{(1)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{E}_n [\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] + \end{aligned}$$

$$O_{\mathbb{P}}\left(\left(\frac{J \log n}{n}\right)^{1/2} + J^{-p} + J \mathfrak{r}_{\text{BR},n} + \|\widehat{\mathbf{w}} - \mathbf{w}\| \left(1 + \left(\frac{J^3 \log n}{n}\right)^{1/2}\right)\right).$$

In the above derivation the probability bound holds uniformly over $x \in \mathcal{X}$ as well. Then the proof is complete. $\square$

SA-5.8 Proof of Theorem SA-3.2

Proof. Since $\widehat{\epsilon}_i := \epsilon_i + \eta_i - \widehat{\eta}_i =: \epsilon_i + u_i$, we can write

$$\begin{aligned} & \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\eta}_{i,1}^2 \psi^\dagger(\widehat{\eta}_i)^2 \psi^\dagger(\widehat{\epsilon}_i)^2] - \mathbb{E}[\mathbf{b}_{p,s}(x_i) \mathbf{b}_{p,s}(x_i)' \eta_{i,1}^2 \sigma^2(x_i, \mathbf{w}_i)] \\ &= \mathbb{E}_n\left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\eta}_{i,1}^2 \psi^\dagger(\widehat{\eta}_i)^2 \left(\psi^\dagger(\epsilon_i + u_i)^2 - \psi^\dagger(\epsilon_i)^2\right)\right] \\ & \quad + \mathbb{E}_n\left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \left(\widehat{\eta}_{i,1}^2 \psi^\dagger(\widehat{\eta}_i)^2 - \eta_{i,1}^2 \psi^\dagger(\eta_i)^2\right) \psi^\dagger(\epsilon_i)^2\right] \\ & \quad + \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 (\psi(y_i, \eta_i)^2 - \sigma^2(x_i, \mathbf{w}_i))] \\ & \quad + \left(\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \sigma^2(x_i, \mathbf{w}_i)] - \mathbb{E}[\mathbf{b}_{p,s}(x_i) \mathbf{b}_{p,s}(x_i)' \eta_{i,1}^2 \sigma^2(x_i, \mathbf{w}_i)]\right) \\ & =: \mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 + \mathbf{V}_4. \end{aligned}$$

We bound each term in the following. The first part of the theorem only concerns $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3$, and the second part needs a bound on $\mathbf{V}_4$ as well where the additional Assumption SA-RP(ii) is used.

Step 1: For $\mathbf{V}_1$, we further write $\mathbf{V}_1 = \mathbf{V}_{11} + \mathbf{V}_{12}$ where

$$\begin{aligned} \mathbf{V}_{11} &:= \mathbb{E}_n\left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 \left(\psi^\dagger(\epsilon_i + u_i)^2 - \psi^\dagger(\epsilon_i)^2\right)\right], \\ \mathbf{V}_{12} &:= \mathbb{E}_n\left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \left(\widehat{\eta}_{i,1}^2 \psi^\dagger(\widehat{\eta}_i)^2 - \eta_{i,1}^2 \psi^\dagger(\eta_i)^2\right) \left(\psi^\dagger(\epsilon_i + u_i)^2 - \psi^\dagger(\epsilon_i)^2\right)\right]. \end{aligned}$$

Let $r_{1,n} = C_1(J \log n/n)^{1/2} + J^{-p-1}$ for a constant $C_1 > 0$. By Assumption SA-SM and Corollary SA-3.1, $\max_{1 \leq i \leq n} |u_i| \leq r_{1,n}$ with arbitrarily large probability for C_1 sufficiently large. For $\mathbf{V}_{11}$, let $\mathcal{J}$ be the set of all discontinuity points of $\psi(\cdot)$. Define $\mathbf{1}_{i,\mathcal{D}} := \mathbf{1}(\epsilon_i \in \mathcal{D})$ and $\mathbf{1}_{i,\mathcal{D}^c} := (1 - \mathbf{1}_{i,\mathcal{D}})$ where $\mathcal{D} := \{a : |a - j| \leq r_{1,n} \text{ for some } j \in \mathcal{J}\}$. Define

$$\mathbf{V}_{111} := \mathbb{E}_n\left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 \left(\psi^\dagger(\epsilon_i + u_i)^2 - \psi^\dagger(\epsilon_i)^2\right) \mathbf{1}_{i,\mathcal{D}}\right],$$

$$\mathbf{V}_{112} := \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 \left(\psi^\dagger(\epsilon_i + u_i)^2 - \psi^\dagger(\epsilon_i)^2 \right) \mathbf{1}_{i,\mathcal{D}^c} \right].$$

By definition of $\mathcal{D}$ and Assumption [SA-SM](#),

$$\|\mathbf{V}_{111}\| \lesssim \|\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \mathbb{E}[\mathbf{1}_{i,\mathcal{D}} | \mathcal{F}_{XW\Delta}]]\| + \|\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' (\mathbf{1}_{i,\mathcal{D}} - \mathbb{E}[\mathbf{1}_{i,\mathcal{D}} | \mathcal{F}_{XW\Delta}])]\|.$$

By Assumption [SA-SM](#) and Lemma SA-3.5 of [Cattaneo et al. \(2024a\)](#), the first term on the right-hand side is $O_{\mathbb{P}}(r_{1,n})$. For the second term, conditional on $\mathcal{F}_{XW\Delta}$, it is an independent sequence with mean zero. Thus, we can apply the argument given in Step 3 below and conclude that the second term is $O_{\mathbb{P}}(\sqrt{r_{1,n} J \log J/n} + J \log J/n)$. In this case, the indicator $\mathbf{1}_{i,\mathcal{D}}$ is trivially bounded uniformly.

On the other hand, by Assumption [SA-SM](#),

$$\|\mathbf{V}_{112}\| \lesssim r_{1,n} \|\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 |\psi^\dagger(\epsilon_i + u_i) + \psi^\dagger(\epsilon_i)|]\|.$$

Since $|c| \leq \frac{1}{2}(1 + c^2)$ for any scalar c , we have

$$\mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 |\psi^\dagger(\epsilon_i)| \right] \leq \frac{1}{2} \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 (1 + \psi^\dagger(\epsilon_i)^2) \right] \lesssim_{\mathbb{P}} 1,$$

by Lemma [SA-3.1](#) and the result in Step 3. In addition, we further write

$$\begin{aligned} & \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 |\psi^\dagger(\epsilon_i + u_i)| \right] \\ &= \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi^\dagger(\eta_i)^2 |\psi^\dagger(\epsilon_i) + (\psi^\dagger(\epsilon_i + u_i) - \psi^\dagger(\epsilon_i))| \right]. \end{aligned}$$

Repeat the previous argument to bound this term. We conclude that $\|\mathbf{V}_{11}\| \lesssim_{\mathbb{P}} r_{1,n}$.

$\mathbf{V}_{12}$ can be treated using the previous argument combined with the argument given in Step 2 and the result in Step 3. It leads to $\|\mathbf{V}_{12}\| \lesssim_{\mathbb{P}} r_{1,n}$.

Step 2: For $\mathbf{V}_2$, by Assumption [SA-SM](#), Corollary [SA-3.1](#) and the argument given later in Step 3, we have

$$\|\mathbf{V}_2\| \leq \max_{1 \leq i \leq n} |\widehat{\eta}_{i,1}^2 \psi^\dagger(\widehat{\eta}_i)^2 - \eta_{i,1}^2 \psi^\dagger(\eta_i)^2| \|\mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \widehat{\mathbf{b}}_{p,s}(x_i)' \psi^\dagger(\epsilon_i)^2]\| \lesssim_{\mathbb{P}} (J \log n/n)^{1/2} + J^{-p-1}.$$

Step 3: For $\mathbf{V}_3$, Assumption [SA-RP\(i\)](#) implies that $\hat{\Delta} \in \Pi_C$ w.p.a. 1 and that $\mathbb{E}[\psi(y_i, \eta_i)^2 - \sigma^2(x_i, \mathbf{w}_i) \mid \mathcal{F}_{XW\Delta}] = 0$. Thus, in view of Lemma [SA-5.3](#), it suffices to show that

$$\sup_{\Delta \in \Pi_C} \left\| \mathbb{E}_n[\mathbf{b}_{p,0}(x_i; \Delta) \mathbf{b}_{p,0}(x_i; \Delta)' \eta_{i,1}^2 (\psi(y_i, \eta_i)^2 - \sigma^2(x_i, \mathbf{w}_i))] \right\| \lesssim_{\mathbb{P}} \left(\frac{J \log J}{n^{\frac{\nu-2}{\nu}}} \right)^{1/2}.$$

For notational simplicity, we write $\varphi_i = \psi(y_i, \eta_i)^2 - \sigma^2(x_i, \mathbf{w}_i)$, $\varphi_i^- = \varphi_i \mathbf{1}(|\varphi_i| \leq M) - \mathbb{E}[\varphi_i \mathbf{1}(|\varphi_i| \leq M) \mid x_i, \mathbf{w}_i]$, $\varphi_i^+ = \varphi_i \mathbf{1}(|\varphi_i| > M) - \mathbb{E}[\varphi_i \mathbf{1}(|\varphi_i| > M) \mid x_i, \mathbf{w}_i]$ for some $M > 0$ to be specified later. Since $\mathbb{E}[\varphi_i \mid x_i, \mathbf{w}_i] = 0$, $\varphi_i = \varphi_i^- + \varphi_i^+$. Then, define a function class

$$\mathcal{G} = \left\{ (x_1, \mathbf{w}_1, \varphi_1) \mapsto b_{p,0,l}(x_1; \Delta) b_{p,0,k}(x_1; \Delta) \eta_{1,1}^2 \varphi_1 : 1 \leq l \leq J(p+1), 1 \leq k \leq J(p+1), \Delta \in \Pi_C \right\}.$$

For $g \in \mathcal{G}$, $\sum_{i=1}^n g(x_i, \mathbf{w}_i, \varphi_i) = \sum_{i=1}^n g(x_i, \mathbf{w}_i, \varphi_i^+) + \sum_{i=1}^n g(x_i, \mathbf{w}_i, \varphi_i^-)$.

For the truncated piece, we have $\sup_{g \in \mathcal{G}} |g(x_i, \mathbf{w}_i, \varphi_i^-)| \lesssim JM$, and

$$\begin{aligned} \sup_{g \in \mathcal{G}} \mathbb{V}[g(x_i, \mathbf{w}_i, \varphi_i^-)] &\lesssim \sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \mathbb{E}[(\varphi_i^-)^2 \mid x_i = x, \mathbf{w}_i = \mathbf{w}] \sup_{\Delta \in \Pi_C} \sup_{1 \leq l, k \leq J(p+1)} \mathbb{E}[b_{p,0,l}^2(x_i; \Delta) b_{p,0,k}^2(x_i; \Delta) \eta_{i,1}^4] \\ &\lesssim JM \sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \mathbb{E}[|\varphi_i| \mid x_i = x, \mathbf{w}_i = \mathbf{w}] \lesssim JM. \end{aligned}$$

The VC condition holds by the same argument given in the proof of Lemma [SA-3.1](#). Then, by Lemma [SA-5.6](#),

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} \left| \mathbb{E}_n[g(x_i, \mathbf{w}_i, \varphi_i^-)] \right| \right] \lesssim \sqrt{\frac{JM \log(JM)}{n}} + \frac{JM \log(JM)}{n}.$$

Regarding the tail, we apply Theorem 2.14.1 of [van der Vaart and Wellner \(1996\)](#) and obtain

$$\begin{aligned} \mathbb{E} \left[\sup_{g \in \mathcal{G}} \left| \mathbb{E}_n[g(x_i, \mathbf{w}_i, \varphi_i^+)] \right| \right] &\lesssim \frac{1}{\sqrt{n}} J \mathbb{E} \left[\sqrt{\mathbb{E}_n[|\varphi_i^+|^2]} \right] \\ &\leq \frac{1}{\sqrt{n}} J (\mathbb{E}[\max_{1 \leq i \leq n} |\varphi_i^+|])^{1/2} (\mathbb{E}[\mathbb{E}_n[|\varphi_i^+|]])^{1/2} \\ &\lesssim \frac{J}{\sqrt{n}} \cdot \frac{n^{\frac{1}{\nu}}}{M^{(\nu-2)/4}}, \end{aligned}$$

where the second line follows from Cauchy-Schwarz inequality and the third line uses the fact that

$$\mathbb{E}[\max_{1 \leq i \leq n} |\varphi_i^+|] \lesssim \mathbb{E}[\max_{1 \leq i \leq n} \psi(y_i, \eta_i)^2] \lesssim n^{2/\nu} \quad \text{and} \quad \mathbb{E}[\mathbb{E}_n[|\varphi_i^+|]] \leq \mathbb{E}[|\varphi_1^+|] \lesssim \frac{\mathbb{E}[|\psi(y_1, \eta_1)|^\nu]}{M^{(\nu-2)/2}}.$$

Then the desired result follows simply by setting $M = J^{\frac{2}{\nu-2}}$ and the sparsity of the basis.

Step 4: For $\mathbf{V}_4$, since by Assumption SA-SM, $\sup_{x \in \mathcal{X}, \mathbf{w} \in \mathcal{W}} \mathbb{E}[\psi(y_i, \eta_i)^2 | x_i = x] \lesssim 1$. Then, by the same argument given in the proof of Lemma SA-3.1,

$$\begin{aligned} \sup_{\Delta \in \Pi_C} \left\| \frac{1}{\sqrt{n}} \mathbb{G}_n[\mathbf{b}_{p,s}(x_i; \Delta) \mathbf{b}_{p,s}(x_i; \Delta)' \eta_{i,1}^2 \sigma^2(x_i, \mathbf{w}_i)] \right\| &\lesssim_{\mathbb{P}} \sqrt{J \log J/n} \quad \text{and} \\ \left\| \mathbb{E}_{\hat{\Delta}} \left[\hat{\mathbf{b}}_{p,s}(x_i) \hat{\mathbf{b}}_{p,s}(x_i)' \eta_{i,1}^2 \psi(y_i, \eta_i)^2 \right] - \mathbb{E} \left[\mathbf{b}_{p,s}(x_i) \mathbf{b}_{p,s}(x_i)' \eta_{i,1}^2 \psi(y_i, \eta_i)^2 \right] \right\| &\lesssim_{\mathbb{P}} \sqrt{J \log J/n} + \mathfrak{r}_{\text{RP}}. \end{aligned}$$

The proof for the first conclusion is complete.

Step 5: The results about $\hat{\Omega}_{\mu^{(v)}}(x)$, $\hat{\Omega}_{\vartheta}(x)$ and $\hat{\Omega}_{\zeta}(x)$ follow by Assumption SA-SM, Assumption SA-HLE(ii), Lemmas SA-5.4 and SA-3.1, and Corollary SA-3.1. In particular,

$$\begin{aligned} \hat{\Omega}_{\vartheta}(x) &= \hat{\eta}_{0,1}(x, \hat{\mathbf{w}})^2 \hat{\Omega}_{\mu^{(0)}}(x), & \bar{\Omega}_{\vartheta}(x) &= \eta_{0,1}(x, \mathbf{w})^2 \bar{\Omega}_{\mu^{(0)}}(x), \\ \hat{\Omega}_{\zeta}(x) &= \hat{\eta}_{0,1}(x, \hat{\mathbf{w}})^2 \hat{\Omega}_{\mu^{(1)}}(x), & \bar{\Omega}_{\zeta}(x) &= \eta_{0,1}(x, \mathbf{w})^2 \bar{\Omega}_{\mu^{(1)}}(x). \end{aligned}$$

Moreover, a first-order expansion and Corollary SA-3.1 give

$$\sup_{x \in \mathcal{X}} |\hat{\eta}_{0,1}(x, \hat{\mathbf{w}})^2 - \eta_{0,1}(x, \mathbf{w})^2| \lesssim_{\mathbb{P}} \left(\frac{J \log n}{n} \right)^{1/2} + J^{-p-1} + \sqrt{d_{\gamma}} \mathfrak{r}_{\gamma} + \|\hat{\mathbf{w}} - \mathbf{w}\|.$$

Assumption SA-HLE(i) and the theorem's rate conditions absorb the first three terms into the stated rates, yielding the transformed-parameter conclusions. For the conclusions involving the population variances, Lemmas SA-5.3, SA-5.4 and SA-3.1 show that replacing $\hat{\mathbf{T}}_s$ by $\mathbf{T}_s$ contributes at most $O_{\mathbb{P}}(J^{1+2v} \mathfrak{r}_{\text{RP}})$, $O_{\mathbb{P}}(J \mathfrak{r}_{\text{RP}})$ and $O_{\mathbb{P}}(J^3 \mathfrak{r}_{\text{RP}})$ to the three respective variance differences, which are absorbed by the stated rates. The proof is complete. □

SA-5.9 Proof of Theorem SA-3.3

Proof. We first show that for each fixed $x \in \mathcal{X}$,

$$\bar{\Omega}_{\mu^{(v)}}(x)^{-1/2} \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \mathbb{G}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] =: \mathbb{G}_n[a_i \psi(y_i, \eta_i)]$$

is asymptotically normal. Conditional on $\mathcal{F}_{XW\Delta}$, the σ -field generated by $\{(x_i, \mathbf{w}_i)\}_{i=1}^n$ and $\widehat{\Delta}$, it is an independent mean-zero sequence over i with variance equal to 1. Then by Berry-Esseen inequality,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\mathbb{G}_n[a_i \psi(y_i, \eta_i)] \leq u \mid \mathcal{F}_{XW\Delta}) - \Phi(u) \right| \leq \min \left(1, \frac{\sum_{i=1}^n \mathbb{E}[|a_i \psi(y_i, \eta_i)|^3 \mid \mathcal{F}_{XW\Delta}]}{n^{3/2}} \right).$$

By Lemmas SA-5.4, SA-3.1 and SA-3.2,

$$\begin{aligned} & \frac{1}{n^{3/2}} \sum_{i=1}^n \mathbb{E} \left[|a_i \psi(y_i, \eta_i)|^3 \mid \mathcal{F}_{XW\Delta} \right] \\ & \lesssim \bar{\Omega}_{\mu^{(v)}}(x)^{-3/2} \frac{1}{n^{3/2}} \sum_{i=1}^n \mathbb{E} \left[|\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)|^3 \mid \mathcal{F}_{XW\Delta} \right] \\ & \lesssim \bar{\Omega}_{\mu^{(v)}}(x)^{-3/2} \frac{1}{n^{3/2}} \sum_{i=1}^n |\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}(x_i)|^3 \\ & \leq \bar{\Omega}_{\mu^{(v)}}(x)^{-3/2} \frac{\sup_{x \in \mathcal{X}} \sup_{z \in \mathcal{X}} |\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}(z)|}{n^{3/2}} \sum_{i=1}^n |\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}(x_i)|^2 \\ & \lesssim_{\mathbb{P}} \frac{1}{J^{3/2+3v}} \cdot \frac{J^{1+v}}{\sqrt{n}} \cdot J^{1+2v} \rightarrow 0 \end{aligned}$$

since $J/n = o(1)$. By Theorem SA-3.2, the above weak convergence still holds if $\bar{\Omega}_{\mu^{(v)}}(x)$ is replaced by $\widehat{\Omega}_{\mu^{(v)}}(x)$. Also, Theorem SA-3.2 and Lemma SA-3.2 imply, under the respective evaluation-point conditions in parts (ii) and (iii),

$$\begin{aligned} \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\Omega}_{\vartheta}(x)}{\bar{\Omega}_{\vartheta}(x)} - 1 \right| & \lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-2/\nu}} \right)^{1/2} + \|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(1), \\ \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\Omega}_{\zeta}(x)}{\bar{\Omega}_{\zeta}(x)} - 1 \right| & \lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-2/\nu}} \right)^{1/2} + \|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(1). \end{aligned}$$

Thus, $\widehat{\Omega}_\vartheta(x) \asymp_{\mathbb{P}} J$ and $\widehat{\Omega}_\zeta(x) \asymp_{\mathbb{P}} J^3$. For $\widehat{\vartheta}(x, \widehat{\mathbf{w}})$, the evaluation-point contribution after Studentization is

$$\frac{\|\widehat{\mathbf{w}} - \mathbf{w}\|}{\sqrt{J/n}} = o_{\mathbb{P}}(1).$$

For $\widehat{\zeta}(x, \widehat{\mathbf{w}})$, Theorem SA-3.1(iii) and $\widehat{\Omega}_\zeta(x) \asymp_{\mathbb{P}} J^3$ imply that the corresponding contribution is bounded in probability by a constant times

$$\frac{\|\widehat{\mathbf{w}} - \mathbf{w}\| \{1 + \sqrt{J^3 \log n/n}\}}{\sqrt{J^3/n}} = \|\widehat{\mathbf{w}} - \mathbf{w}\| \left\{ \sqrt{n/J^3} + \sqrt{\log n} \right\} = o_{\mathbb{P}}(1),$$

where the last equality follows from the rate condition in part (iii). The remaining Bahadur remainders are negligible under the maintained rate conditions, and hence the three desired conclusions follow. $\square$

SA-5.10 Proof of Theorem SA-3.4

Proof. We let $\widehat{\beta}_0$ and $\widehat{r}_{0,v}$ be defined as in Lemma SA-5.5. Let $\check{\eta}_i = \eta(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0)$. To retain the leading approximation error explicitly, first note the exact decomposition

$$\begin{aligned} & \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \eta^{(1)} \left(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0 \right) \psi(y_i, \check{\eta}_i) \right] \\ &= \mathbb{E}_n [\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] + \mathbb{E}_n [\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i)] \\ &+ \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \left\{ \eta^{(1)} \left(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0 \right) \psi(y_i, \check{\eta}_i) - \eta_{i,1} \psi(y_i, \eta_i) \right. \right. \\ &\quad \left. \left. - \mathbb{E} \left[\eta^{(1)} \left(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0 \right) \psi(y_i, \check{\eta}_i) - \eta_{i,1} \psi(y_i, \eta_i) \mid \mathcal{F}_{XW\Delta} \right] \right\} \right] \\ &+ \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \left\{ \eta^{(1)} \left(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0 \right) - \eta_{i,1} \right\} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i) \right]. \end{aligned}$$

The last line is uniformly $O_{\mathbb{P}}(J^{-2p-2})$ before projection. The same conditional maximal-inequality argument used in the proof of Lemma SA-3.4 gives

$$\begin{aligned} & \sup_{x \in \mathcal{X}} \left| \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n \left[\widehat{\mathbf{b}}_{p,s}(x_i) \left\{ \eta^{(1)} \left(\widehat{\mathbf{b}}_{p,s}(x_i)' \widehat{\beta}_0 + \mathbf{w}_i' \gamma_0 \right) \psi(y_i, \check{\eta}_i) \right. \right. \right. \\ & \quad \left. \left. \left. - \eta_{i,1} \psi(y_i, \eta_i) - \mathbb{E}[\cdot \mid \mathcal{F}_{XW\Delta}] \right\} \right] \right| \\ & \lesssim_{\mathbb{P}} J^v \left\{ J^{-\frac{p+1}{2}} \sqrt{\frac{J \log J}{n}} + \frac{J \log J}{n} \right\}. \end{aligned}$$

where $\mathbb{E}[\cdot \mid \mathcal{F}_{XW\Delta}]$ denotes the conditional expectation of the two preceding score terms inside braces. By Lemmas SA-5.5 and SA-3.1, Theorem SA-3.1 and the results given in the proof of Lemma SA-3.4, the deterministic J^{-p-1} component is therefore represented by the explicit conditional-mean and approximation-error terms below, and

$$\begin{aligned} \widehat{\mu}^{(v)}(x) - \mu_0^{(v)}(x) &= \widehat{\mathbf{b}}_{p,s}^{(v)}(x)'(\widehat{\beta} - \widehat{\beta}_0) - \widehat{r}_{0,v}(x) \\ &= -\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] - \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i)] \\ &\quad - \widehat{r}_{0,v}(x) \\ &\quad + O_{\mathbb{P}} \left(J^v \left\{ \left(\frac{J \log n}{n} \right)^{3/4} \ell_{\gamma,n} + J^{-\frac{p+1}{2}} \sqrt{\frac{J}{n}} \ell_{\gamma,n} + \frac{J \ell_{\gamma,n}^2}{n} + \sqrt{d_{\gamma}} \mathfrak{r}_{\gamma} \right\} \right), \end{aligned}$$

The $O_{\mathbb{P}}(\cdot)$ in the last line holds uniformly over $x \in \mathcal{X}$ and is $o_{\mathbb{P}}(J^v(\sqrt{J/n} + J^{-p-1}))$ under the maintained rate conditions. Thus, the integral of its square is $o_{\mathbb{P}}(J^{1+2v}/n + J^{-2(p+1-v)})$. Then,

$$\begin{aligned} \text{AISE}_{\mu^{(v)}} &= \int_{\mathcal{X}} \left(\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right. \\ &\quad \left. + \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i)] + \widehat{r}_{0,v}(x) \right)^2 \omega(x) dx. \end{aligned}$$

Next, taking conditional expectation given $\mathbf{X}$, $\mathbf{W}$ and $\widehat{\Delta}$ and using the argument in the proof of Lemma SA-3.1 again, we have

$$\begin{aligned} \mathbb{E}[\text{AISE}_{\mu^{(v)}} | \mathbf{X}, \mathbf{W}, \widehat{\Delta}] &= \frac{1}{n} \text{trace} \left(\mathbf{Q}_0^{-1} \Sigma_0 \mathbf{Q}_0^{-1} \int_{\mathcal{X}} \mathbf{b}_{p,s}^{(v)}(x) \mathbf{b}_{p,s}^{(v)}(x)' \omega(x) dx \right) + o_{\mathbb{P}}(J^{2v+1}/n) \\ &\quad + \int_{\mathcal{X}} \left(\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \widehat{\beta}_0 - \mu_0^{(v)}(x) \right)^2 \omega(x) dx \\ &\quad + \int_{\mathcal{X}} \left(\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i)] \right)^2 \omega(x) dx \\ &\quad + 2 \int_{\mathcal{X}} \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \Psi(x_i, \mathbf{w}_i; \check{\eta}_i)] \widehat{r}_{0,v}(x) \omega(x) dx. \end{aligned}$$

By Assumption SA-SM, $\Psi(x_i, \mathbf{w}_i; \check{\eta}_i) = -\Psi_1(x_i, \mathbf{w}_i; \eta_i) \eta_{i,1} \widehat{r}_0(x_i) + O_{\mathbb{P}}(J^{-2p-2})$ where $O_{\mathbb{P}}(\cdot)$ holds uniformly over i . The terms in the last three lines correspond to the integrated squared bias. Also, using the same argument in the proof of Lemma SA-3.1, $\mathbb{E}_n[\cdot]$ in the last two lines can be safely replaced by $\mathbb{E}_{\widehat{\Delta}}[\cdot]$, which only introduces some additional approximation error of order $o_{\mathbb{P}}(J^{-2p-2+2v})$.

For any $\Delta \in \Pi_C$, let $\tau_x^L(\Delta)$ and $h_x(\Delta)$ denote, respectively, the left endpoint and width of the interval containing x , and define

$$q_{0,v}(x; \Delta) := \frac{\mu_0^{(p+1)}(x)}{(p+1-v)!} h_x(\Delta)^{p+1-v} \mathcal{E}_{p+1-v} \left(\frac{x - \tau_x^L(\Delta)}{h_x(\Delta)} \right).$$

For $s = 0$, apply Taylor's theorem on each bin around its left endpoint and decompose the order- $(p+1)$ term using $z^{p+1} = \mathcal{E}_{p+1}(z) + \{z^{p+1} - \mathcal{E}_{p+1}(z)\}$. The term in braces is a polynomial of degree p and can therefore be absorbed separately in each bin into the piecewise-polynomial approximation. Differentiating v times and using the uniform bound on $\mu_0^{(p+2)}$ gives, for $\Delta = \hat{\Delta}$ and $\Delta = \Delta_0$,

$$r_{0,v}(x; \Delta) = q_{0,v}(x; \Delta) + \mathbf{b}_{p,0}^{(v)}(x; \Delta)' \mathbf{a}(\Delta) + o_{\mathbb{P}}(J^{-p-1+v}).$$

For $s > 0$, first note that the additional condition in the theorem and Assumption [SA-RP\(ii\)](#) imply

$$\max_{2 \leq j \leq J} |\hat{h}_j - \hat{h}_{j-1}| \leq \max_{2 \leq j \leq J} |h_j - h_{j-1}| + 2 \max_{1 \leq j \leq J} |\hat{h}_j - h_j| = o_{\mathbb{P}}(J^{-1}).$$

Under this condition, the proof of Lemma SA-6.1 in [Cattaneo et al. \(2020\)](#) shows that there exists a maximally smooth degree- p spline whose v th-order approximation error is $q_{0,v}(x; \Delta) + o_{\mathbb{P}}(J^{-p-1+v})$ uniformly over $x \in \mathcal{X}$, for $\Delta = \hat{\Delta}$ and $\Delta = \Delta_0$. Indeed, the uniform bound on $\mu_0^{(p+2)}$ makes the Taylor remainder $O(J^{-p-2+v}) = o(J^{-p-1+v})$, while the contribution of unequal widths is bounded by $J^{-p+v} \max_{2 \leq j \leq J} |h_j - h_{j-1}| = o(J^{-p-1+v})$ (and analogously in probability for $\hat{\Delta}$). Since this approximant also belongs to the spline space for $0 < s < p$, the argument following Equation (SA-5.6) in the proof of Theorem SA-3.4 in [Cattaneo et al. \(2024a\)](#), applied to the least-squares projection defining $\beta_0(\Delta)$, yields

$$r_{0,v}(x; \Delta) = q_{0,v}(x; \Delta) + \mathbf{b}_{p,s}^{(v)}(x; \Delta)' \mathbf{a}(\Delta) + o_{\mathbb{P}}(J^{-p-1+v})$$

uniformly over $x \in \mathcal{X}$, where the approximation remainders for both $s = 0$ and $s > 0$ are nonrandom for $\Delta = \Delta_0$. The spline components in the two approximation displays cancel after weighted projection. Indeed, if

$$\mathbf{Q}(\Delta) := \mathbb{E}_{\Delta} [\mathbf{b}_{p,s}(x_i; \Delta) \mathbf{b}_{p,s}(x_i; \Delta)' \kappa(x_i, \mathbf{w}_i)],$$

where $\mathbb{E}_\Delta[\cdot]$ treats Δ as fixed. Then

$$\begin{aligned} & r_{0,v}(x; \Delta) - \mathbf{b}_{p,s}^{(v)}(x; \Delta)' \mathbf{Q}(\Delta)^{-1} \mathbb{E}_\Delta[\mathbf{b}_{p,s}(x_i; \Delta) \varkappa(x_i, \mathbf{w}_i) r_{0,0}(x_i; \Delta)] \\ &= q_{0,v}(x; \Delta) - \mathbf{b}_{p,s}^{(v)}(x; \Delta)' \mathbf{Q}(\Delta)^{-1} \mathbb{E}_\Delta[\mathbf{b}_{p,s}(x_i; \Delta) \varkappa(x_i, \mathbf{w}_i) q_{0,0}(x_i; \Delta)] + o_{\mathbb{P}}(J^{-p-1+v}). \end{aligned}$$

Combining this cancellation with the preceding Taylor expansion, the conditional Gram-matrix argument, and Assumption [SA-RP\(ii\)](#), the integrated squared bias is

$$\int_{\mathcal{X}} \left(q_{0,v}(x; \Delta_0) - \mathbf{b}_{p,s}^{(v)}(x)' \mathbf{Q}_0^{-1} \mathbb{E}[\mathbf{b}_{p,s}(x_i) \varkappa(x_i, \mathbf{w}_i) q_{0,0}(x_i; \Delta_0)] \right)^2 \omega(x) dx + o_{\mathbb{P}}(J^{-2p-2+2v}).$$

Since $q_{0,v}(x; \Delta_0) = r_{0,v}^*(x)$ by definition, the expression of $\mathcal{B}_n(p, s, v)$ given in the theorem follows.

Finally, consider the contribution of estimating the evaluation point. For part (ii), the stated condition gives $\|\widehat{\mathbf{w}} - \mathbf{w}\| = o_{\mathbb{P}}(\sqrt{J/n} + J^{-p-1})$, which makes its squared contribution and its cross-products negligible relative to $J/n + J^{-2p-2}$. For part (iii), let

$$L_{\zeta,n} := \sqrt{J^3/n + J^{-2p}} \quad \text{and} \quad C_{\zeta,n} := 1 + \sqrt{J^3 \log n/n}.$$

Theorem [SA-3.1\(iii\)](#) bounds the evaluation-point contribution uniformly over x by

$$O_{\mathbb{P}}(\|\widehat{\mathbf{w}} - \mathbf{w}\| C_{\zeta,n}) = o_{\mathbb{P}}(L_{\zeta,n}).$$

Its integrated square is therefore $o_{\mathbb{P}}(J^3/n + J^{-2p})$, and its cross-products with the leading stochastic and bias terms have the same order by Cauchy–Schwarz. The remaining terms are controlled as in part (i), proving parts (ii) and (iii). $\square$

SA-5.11 Proof of Theorem [SA-3.5](#)

Proof. The proof is divided into several steps.

Step 1: Note that

$$\sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)/n}} - \frac{\widehat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)/n}} \right|$$

$$\begin{aligned}
&\leq \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)/n}} \right| \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\Omega}_{\mu^{(v)}}(x)^{1/2} - \bar{\Omega}_{\mu^{(v)}}(x)^{1/2}}{\widehat{\Omega}_{\mu^{(v)}}(x)^{1/2}} \right| \\
&\lesssim \mathbb{P} \left(\sqrt{\log n} + \sqrt{n} J^{-p-1-1/2} \right) \left(J^{-p-1} + \sqrt{\frac{J \log n}{n^{1-\frac{2}{\nu}}}} \right)
\end{aligned}$$

where the last step uses Lemma SA-3.2 and Corollary SA-3.1. Then, in view of Lemmas SA-5.5, SA-3.4, Theorems SA-3.1, SA-3.2 and the rate restriction given in the lemma, we have

$$\sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\widehat{\Omega}_{\mu^{(v)}}(x)/n}} + \frac{\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1}}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \mathbb{G}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \eta_{i,1} \psi(y_i, \eta_i)] \right| = o_{\mathbb{P}}(a_n^{-1}).$$

Step 2: Let us write $\mathcal{K}(x, x_i) = \bar{\Omega}_{\mu^{(v)}}(x)^{-1/2} \widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}(x_i)$ (the dependence of $\widehat{\mathbf{b}}_{p,s}^{(v)}(x)$, $\bar{\mathbf{Q}}$ and $\bar{\Omega}_{\mu^{(v)}}(x)$ on $\mathbf{X}$, $\mathbf{W}$ and $\widehat{\Delta}$ is omitted for simplicity), and $\tilde{\sigma}^2(x_i, \mathbf{w}_i) = \mathbb{E}[\psi^\dagger(\epsilon_i)^2 | x_i, \mathbf{w}_i]$. Now we rearrange $\{x_i\}_{i=1}^n$ as a sequence of order statistics $\{x_{(i)}\}_{i=1}^n$, i.e., $x_{(1)} \leq \dots \leq x_{(n)}$. Accordingly, $\{\epsilon_i\}_{i=1}^n$, $\{\mathbf{w}_i\}_{i=1}^n$ and $\{\tilde{\sigma}^2(x_i, \mathbf{w}_i)\}_{i=1}^n$ are ordered as concomitants $\{\epsilon_{[i]}\}_{i=1}^n$, $\{\mathbf{w}_{[i]}\}_{i=1}^n$ and $\{\tilde{\sigma}_{[i]}^2\}_{i=1}^n$ where $\tilde{\sigma}_{[i]}^2 = \tilde{\sigma}^2(x_{(i)}, \mathbf{w}_{[i]})$. Clearly, conditional on $\mathcal{F}_{XW\Delta}$ (the σ -field generated by $\{(x_i, \mathbf{w}_i)\}$ and $\widehat{\Delta}$), $\{\psi^\dagger(\epsilon_{[i]})\}_{i=1}^n$ is still an independent mean-zero sequence. Then by Assumptions SA-DGP, SA-SM and the result of Sakhanenko (1991), there exists a sequence of i.i.d. standard normal random variables $\{\zeta_{[i]}\}_{i=1}^n$ such that

$$\begin{aligned}
\max_{1 \leq \ell \leq n} |S_\ell| &:= \max_{1 \leq \ell \leq n} \left| \sum_{i=1}^{\ell} \eta^{(1)}(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0) \psi^\dagger(\eta(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0)) \psi^\dagger(\epsilon_{[i]}) \right. \\
&\quad \left. - \sum_{i=1}^{\ell} \eta^{(1)}(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0) \psi^\dagger(\eta(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0)) \tilde{\sigma}_{[i]} \zeta_{[i]} \right| \lesssim_{\mathbb{P}} n^{\frac{1}{\nu}}.
\end{aligned}$$

Then, using summation by parts,

$$\begin{aligned}
&\sup_{x \in \mathcal{X}} \left| \sum_{i=1}^n \mathcal{K}(x, x_{(i)}) \eta^{(1)}(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0) \psi^\dagger(\eta(\mu_0(x_{(i)}) + \mathbf{w}'_{[i]} \gamma_0)) (\psi^\dagger(\epsilon_{[i]}) - \tilde{\sigma}_{[i]} \zeta_{[i]}) \right| \\
&= \sup_{x \in \mathcal{X}} \left| \mathcal{K}(x, x_{(n)}) S_n - \sum_{i=1}^{n-1} S_i (\mathcal{K}(x, x_{(i+1)}) - \mathcal{K}(x, x_{(i)})) \right| \\
&\leq \sup_{x \in \mathcal{X}} \max_{1 \leq i \leq n} |\mathcal{K}(x, x_i)| |S_n| + \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1}}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \sum_{i=1}^{n-1} S_i (\widehat{\mathbf{b}}_{p,s}(x_{(i+1)}) - \widehat{\mathbf{b}}_{p,s}(x_{(i)})) \right|
\end{aligned}$$

$$\leq \sup_{x \in \mathcal{X}} \max_{1 \leq i \leq n} |\mathcal{K}(x, x_i)| |S_n| + \sup_{x \in \mathcal{X}} \left\| \frac{\bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}^{(v)}(x)}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \right\|_1 \left\| \sum_{i=1}^{n-1} S_i \left(\widehat{\mathbf{b}}_{p,s}(x_{(i+1)}) - \widehat{\mathbf{b}}_{p,s}(x_{(i)}) \right) \right\|_{\infty}.$$

By Lemmas SA-5.4, SA-3.1 and SA-3.2, $\sup_{x \in \mathcal{X}} \sup_{x_i \in \mathcal{X}} |\mathcal{K}(x, x_i)| \lesssim_{\mathbb{P}} \sqrt{J}$, and

$$\sup_{x \in \mathcal{X}} \left\| \frac{\bar{\mathbf{Q}}^{-1} \widehat{\mathbf{b}}_{p,s}^{(v)}(x)}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \right\|_1 \lesssim_{\mathbb{P}} 1.$$

Then, notice that

$$\max_{1 \leq l \leq K_{p,s}} \left| \sum_{i=1}^{n-1} \left(\widehat{b}_{p,s,l}(x_{(i+1)}) - \widehat{b}_{p,s,l}(x_{(i)}) \right) S_i \right| \leq \max_{1 \leq l \leq K_{p,s}} \sum_{i=1}^{n-1} \left| \widehat{b}_{p,s,l}(x_{(i+1)}) - \widehat{b}_{p,s,l}(x_{(i)}) \right| \max_{1 \leq \ell \leq n} |S_{\ell}|.$$

By construction of the ordering, $\max_{1 \leq l \leq K_{p,s}} \sum_{i=1}^{n-1} \left| \widehat{b}_{p,s,l}(x_{(i+1)}) - \widehat{b}_{p,s,l}(x_{(i)}) \right| \lesssim \sqrt{J}$. Under the rate restriction in the theorem, this suffices to show that for any $\xi > 0$,

$$\mathbb{P} \left(\sup_{x \in \mathcal{X}} \left| \mathbb{G}_n[\mathcal{K}(x, x_i) \eta^{(1)}(\mu_0(x_i) + \mathbf{w}_i' \boldsymbol{\gamma}_0)(\psi(y_i, \eta_i) - \sigma(x_i, \mathbf{w}_i) \zeta_i)] \right| > \xi a_n^{-1} \middle| \mathcal{F}_{XW\Delta} \right) = o_{\mathbb{P}}(1),$$

where we recover the original ordering. Here, $=_{d|\mathcal{F}_{XW\Delta}}$ denotes equality in distribution conditional on $\mathcal{F}_{XW\Delta}$. Since $\mathbb{G}_n[\widehat{\mathbf{b}}_{p,s}(x_i) \zeta_i \sigma(x_i, \mathbf{w}_i) \eta_{i,1}] =_{d|\mathcal{F}_{XW\Delta}} \mathbf{N}(0, \bar{\boldsymbol{\Sigma}})$, the above steps construct the following approximating process:

$$\bar{Z}_{\mu^{(v)},p}(x) := \frac{\widehat{\mathbf{b}}_{p,s}^{(v)}(x)' \bar{\mathbf{Q}}^{-1}}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} \bar{\boldsymbol{\Sigma}}^{1/2} \mathbf{N}_{K_{p,s}}.$$

Step 3: Suppose that Assumption [SA-RP\(ii\)](#) also holds. Note that

$$\begin{aligned}
& \sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x) - Z_{\mu^{(v)},p}(x)| \\
& \leq \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mathbf{b}}^{(v)}(x)'(\bar{\mathbf{Q}}^{-1} - \mathbf{Q}_0^{-1})\bar{\Sigma}^{1/2}\mathbf{N}_{K_{p,s}}}{\sqrt{\Omega_{\mu^{(v)}}(x)}} \right| \\
& \quad + \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mathbf{b}}^{(v)}(x)'\mathbf{Q}_0^{-1}}{\sqrt{\Omega_{\mu^{(v)}}(x)}} (\bar{\Sigma}^{1/2} - \Sigma_0^{1/2})\mathbf{N}_{K_{p,s}} \right| \\
& \quad + \sup_{x \in \mathcal{X}} \left| \frac{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)'(\widehat{\mathbf{T}}_s - \mathbf{T}_s)'\mathbf{Q}_0^{-1}}{\sqrt{\Omega_{\mu^{(v)}}(x)}} \Sigma_0^{1/2}\mathbf{N}_{K_{p,s}} \right| \\
& \quad + \sup_{x \in \mathcal{X}} \left| \left(\frac{1}{\sqrt{\bar{\Omega}_{\mu^{(v)}}(x)}} - \frac{1}{\sqrt{\Omega_{\mu^{(v)}}(x)}} \right) \widehat{\mathbf{b}}_{p,0}^{(v)}(x)'\widehat{\mathbf{T}}_s\bar{\mathbf{Q}}^{-1}\bar{\Sigma}^{1/2}\mathbf{N}_{K_{p,s}} \right| \\
& = I + II + III + IV,
\end{aligned}$$

where each term on the right-hand side is a mean-zero Gaussian process conditional on $\mathcal{F}_{XW\Delta}$. By Theorem [SA-3.2](#) (see Steps 4 and 5 of its proof), $\sup_{x \in \mathcal{X}} |\bar{\Omega}_{\mu^{(v)}}(x) - \Omega_{\mu^{(v)}}(x)| \lesssim_{\mathbb{P}} J^{1+2v}(\sqrt{J \log n/n} + \mathfrak{r}_{\text{RP}})$. By a similar calculation given in Step 1 and the rate condition imposed, the last term is $o_{\mathbb{P}}(a_n^{-1})$. By Lemmas [SA-5.3](#) and [SA-3.1](#), and the argument in the proof of Lemma [SA-5.4](#) together with Theorem X.3.8 of [Bhatia \(2013\)](#),

$$\begin{aligned}
\|\bar{\mathbf{Q}}^{-1} - \mathbf{Q}_0^{-1}\| & \lesssim_{\mathbb{P}} \sqrt{J \log J/n} + \mathfrak{r}_{\text{RP}}, & \|\widehat{\mathbf{T}}_s - \mathbf{T}_s\| & \lesssim_{\mathbb{P}} \mathfrak{r}_{\text{RP}}, \\
\|\bar{\Sigma}^{1/2} - \Sigma_0^{1/2}\| & \lesssim_{\mathbb{P}} \sqrt{J \log J/n} + \mathfrak{r}_{\text{RP}}.
\end{aligned}$$

By Gaussian Maximal Inequality ([van der Vaart and Wellner, 1996](#), Corollary 2.2.8),

$$\begin{aligned}
\mathbb{E} \left[I + II + III \middle| \mathcal{F}_{XW\Delta} \right] & \lesssim_{\mathbb{P}} \sqrt{\log J} \left(\|\bar{\Sigma}^{1/2} - \Sigma_0^{1/2}\| + \|\bar{\mathbf{Q}}^{-1} - \mathbf{Q}_0^{-1}\| + \|\widehat{\mathbf{T}}_s - \mathbf{T}_s\| \right) \\
& \lesssim_{\mathbb{P}} \sqrt{\log J} \left\{ \sqrt{\frac{J \log J}{n}} + \mathfrak{r}_{\text{RP}} \right\} = o_{\mathbb{P}}(a_n^{-1}).
\end{aligned}$$

where the last line follows from the imposed rate restriction. Then the proof for part (i) is complete.

For part (ii), the evaluation-point remainder after Studentization is

$$\frac{\|\widehat{\mathbf{w}} - \mathbf{w}\|}{\sqrt{J/n}} = o_{\mathbb{P}}(a_n^{-1}).$$

For part (iii), Theorem SA-3.1(iii) and $\widehat{\Omega}_\zeta(x) \asymp_{\mathbb{P}} J^3$ give

$$\sup_{x \in \mathcal{X}} \frac{\|\widehat{\mathbf{w}} - \mathbf{w}\| \{1 + \sqrt{J^3 \log n/n}\}}{\sqrt{\widehat{\Omega}_\zeta(x)/n}} \lesssim_{\mathbb{P}} \|\widehat{\mathbf{w}} - \mathbf{w}\| \left\{ \sqrt{n/J^3} + \sqrt{\log n} \right\} = o_{\mathbb{P}}(a_n^{-1}).$$

The definition of $\widehat{\Omega}_\zeta(x)$, a first-order expansion of the transformation, and the proof of Theorem SA-3.2 give

$$\sup_{x \in \mathcal{X}} \left| \frac{\widehat{\Omega}_\zeta(x)}{\bar{\Omega}_\zeta(x)} - 1 \right| \lesssim_{\mathbb{P}} J^{-p-1} + \left(\frac{J \log n}{n^{1-2/\nu}} \right)^{1/2} + \sqrt{d_\gamma} \mathfrak{r}_\gamma + \|\widehat{\mathbf{w}} - \mathbf{w}\|.$$

The right-hand side multiplied by $\sqrt{\log n}$ is $o_{\mathbb{P}}(a_n^{-1})$ under the rate conditions in the theorem and the evaluation-point condition in part (iii). Since the supremum of the leading Studentized process is $O_{\mathbb{P}}(\sqrt{\log n})$, replacing $\bar{\Omega}_\zeta(x)$ by $\widehat{\Omega}_\zeta(x)$ therefore contributes $o_{\mathbb{P}}(a_n^{-1})$ uniformly in x . All other remainder terms are controlled by the common rate conditions. The leading variance term for $\widehat{\vartheta}(x, \widehat{\mathbf{w}})$ or $\widehat{\zeta}(x, \widehat{\mathbf{w}})$ differs from that for $\widehat{\mu}(x)$ or $\widehat{\mu}^{(1)}(x)$ only by the stated transformation, so the preceding coupling argument completes parts (ii) and (iii). $\square$

SA-5.12 Proof of Theorem SA-3.6

Proof. This conclusion follows from Lemmas SA-5.4, SA-3.1, Theorem SA-3.2 and Gaussian Maximal Inequality as applied in Step 3 in the proof of Theorem SA-3.5. $\square$

SA-5.13 Proof of Theorem SA-3.7

Proof. We first show that

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| \leq u \right) - \mathbb{P} \left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u \right) \right| = o(1).$$

By Theorem SA-3.5, there exists a sequence of constants ξ_n such that $\xi_n = o(1)$ and

$$\mathbb{P} \left(\left| \sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| - \sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \right| > \xi_n/a_n \right) = o(1).$$

Then,

$$\mathbb{P} \left(\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| \leq u \right) \leq \mathbb{P} \left(\left\{ \sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| \leq u \right\} \cap \left\{ \left| \sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| - \sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \right| \leq \xi_n/a_n \right\} \right) + o(1)$$

$$\begin{aligned}
&\leq \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u + \xi_n/a_n\right) + o(1) \\
&\leq \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u\right) + \sup_{u \in \mathbb{R}} \mathbb{E}\left[\mathbb{P}\left(\left|\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| - u\right| \leq \xi_n/a_n \middle| \widehat{\Delta}\right)\right] \\
&\leq \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u\right) + \mathbb{E}\left[\sup_{u \in \mathbb{R}} \mathbb{P}\left(\left|\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| - u\right| \leq \xi_n/a_n \middle| \widehat{\Delta}\right)\right] + o(1).
\end{aligned}$$

Apply the Anti-Concentration Inequality conditional on $\widehat{\Delta}$ (Chernozhukov et al., 2014) to the second term:

$$\begin{aligned}
\sup_{u \in \mathbb{R}} \mathbb{P}\left(\left|\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| - u\right| \leq \xi_n/a_n \middle| \widehat{\Delta}\right) &\leq 4\xi_n a_n^{-1} \mathbb{E}\left[\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \middle| \widehat{\Delta}\right] + o(1) \\
&\lesssim_{\mathbb{P}} \xi_n a_n^{-1} \sqrt{\log J} + o(1) \rightarrow 0
\end{aligned}$$

where the last step uses Gaussian Maximal Inequality (see van der Vaart and Wellner, 1996, Corollary 2.2.8). By Dominated Convergence Theorem,

$$\mathbb{E}\left[\sup_{u \in \mathbb{R}} \mathbb{P}\left(\left|\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| - u\right| \leq \xi_n/a_n \middle| \widehat{\Delta}\right)\right] = o(1).$$

The other side of the inequality follows similarly.

By similar argument, using Theorem SA-3.6, we have

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x \in \mathcal{X}} |\widehat{Z}_{\mu^{(v)},p}(x)| \leq u \middle| \mathbf{D}, \widehat{\Delta}\right) - \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u \middle| \widehat{\Delta}\right) \right| = o_{\mathbb{P}}(1).$$

Then, it remains to show that

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u\right) - \mathbb{P}\left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u \middle| \widehat{\Delta}\right) \right| = o_{\mathbb{P}}(1). \quad (\text{SA-5.4})$$

We can write

$$Z_{\mu^{(v)},p}(x) = \frac{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)'}{\sqrt{\widehat{\mathbf{b}}_{p,0}^{(v)}(x)' \mathbf{V}_0 \widehat{\mathbf{b}}_{p,0}^{(v)}(x)}} \check{\mathbf{N}}_{K_{p,0}}$$

where $\mathbf{V}_0 = \mathbf{T}_s' \mathbf{Q}_0^{-1} \Sigma_0 \mathbf{Q}_0^{-1} \mathbf{T}_s$ and $\check{\mathbf{N}}_{K_{p,0}} := \mathbf{T}_s' \mathbf{Q}_0^{-1} \Sigma_0^{1/2} \mathbf{N}_{K_{p,s}}$ is a $K_{p,0}$ -dimensional Gaussian random vector. Importantly, by this construction, $\check{\mathbf{N}}_{K_{p,0}}$ and $\mathbf{V}_0$ do not depend on $\widehat{\Delta}$ and x , and they are only determined by the deterministic partition Δ_0 .

First consider $v = 0$. For any two partitions $\Delta_1, \Delta_2 \in \Pi_C$, for any $x \in \mathcal{X}$, there exists $\check{x} \in \mathcal{X}$ such that

$$\mathbf{b}_{p,0}^{(0)}(x; \Delta_1) = \mathbf{b}_{p,0}^{(0)}(\check{x}; \Delta_2),$$

and vice versa. Therefore, the following two events are equivalent: $\{\omega : \sup_{x \in \mathcal{X}} |Z_p(x; \Delta_1)| \leq u\} = \{\omega : \sup_{x \in \mathcal{X}} |Z_p(x; \Delta_2)| \leq u\}$ for any u . Thus,

$$\mathbb{E} \left[\mathbb{P} \left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u \middle| \widehat{\Delta} \right) \right] = \mathbb{P} \left(\sup_{x \in \mathcal{X}} |Z_{\mu^{(v)},p}(x)| \leq u \middle| \widehat{\Delta} \right) + o_{\mathbb{P}}(1).$$

Then for $v = 0$, the desired result follows.

For $v > 0$, simply notice that $\widehat{\mathbf{b}}_{p,0}^{(v)}(x) = \widehat{\mathfrak{T}}_v \widehat{\mathbf{b}}_{p,0}(x)$ for some transformation matrix $\widehat{\mathfrak{T}}_v$. Clearly, $\widehat{\mathfrak{T}}_v$ takes a similar structure as $\widehat{\mathbf{T}}_s$: each row and each column only have a finite number of nonzeros. Each nonzero element is simply $\hat{h}_j^{-v}$ up to some constants. Under Assumption [SA-RP\(ii\)](#), let $\mathfrak{T}_v$ be the fixed analogue of $\widehat{\mathfrak{T}}_v$ obtained by replacing $\hat{h}_j$ with h_j . Since $\hat{h}_j \asymp_{\mathbb{P}} h_j \asymp J^{-1}$, the mean value theorem and the finite number of nonzero elements in each row and column give

$$\|\widehat{\mathfrak{T}}_v - \mathfrak{T}_v\| \lesssim_{\mathbb{P}} J^v \mathbf{r}_{\text{RP}}, \quad \sup_{x \in \mathcal{X}} \|(\widehat{\mathfrak{T}}_v - \mathfrak{T}_v) \widehat{\mathbf{b}}_{p,0}(x)\| \lesssim_{\mathbb{P}} J^{v+1/2} \mathbf{r}_{\text{RP}}.$$

Write $Z_{\mu^{(v)},p}(x; \mathfrak{T})$ for the displayed representation of $Z_{\mu^{(v)},p}(x)$ above with $\widehat{\mathbf{b}}_{p,0}^{(v)}(x)$ replaced by $\mathfrak{T} \widehat{\mathbf{b}}_{p,0}(x)$ in both the numerator and denominator, so $Z_{\mu^{(v)},p}(x) = Z_{\mu^{(v)},p}(x; \widehat{\mathfrak{T}}_v)$. Lemma [SA-3.2](#), a first-order expansion of the denominator, and Gaussian Maximal Inequality yield

$$\mathbb{E} \left[\sup_{x \in \mathcal{X}} \left| Z_{\mu^{(v)},p}(x; \widehat{\mathfrak{T}}_v) - Z_{\mu^{(v)},p}(x; \mathfrak{T}_v) \right| \middle| \widehat{\Delta} \right] \lesssim_{\mathbb{P}} \mathbf{r}_{\text{RP}} \sqrt{\log J} = o_{\mathbb{P}}((\log J)^{-1/2}),$$

where the last equality follows from the rate restriction in the theorem. Thus, $\widehat{\mathfrak{T}}_v$ can be replaced by $\mathfrak{T}_v$ without affecting the approximation rate, and the desired result follows by repeating the argument for $v = 0$.

Finally, the result for $T_{\vartheta,p}(x)$ ($T_{\zeta,p}(x)$) follows by the fact that $Z_{\vartheta,p}(x)$ and $\widehat{Z}_{\vartheta,p}(x)$ ($Z_{\zeta,p}(x)$ and $\widehat{Z}_{\zeta,p}(x)$) differ from $Z_{\mu^{(v)},p}(x)$ and $\widehat{Z}_{\mu^{(v)},p}(x)$ up to a sign only. $\square$

SA-5.14 Proof of Theorem SA-3.8

Proof. We only consider $\hat{T}_{\mu^{(v)},p}(x)$. The results in part (ii) and part (iii) follow similarly.

Let $\xi_{1,n} = o(1)$, $\xi_{2,n} = o(1)$ and $\xi_{3,n} = o(1)$. Then,

$$\begin{aligned} \mathbb{P} \left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| \leq \mathbf{c}_{\mu^{(v)}} \right] &\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| \leq \mathbf{c}_{\mu^{(v)}} + \xi_{1,n}/a_n \right] + o(1) \\ &\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| \leq c^0(1 - \alpha + \xi_{3,n}) + (\xi_{1,n} + \xi_{2,n})/a_n \right] + o(1) \\ &\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| \leq c^0(1 - \alpha + \xi_{3,n}) \right] + o(1) \rightarrow 1 - \alpha, \end{aligned}$$

where $c^0(1 - \alpha + \xi_{3,n})$ denotes the $(1 - \alpha + \xi_{3,n})$ -quantile of $\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)|$ conditional on $\mathcal{F}_{XW\Delta}$ (the σ -field generated by $\mathbf{X}$, $\mathbf{W}$ and the partition $\hat{\Delta}$), the first inequality holds by Theorem SA-3.5, the second by Lemma A.1 of Belloni et al. (2015), and the third by Anti-Concentration Inequality in Chernozhukov et al. (2014). The other side of the bound follows similarly. $\square$

SA-5.15 Proof of Theorem SA-3.9

Proof. We only consider the proof for part (i). The results in part (ii) and part (iii) follow similarly. For those parts, the displayed function-level convergence conditions absorb the triangular-array dependence of $\bar{\gamma}$, while the imposed lim inf conditions ensure that the fixed-alternative signal used in the power argument remains bounded away from zero uniformly in n .

Throughout this proof, we let $\xi_{1,n} = o(1)$, $\xi_{2,n} = o(1)$ and $\xi_{3,n} = o(1)$ be sequences of vanishing constants. Moreover, let A_n be a sequence of diverging constants such that $\sqrt{\log J} A_n \lesssim \sqrt{\frac{n}{J^{1+2v}}}$. Note that

$$\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)},p}(x)| \leq \sup_{x \in \mathcal{X}} \left| \frac{\hat{\mu}^{(v)}(x) - \mu_0^{(v)}(x)}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| + \sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right|.$$

Therefore, under $\dot{H}_0^{\mu^{(v)}}$,

$$\begin{aligned} \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)},p}(x)| > \mathbf{c}_{\mu^{(v)}} \right] &\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| > \mathbf{c}_{\mu^{(v)}} - \sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| \right] \\ &\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| > \mathbf{c}_{\mu^{(v)}} - \xi_{1,n}/a_n - \sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| \right] + o(1) \end{aligned}$$

$$\begin{aligned}
&\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| > c^0(1 - \alpha - \xi_{3,n}) - (\xi_{1,n} + \xi_{2,n})/a_n - \right. \\
&\quad \left. \sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| \right] + o(1) \\
&\leq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| > c^0(1 - \alpha - \xi_{3,n}) \right] + o(1) \\
&= \alpha + o(1)
\end{aligned}$$

where $c^0(1 - \alpha - \xi_{3,n})$ denotes the $(1 - \alpha - \xi_{3,n})$ -quantile of $\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)|$ conditional on $\mathcal{F}_{XW\Delta}$ (the σ -field generated by $\mathbf{X}$, $\mathbf{W}$ and $\hat{\Delta}$), the second inequality holds by Theorem SA-3.5, the third by Lemma A.1 of Belloni et al. (2015), the fourth by the fact that $\sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| = o_{\mathbb{P}}(\frac{1}{\sqrt{\log J}})$ and Anti-Concentration Inequality in Chernozhukov et al. (2014). The other side of the bound follows similarly.

On the other hand, under $\dot{\mathbf{H}}_A^{\mu^{(v)}}$,

$$\begin{aligned}
&\mathbb{P} \left[\sup_{x \in \mathcal{X}} |\dot{T}_{\mu^{(v)},p}(x)| > \mathbf{c}_{\mu^{(v)}} \right] \\
&= \mathbb{P} \left[\sup_{x \in \mathcal{X}} \left| T_{\mu^{(v)},p}(x) + \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} + \frac{m^{(v)}(x; \bar{\boldsymbol{\theta}}) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| > \mathbf{c}_{\mu^{(v)}} \right] \\
&\geq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| < \sup_{x \in \mathcal{X}} \left| \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \bar{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} + \frac{m^{(v)}(x; \bar{\boldsymbol{\theta}}) - m^{(v)}(x; \tilde{\boldsymbol{\theta}})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} \right| - \mathbf{c}_{\mu^{(v)}} \right] \\
&\geq \mathbb{P} \left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| \leq \sqrt{\log J} A_n - \xi_{1,n}/a_n \right] - o(1) \\
&\geq 1 - o(1).
\end{aligned}$$

where the fourth line holds by Lemma SA-3.2, Theorem SA-3.2, Theorem SA-3.5, the fixed-separation condition, the condition that $J^v \sqrt{J \log J/n} = o(1)$ and the definition of A_n , and the last by the Talagrand-Samorodnitsky Concentration Inequality (van der Vaart and Wellner, 1996, Proposition A.2.7). $\square$

SA-5.16 Proof of Theorem SA-3.10

Proof. We only consider the proof for part (i). The results in part (ii) and part (iii) follow similarly. For those parts, the displayed function-level convergence conditions absorb the triangular-array

dependence of $\bar{\gamma}$, while the imposed $\liminf$ conditions ensure that the one-sided signal used in the power argument remains bounded away from zero uniformly in n .

Throughout this proof, the definitions of A_n , $\xi_{1,n}$, $\xi_{2,n}$ and $\xi_{3,n}$ are the same as in the proof of Theorem SA-3.9. Under $\ddot{H}_0^{\mu^{(v)}}$,

$$\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)},p}(x) \leq \sup_{x \in \mathcal{X}} T_{\mu^{(v)},p}(x) + \sup_{x \in \mathcal{X}} \frac{|m^{(v)}(x; \bar{\theta}) - m^{(v)}(x; \tilde{\theta})|}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}}.$$

Then,

$$\begin{aligned} \mathbb{P}\left[\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)},p}(x) > \mathfrak{c}_{\mu^{(v)}}\right] &\leq \mathbb{P}\left[\sup_{x \in \mathcal{X}} T_{\mu^{(v)},p}(x) > \mathfrak{c}_{\mu^{(v)}} - \sup_{x \in \mathcal{X}} \frac{|m^{(v)}(x; \bar{\theta}) - m^{(v)}(x; \tilde{\theta})|}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}}\right] \\ &\leq \mathbb{P}\left[\sup_{x \in \mathcal{X}} \bar{Z}_{\mu^{(v)},p}(x) > \mathfrak{c}_{\mu^{(v)}} - \xi_{1,n}/a_n\right] + o(1) \\ &\leq \mathbb{P}\left[\sup_{x \in \mathcal{X}} \bar{Z}_{\mu^{(v)},p}(x) > c^0(1 - \alpha - \xi_{3,n}) - (\xi_{1,n} + \xi_{2,n})/a_n\right] + o(1) \\ &\leq \mathbb{P}\left[\sup_{x \in \mathcal{X}} \bar{Z}_{\mu^{(v)},p}(x) > c^0(1 - \alpha - \xi_{3,n})\right] + o(1) \\ &= \alpha + o(1) \end{aligned}$$

where $c^0(1 - \alpha - \xi_{3,n})$ denotes the $(1 - \alpha - \xi_{3,n})$ -quantile of $\sup_{x \in \mathcal{X}} \bar{Z}_{\mu^{(v)},p}(x)$ conditional on $\mathcal{F}_{XW\Delta}$ (the σ -field generated by $\mathbf{X}$, $\mathbf{W}$ and $\hat{\Delta}$), the second line holds by Theorem SA-3.5, the third by Lemma A.1 of Belloni et al. (2015), the fourth by Anti-Concentration Inequality in Chernozhukov et al. (2014).

On the other hand, under $\ddot{H}_A^{\mu^{(v)}}$,

$$\begin{aligned} \mathbb{P}\left[\sup_{x \in \mathcal{X}} \ddot{T}_{\mu^{(v)},p}(x) > \mathfrak{c}_{\mu^{(v)}}\right] &= \mathbb{P}\left[\sup_{x \in \mathcal{X}} \left(T_{\mu^{(v)},p}(x) + \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\theta})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} - \mathfrak{c}_{\mu^{(v)}}\right) > 0\right] \\ &\geq \mathbb{P}\left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| < \sup_{x \in \mathcal{X}} \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\theta})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} - \mathfrak{c}_{\mu^{(v)}}, \right. \\ &\quad \left. \sup_{x \in \mathcal{X}} \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\theta})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} > \mathfrak{c}_{\mu^{(v)}}\right] \\ &\geq \mathbb{P}\left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| < \sup_{x \in \mathcal{X}} \frac{\mu_0^{(v)}(x) - m^{(v)}(x; \tilde{\theta})}{\sqrt{\hat{\Omega}_{\mu^{(v)}}(x)/n}} - \mathfrak{c}_{\mu^{(v)}}\right] - o(1) \end{aligned}$$

$$\begin{aligned}
&\geq \mathbb{P}\left[\sup_{x \in \mathcal{X}} |T_{\mu^{(v)},p}(x)| < \sqrt{\log J} A_n\right] - o(1) \\
&\geq \mathbb{P}\left[\sup_{x \in \mathcal{X}} |\bar{Z}_{\mu^{(v)},p}(x)| < \sqrt{\log J} A_n - \xi_{1,n}/a_n\right] - o(1) \\
&\geq 1 - o(1)
\end{aligned}$$

where the third line holds by Lemma SA-3.2, Theorem SA-3.2, Lemma A.1 of Belloni et al. (2015), the fixed-separation condition, the assumption that $\sup_{x \in \mathcal{X}} |m^{(v)}(x; \tilde{\boldsymbol{\theta}}) - m^{(v)}(x; \bar{\boldsymbol{\theta}})| = o_{\mathbb{P}}(1)$ and $J^v \sqrt{J \log J/n} = o(1)$, the fourth by definition of A_n , and the fifth by Theorem SA-3.5, and the last by Proposition A.2.7 in van der Vaart and Wellner (1996).

□

SA-5.17 Proof of Lemma SA-3.6

Proof. Step 1: Reparametrization and key facts. By construction,

$$\mathbb{E}_n[\boldsymbol{\varkappa}_i \widehat{\mathbf{b}}_i \tilde{\mathbf{w}}_i'] = 0. \quad (\text{SA-5.5})$$

The one-to-one reparametrization

$$\boldsymbol{\theta} = \boldsymbol{\beta} + \bar{\mathbf{Q}}^{-1} \mathbb{E}_n[\widehat{\mathbf{b}}_i \mathbf{w}_i' \boldsymbol{\varkappa}_i] \boldsymbol{\gamma}$$

satisfies

$$\widehat{\mathbf{b}}_i' \boldsymbol{\beta} + \mathbf{w}_i' \boldsymbol{\gamma} = \widehat{\mathbf{b}}_i' \boldsymbol{\theta} + \tilde{\mathbf{w}}_i' \boldsymbol{\gamma}.$$

Thus, if $\widehat{\boldsymbol{\theta}}$ denotes the transform of $(\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$, then $(\widehat{\boldsymbol{\theta}}, \widehat{\boldsymbol{\gamma}})$ minimizes

$$Q_n^*(\boldsymbol{\theta}, \boldsymbol{\gamma}) = \mathbb{E}_n[\rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\theta} + \tilde{\mathbf{w}}_i' \boldsymbol{\gamma}))].$$

By Lemma SA-5.5,

$$\sup_{x \in \mathcal{X}} |\widehat{\tau}_{0,0}(x)| \lesssim_{\mathbb{P}} J^{-p-1}.$$

A coordinatewise approximation error of order c_J implies $\mathfrak{a}_{\pi,J} \lesssim \sqrt{d_\gamma} c_J$, while $\mathfrak{a}_{\pi,J} \lesssim c_J$ requires a uniform directional, or vector-valued, smoothness condition.

The weights $\varkappa_i$ are bounded and bounded away from zero. Hence, the sparsity of the binscatter basis and the bounded absolute row-sum norm of the inverse weighted Gram matrix give, coordinatewise,

$$\max_{1 \leq i \leq n} |[\boldsymbol{\pi}_w(x_i)]_k| \lesssim_{\mathbb{P}} \max_{1 \leq i \leq n} |w_{i,k}| \lesssim 1, \quad 1 \leq k \leq d_\gamma.$$

Compact support and $\|\tilde{\mathbf{w}}_i\| \leq \sqrt{d_\gamma} \|\tilde{\mathbf{w}}_i\|_\infty$ then give

$$\max_{1 \leq i \leq n} \|\tilde{\mathbf{w}}_i\|_\infty \lesssim_{\mathbb{P}} 1, \quad \max_{1 \leq i \leq n} \|\tilde{\mathbf{w}}_i\| \lesssim_{\mathbb{P}} \sqrt{d_\gamma}. \quad (\text{SA-5.6})$$

By condition (i), w.p.a. 1,

$$0 < \underline{\lambda}_{w,n} \leq \lambda_{\min}(\hat{\mathbf{G}}_w) \leq \lambda_{\max}(\hat{\mathbf{G}}_w) \leq \bar{\lambda}_{w,n}, \quad \hat{\mathbf{G}}_w = \mathbb{E}_n[\varkappa_i \tilde{\mathbf{w}}_i \tilde{\mathbf{w}}_i'],$$

so that

$$\mathbb{E}_n[\varkappa_i \|\tilde{\mathbf{w}}_i\|^2] = \text{trace}(\hat{\mathbf{G}}_w) \leq d_\gamma \bar{\lambda}_{w,n}.$$

Condition (ii) gives

$$\left\| \mathbb{E}_n[\varkappa_i \tilde{\mathbf{w}}_i \hat{r}_{0,0}(x_i)] - \mathbb{E}_{\hat{\Delta}}[\varkappa_i \{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\} \hat{r}_{0,0}(x_i)] \right\| \lesssim_{\mathbb{P}} J^{-p-1} \sqrt{\bar{\lambda}_{w,n} d_\gamma / n}.$$

By iterated expectations, Cauchy–Schwarz, the definition of $\mathbf{a}_{\pi,J}$, and Euclidean duality,

$$\begin{aligned} & \left\| \mathbb{E}_{\hat{\Delta}}[\varkappa_i \{\mathbf{w}_i - \boldsymbol{\pi}_{w,J}(x_i)\} \hat{r}_{0,0}(x_i)] \right\| \\ &= \sup_{\|\boldsymbol{\alpha}\|=1} \left| \mathbb{E}_{\hat{\Delta}}[\mathbb{E}[\varkappa_i | x_i] \boldsymbol{\alpha}' \{\boldsymbol{\pi}_{w,0}(x_i) - \boldsymbol{\pi}_{w,J}(x_i)\} \hat{r}_{0,0}(x_i)] \right| \lesssim_{\mathbb{P}} J^{-p-1} \mathbf{a}_{\pi,J}. \end{aligned}$$

Consequently,

$$\left\| \mathbb{E}_n[\varkappa_i \tilde{\mathbf{w}}_i \hat{r}_{0,0}(x_i)] \right\| \lesssim_{\mathbb{P}} J^{-p-1} \mathbf{a}_{\pi,J} + J^{-p-1} \sqrt{\bar{\lambda}_{w,n} d_\gamma / n}. \quad (\text{SA-5.7})$$

Since $J^{-p-1} = o(1)$, the second term in (SA-5.7) is $o_{\mathbb{P}}(\sqrt{\bar{\lambda}_{w,n} d_\gamma / n})$.

Equation (SA-5.5) implies

$$\mathbb{E}_n[\varkappa_i \{u_i(\boldsymbol{\theta}) + \tilde{\mathbf{w}}_i' \mathbf{h}\}^2] = \mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}' \hat{\mathbf{G}}_w \mathbf{h}. \quad (\text{SA-5.8})$$

It remains to establish a preliminary rate for the full index and to prove $\|\widehat{\gamma} - \gamma_0\| = O_{\mathbb{P}}(\mathfrak{r}_{\gamma,n})$.

Step 2: Preliminary rate. Let $\mathcal{F}_{XW\Delta}$ be the σ -field generated by $\{(x_i, \mathbf{w}_i)\}_{i=1}^n$ and $\widehat{\Delta}$, and set

$$\mathbb{G}_n[f_i] = \sqrt{n} \mathbb{E}_n[f_i - \mathbb{E}[f_i \mid \mathcal{F}_{XW\Delta}]].$$

For this step, write

$$q_i = u_i(\boldsymbol{\theta}) + \widetilde{\mathbf{w}}_i' \mathbf{h}.$$

Since $\widehat{\mathbf{b}}_i' \boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}_i' \gamma_0 = \varpi_i - \widehat{r}_{0,0}(x_i)$, absolute continuity of the loss gives

$$\begin{aligned} & \mathbb{E} \left[\rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\theta} + \widetilde{\mathbf{w}}_i'(\gamma_0 + \mathbf{h}))) - \rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}_i' \gamma_0)) \mid \mathcal{F}_{XW\Delta} \right] \\ &= \int_0^{q_i} \Psi(x_i, \mathbf{w}_i; \eta(\varpi_i - \widehat{r}_{0,0}(x_i) + t)) \eta^{(1)}(\varpi_i - \widehat{r}_{0,0}(x_i) + t) dt. \end{aligned}$$

Because $\Psi(x_i, \mathbf{w}_i; \eta_i) = 0$, the derivative of the integrand with respect to t , evaluated at $\widehat{r}_{0,0}(x_i) = t = 0$, is $\varkappa_i$. Assumption [SA-SM](#) therefore gives, uniformly on a sufficiently small neighborhood,

$$\begin{aligned} & \Psi(x_i, \mathbf{w}_i; \eta(\varpi_i - \widehat{r}_{0,0}(x_i) + t)) \eta^{(1)}(\varpi_i - \widehat{r}_{0,0}(x_i) + t) \\ &= \varkappa_i \{t - \widehat{r}_{0,0}(x_i)\} + R_{i,t}, \quad |R_{i,t}| \lesssim \{|t| + |\widehat{r}_{0,0}(x_i)|\}^2. \end{aligned}$$

Integrating and averaging yields

$$\begin{aligned} & \mathbb{E}_n \left[\mathbb{E} \left[\rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\theta} + \widetilde{\mathbf{w}}_i'(\gamma_0 + \mathbf{h}))) - \rho(y_i; \eta(\widehat{\mathbf{b}}_i' \boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}_i' \gamma_0)) \mid \mathcal{F}_{XW\Delta} \right] \right] \\ &= \frac{1}{2} \mathbb{E}_n[\varkappa_i q_i^2] - \mathbb{E}_n[\varkappa_i \widehat{r}_{0,0}(x_i) q_i] + \mathbb{E}_n \left[\int_0^{q_i} R_{i,t} dt \right]. \end{aligned}$$

By Cauchy–Schwarz and Lemma [SA-5.5](#),

$$|\mathbb{E}_n[\varkappa_i \widehat{r}_{0,0}(x_i) q_i]| \lesssim_{\mathbb{P}} J^{-p-1} \mathbb{E}_n[\varkappa_i q_i^2]^{1/2}.$$

Moreover,

$$\left| \mathbb{E}_n \left[\int_0^{q_i} R_{i,t} dt \right] \right| \lesssim_{\mathbb{P}} \{J^{-p-1} + \max_i |q_i|\} \mathbb{E}_n[q_i^2] + J^{-2p-2} \mathbb{E}_n[|q_i|].$$

Since $\varkappa_i$ is bounded and bounded away from zero, the first term on the right is absorbed by the quadratic term when the neighborhood is sufficiently small, while the second is bounded by

$CJ^{-p-1}\mathbb{E}_n[\varkappa_i q_i^2]^{1/2}$. Finally, (SA-5.8) gives

$$\mathbb{E}_n[\varkappa_i q_i^2] = \mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}'\widehat{\mathbf{G}}_w\mathbf{h}.$$

Combining these bounds,

$$\begin{aligned} & \mathbb{E}_n \left[\mathbb{E} \left[\rho(y_i; \eta(\widehat{\mathbf{b}}'_i \boldsymbol{\theta} + \widetilde{\mathbf{w}}'_i(\gamma_0 + \mathbf{h}))) - \rho(y_i; \eta(\widehat{\mathbf{b}}'_i \boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}'_i \gamma_0)) \mid \mathcal{F}_{XW\Delta} \right] \right] \\ & \geq c \left\{ \mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}'\widehat{\mathbf{G}}_w\mathbf{h} \right\} - CJ^{-p-1} \left\{ \mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}'\widehat{\mathbf{G}}_w\mathbf{h} \right\}^{1/2}. \end{aligned}$$

Denote the criterion-increment class in the preceding display by $\mathcal{G}$. Since $d_\gamma = o(J)$, it is indexed by $K_{p,s} + d_\gamma \asymp J$ parameters. On the ball

$$\mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}'\widehat{\mathbf{G}}_w\mathbf{h} \leq M^2 \mathfrak{r}_{\boldsymbol{\theta},n}^2,$$

its conditional L_2 radius is $\bar{\sigma}_G \lesssim M \mathfrak{r}_{\boldsymbol{\theta},n}$, and an envelope satisfies

$$\bar{G}_i \lesssim M \mathfrak{r}_{\boldsymbol{\theta},n} \left(\sqrt{J} + \sqrt{\frac{d_\gamma}{\underline{\lambda}_{w,n}}} \right) (1 + |\psi(y_i, \eta_i)|).$$

With $\delta_G = \bar{\sigma}_G / \|\bar{G}\|_{L_2(\mathbb{P})}$, the localized entropy bound is

$$\sup_{\mathbb{Q}} \log N(\mathcal{G}, L_2(\mathbb{Q}), \varepsilon \|\bar{G}\|_{\mathbb{Q},2}) \lesssim J \log(C\delta_G/\varepsilon), \quad 0 < \varepsilon \leq \delta_G.$$

Assumption SA-SM(iii) and condition (iv) give

$$\frac{J}{n} \left\| \max_{1 \leq i \leq n} \bar{G}_i \right\|_{L_2(\mathbb{P})} = o_{\mathbb{P}}(\mathfrak{r}_{\boldsymbol{\theta},n}^2).$$

Hence, Lemma SA-5.6 yields

$$\begin{aligned} & \sup_{\mathbb{E}_n[\varkappa_i u_i(\boldsymbol{\theta})^2] + \mathbf{h}'\widehat{\mathbf{G}}_w\mathbf{h} \leq M^2 \mathfrak{r}_{\boldsymbol{\theta},n}^2} \left| \frac{1}{\sqrt{n}} \mathbb{G}_n \left[\rho(y_i; \eta(\widehat{\mathbf{b}}'_i \boldsymbol{\theta} + \widetilde{\mathbf{w}}'_i(\gamma_0 + \mathbf{h}))) - \rho(y_i; \eta(\widehat{\mathbf{b}}'_i \boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}'_i \gamma_0)) \right] \right| \\ & \lesssim_{\mathbb{P}} M \mathfrak{r}_{\boldsymbol{\theta},n} \sqrt{J/n} + o_{\mathbb{P}}(\mathfrak{r}_{\boldsymbol{\theta},n}^2). \end{aligned}$$

Since $\mathfrak{r}_{\theta,n} = \sqrt{J/n} + J^{-p-1}$, convexity and the preceding two displays imply

$$\mathbb{E}_n[\mathfrak{x}_i u_i(\hat{\boldsymbol{\theta}})^2] + (\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0)' \hat{\mathbf{G}}_w (\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0) = O_{\mathbb{P}}(\mathfrak{r}_{\theta,n}^2). \quad (\text{SA-5.9})$$

In particular, $\mathbb{E}_n[u_i(\hat{\boldsymbol{\theta}})^2]^{1/2} = O_{\mathbb{P}}(\mathfrak{r}_{\theta,n})$. The local support and normalization of $\hat{\mathbf{b}}_i$, together with the Gram bound, also give

$$\max_{1 \leq i \leq n} |u_i(\hat{\boldsymbol{\theta}})| \lesssim_{\mathbb{P}} \sqrt{J} \mathfrak{r}_{\theta,n} = o_{\mathbb{P}}(1),$$

where the last equality follows from condition (iii). Consequently, for every $\varepsilon > 0$ there exists a sufficiently large fixed $C_T = C_T(\varepsilon) < \infty$ such that

$$\mathcal{T}_n(C_T) := \left\{ \boldsymbol{\theta} : \max_i |u_i(\boldsymbol{\theta})| \leq C_T \sqrt{J} \mathfrak{r}_{\theta,n}, \quad \mathbb{E}_n[u_i(\boldsymbol{\theta})^2]^{1/2} \leq C_T \mathfrak{r}_{\theta,n} \right\}$$

and

$$\liminf_{n \rightarrow \infty} \mathbb{P}[\hat{\boldsymbol{\theta}} \in \mathcal{T}_n(C_T)] \geq 1 - \varepsilon.$$

Step 3: Conditional mean. For $\boldsymbol{\theta} \in \mathcal{T}_n(C_T)$ and $\mathbf{h} = \boldsymbol{\gamma} - \boldsymbol{\gamma}_0$, define

$$\delta_i(\boldsymbol{\theta}, \mathbf{h}) = \rho(y_i; \eta(\hat{\mathbf{b}}'_i \boldsymbol{\theta} + \tilde{\mathbf{w}}'_i(\boldsymbol{\gamma}_0 + \mathbf{h}))) - \rho(y_i; \eta(\hat{\mathbf{b}}'_i \boldsymbol{\theta} + \tilde{\mathbf{w}}'_i \boldsymbol{\gamma}_0)).$$

Because $\hat{\mathbf{b}}'_i \boldsymbol{\theta}_0 + \tilde{\mathbf{w}}'_i \boldsymbol{\gamma}_0 = \varpi_i - \hat{r}_{0,0}(x_i)$,

$$\begin{aligned} \mathbb{E}[\delta_i(\boldsymbol{\theta}, \mathbf{h}) \mid \mathcal{F}_{XW\Delta}] &= \int_0^{\tilde{\mathbf{w}}'_i \mathbf{h}} \Psi(x_i, \mathbf{w}_i; \eta(\varpi_i - \hat{r}_{0,0}(x_i) + u_i(\boldsymbol{\theta}) + t)) \\ &\quad \times \eta^{(1)}(\varpi_i - \hat{r}_{0,0}(x_i) + u_i(\boldsymbol{\theta}) + t) dt. \end{aligned}$$

Expanding the integrand around ϖ_i gives the leading terms

$$\frac{1}{2} \mathbf{h}' \hat{\mathbf{G}}_w \mathbf{h} - \mathbf{h}' \mathbb{E}_n[\mathfrak{x}_i \tilde{\mathbf{w}}_i \hat{r}_{0,0}(x_i)] + \mathbb{E}_n[\mathfrak{x}_i u_i(\boldsymbol{\theta}) \tilde{\mathbf{w}}'_i \mathbf{h}].$$

The last term is zero by (SA-5.5). By condition (iii), whose product-moment bound follows under the primitive condition in Remark SA-3.12, uniformly over $\boldsymbol{\theta} \in \mathcal{T}_n(C_T)$ and the local values of $\mathbf{h}$

used below, the remaining terms are bounded by

$$C\sqrt{\bar{\lambda}_{w,n}}\|\mathbf{h}\|\mathfrak{r}_{\theta,n}^2 + o_{\mathbb{P}}(\bar{\lambda}_{w,n}\|\mathbf{h}\|^2).$$

Therefore, by (SA-5.7),

$$\begin{aligned} \mathbb{E}_n[\mathbb{E}[\delta_i(\boldsymbol{\theta}, \mathbf{h}) \mid \mathcal{F}_{XW\Delta}]] &\geq \frac{\bar{\lambda}_{w,n}}{4}\|\mathbf{h}\|^2 \\ &\quad - C\|\mathbf{h}\| \left(J^{-p-1}\mathfrak{a}_{\pi,J} + J^{-p-1}\sqrt{\bar{\lambda}_{w,n}d_{\gamma}/n} + \sqrt{\bar{\lambda}_{w,n}}\mathfrak{r}_{\theta,n}^2 \right). \end{aligned} \quad (\text{SA-5.10})$$

Step 4: Centered empirical process. Decompose the objective increment as

$$\delta_i(\boldsymbol{\theta}, \mathbf{h}) = \delta_i^0(\mathbf{h}) + \delta_i^u(\boldsymbol{\theta}, \mathbf{h}),$$

where

$$\begin{aligned} \delta_i^0(\mathbf{h}) &= \rho(y_i; \eta(\widehat{\mathbf{b}}_i'\boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}_i'(\gamma_0 + \mathbf{h}))) - \rho(y_i; \eta(\widehat{\mathbf{b}}_i'\boldsymbol{\theta}_0 + \widetilde{\mathbf{w}}_i'\gamma_0)), \\ \delta_i^u(\boldsymbol{\theta}, \mathbf{h}) &= \delta_i(\boldsymbol{\theta}, \mathbf{h}) - \delta_i^0(\mathbf{h}). \end{aligned}$$

Denote the first class by $\mathcal{G}_0$ and the second class by $\mathcal{G}_u$.

For $\mathcal{G}_0$, it is indexed only by $\mathbf{h} \in \mathbb{R}^{d_{\gamma}}$ and has envelope

$$\bar{F}_0 \lesssim M\mathfrak{r}_{\gamma,n}\|\widetilde{\mathbf{w}}_i\|(1 + |\psi(y_i, \eta_i)|).$$

For every $\|\boldsymbol{\alpha}\| = 1$, Assumption SA-SM and the eigenvalue bound give

$$\mathbb{E}_n[\mathbb{E}[(\boldsymbol{\alpha}'\widetilde{\mathbf{w}}_i)^2(1 + |\psi(y_i, \eta_i)|)^2 \mid \mathcal{F}_{XW\Delta}]] \lesssim \boldsymbol{\alpha}'\widehat{\mathbf{G}}_w\boldsymbol{\alpha} \leq \bar{\lambda}_{w,n}.$$

Thus, a conditional L_2 radius for $\mathcal{G}_0$ is $\bar{\sigma}_0 \lesssim M\mathfrak{r}_{\gamma,n}\sqrt{\bar{\lambda}_{w,n}}$. Moreover, the L_2 norm of the pointwise envelope satisfies

$$\begin{aligned} \|\bar{F}_0\|_{L_2(\mathbb{P})}^2 &\lesssim M^2\mathfrak{r}_{\gamma,n}^2\mathbb{E}_n[\mathbb{E}[\|\widetilde{\mathbf{w}}_i\|^2(1 + |\psi(y_i, \eta_i)|)^2 \mid \mathcal{F}_{XW\Delta}]] \\ &\lesssim M^2\mathfrak{r}_{\gamma,n}^2\text{trace}(\widehat{\mathbf{G}}_w) \leq M^2\mathfrak{r}_{\gamma,n}^2d_{\gamma}\bar{\lambda}_{w,n}. \end{aligned}$$

Let $\delta_0 = \bar{\sigma}_0 / \|\bar{F}_0\|_{L_2(\mathbb{P})}$. The localized entropy bound is

$$\sup_{\mathbb{Q}} \log N(\mathcal{G}_0, L_2(\mathbb{Q}), \varepsilon \|\bar{F}_0\|_{\mathbb{Q}, 2}) \lesssim d_\gamma \log(C\delta_0/\varepsilon), \quad 0 < \varepsilon \leq \delta_0,$$

and hence $J(\delta_0, \mathcal{G}_0, \bar{F}_0) \lesssim \delta_0 \sqrt{d_\gamma}$. Lemma SA-5.6 gives

$$\sup_{\|\mathbf{h}\| \leq M\mathbf{r}_{\gamma,n}} \left| \frac{1}{\sqrt{n}} \mathbb{G}_n[\delta_i^0(\mathbf{h})] \right| \lesssim_{\mathbb{P}} M\mathbf{r}_{\gamma,n} \sqrt{\bar{\lambda}_{w,n} d_\gamma / n} + \frac{d_\gamma}{n} \left\| \max_{1 \leq i \leq n} \bar{F}_0(x_i, \mathbf{w}_i, y_i) \right\|_{L_2(\mathbb{P})}. \quad (\text{SA-5.11})$$

Assumption SA-SM(iii) implies $\|\max_{1 \leq i \leq n} (1 + |\psi(y_i, \eta_i)|)\|_{L_2(\mathbb{P})} \lesssim n^{1/\nu}$. Hence,

$$\frac{d_\gamma}{n} \left\| \max_{1 \leq i \leq n} \bar{F}_0(x_i, \mathbf{w}_i, y_i) \right\|_{L_2(\mathbb{P})} \lesssim_{\mathbb{P}} M\mathbf{r}_{\gamma,n} \frac{d_\gamma^{3/2}}{n^{1-1/\nu}}.$$

Condition (iv) gives

$$\frac{d_\gamma^{3/2}}{n^{1-1/\nu}} = o_{\mathbb{P}}(\underline{\lambda}_{w,n} \mathbf{r}_{\gamma,n}). \quad (\text{SA-5.12})$$

For the second class $\mathcal{G}_u$, decompose it according to the globally Lipschitz and step-function components of the piecewise-Lipschitz score. The globally Lipschitz component gives the pointwise bound

$$|u_i(\boldsymbol{\theta})| |\tilde{\mathbf{w}}'_i \mathbf{h}| (1 + |\psi(y_i, \eta_i)|) + |u_i(\boldsymbol{\theta})| |\tilde{\mathbf{w}}'_i \mathbf{h}| \{|u_i(\boldsymbol{\theta})| + |\tilde{\mathbf{w}}'_i \mathbf{h}|\}.$$

The localized entropy argument for this component gives a centered-process bound of order

$$M\mathbf{r}_{\gamma,n} \sqrt{\bar{\lambda}_{w,n} \mathbf{r}_{\theta,n} \sqrt{J/n}} + o_{\mathbb{P}}(\underline{\lambda}_{w,n} \mathbf{r}_{\gamma,n}^2).$$

For each step-function component, the crossing contribution is bounded by $\min\{|u_i(\boldsymbol{\theta})|, |\tilde{\mathbf{w}}'_i \mathbf{h}|\}$ and can be nonzero only if y_i lies in an interval of length proportional to $|u_i(\boldsymbol{\theta})| + |\tilde{\mathbf{w}}'_i \mathbf{h}|$ around the corresponding discontinuity threshold. Assumption SA-DGP(iii) therefore bounds its conditional second moment by a constant multiple of $|u_i(\boldsymbol{\theta})| |\tilde{\mathbf{w}}'_i \mathbf{h}| \min\{|u_i(\boldsymbol{\theta})|, |\tilde{\mathbf{w}}'_i \mathbf{h}|\}$. For the sum of the finitely many crossing components, an envelope may be chosen as

$$\bar{F}_u \lesssim M\mathbf{r}_{\gamma,n} \|\tilde{\mathbf{w}}_i\|.$$

The first bound in condition (iii), Cauchy–Schwarz, and the eigenvalue bound give

$$\mathbb{E}_n[|u_i(\boldsymbol{\theta})|(\tilde{\mathbf{w}}_i' \mathbf{h})^2] \lesssim_{\mathbb{P}} M^2 \bar{\lambda}_{w,n} \mathfrak{r}_{\gamma,n}^2 \mathfrak{r}_{\theta,n},$$

so we may take

$$\bar{\sigma}_u = \min \left\{ \|\bar{F}_u\|_{L_2(\mathbb{P})}, CM \mathfrak{r}_{\gamma,n} \sqrt{\bar{\lambda}_{w,n} \mathfrak{r}_{\theta,n}} \right\}, \quad \delta_u = \frac{\bar{\sigma}_u}{\|\bar{F}_u\|_{L_2(\mathbb{P})}},$$

as the conditional L_2 radius and its ratio to the envelope norm. The threshold-crossing components are VC-type with index proportional to $K_{p,s} + d_\gamma \asymp J$, and hence the conservative entropy bound is

$$\sup_{\mathbb{Q}} \log N(\mathcal{G}_u, L_2(\mathbb{Q}), \varepsilon \|\bar{F}_u\|_{\mathbb{Q},2}) \lesssim J \log(C/\varepsilon), \quad 0 < \varepsilon \leq \delta_u.$$

The envelope and eigenvalue bounds give $\delta_u \gtrsim_{\mathbb{P}} \min\{1, \sqrt{\mathfrak{r}_{\theta,n}/d_\gamma}\}$. Together with $d_\gamma = o(J)$ and $\mathfrak{r}_{\theta,n} \geq \sqrt{J/n}$, this yields $\log(C/\delta_u) \lesssim_{\mathbb{P}} \log n$. Therefore,

$$J(\delta_u, \mathcal{G}_u, \bar{F}_u) \lesssim_{\mathbb{P}} \delta_u \sqrt{J \log n}, \quad \frac{J \log n}{n} \left\| \max_{1 \leq i \leq n} \bar{F}_u(x_i, \mathbf{w}_i, y_i) \right\|_{L_2(\mathbb{P})} \lesssim_{\mathbb{P}} M \mathfrak{r}_{\gamma,n} \frac{J \sqrt{d_\gamma} \log n}{n}.$$

Applying Lemma SA-5.6 separately to the smooth and crossing components gives

$$\begin{aligned} & \sup_{\substack{\boldsymbol{\theta} \in \mathcal{T}_n(C_T) \\ \|\mathbf{h}\| \leq M \mathfrak{r}_{\gamma,n}}} \left| \frac{1}{\sqrt{n}} \mathbb{G}_n[\delta_i^u(\boldsymbol{\theta}, \mathbf{h})] \right| \\ & \lesssim_{\mathbb{P}} M \mathfrak{r}_{\gamma,n} \sqrt{\frac{\bar{\lambda}_{w,n} J \mathfrak{r}_{\theta,n} \log n}{n}} + M \mathfrak{r}_{\gamma,n} \frac{J \sqrt{d_\gamma} \log n}{n} + o_{\mathbb{P}}(\underline{\lambda}_{w,n} \mathfrak{r}_{\gamma,n}^2). \end{aligned} \tag{SA-5.13}$$

Step 5: Conclusion. Since $J^{-p-1} = o(1)$, its product with $\sqrt{\bar{\lambda}_{w,n} d_\gamma/n}$ is absorbed by the leading stochastic term. Combining (SA-5.10)–(SA-5.13), uniformly over $\boldsymbol{\theta} \in \mathcal{T}_n(C_T)$ and $\|\mathbf{h}\| = M \mathfrak{r}_{\gamma,n}$,

$$\begin{aligned} \mathbb{E}_n[\delta_i(\boldsymbol{\theta}, \mathbf{h})] & \geq \frac{\underline{\lambda}_{w,n}}{8} \|\mathbf{h}\|^2 \\ & \quad - C \|\mathbf{h}\| \left(\sqrt{\bar{\lambda}_{w,n} d_\gamma/n} + J^{-p-1} \mathbf{a}_{\pi,J} + \sqrt{\bar{\lambda}_{w,n} \mathfrak{r}_{\theta,n}^2} \right) \\ & \quad - C \|\mathbf{h}\| \left(\sqrt{\bar{\lambda}_{w,n} J \mathfrak{r}_{\theta,n} \log n/n} + J \sqrt{d_\gamma} \log n/n \right) + o_{\mathbb{P}}(\underline{\lambda}_{w,n} \|\mathbf{h}\|^2). \end{aligned}$$

Fix $\varepsilon > 0$ and choose $C_T = C_T(\varepsilon)$ as in Step 2. By the definition of $\mathbf{r}_{\gamma,n}$, there exists a sufficiently large fixed $M = M(\varepsilon, C_T)$ such that the probability that the right-hand side is positive uniformly over $\boldsymbol{\theta} \in \mathcal{T}_n(C_T)$ has limit inferior at least $1 - \varepsilon$. On this event, convexity gives

$$\arg \min_{\boldsymbol{\gamma}} Q_n^*(\boldsymbol{\theta}, \boldsymbol{\gamma}) \subseteq \{\boldsymbol{\gamma} : \|\boldsymbol{\gamma} - \boldsymbol{\gamma}_0\| \leq M\mathbf{r}_{\gamma,n}\}$$

uniformly over $\boldsymbol{\theta} \in \mathcal{T}_n(C_T)$. Combining this event with Step 2 and then letting ε be arbitrary gives

$$\|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}_0\| = O_{\mathbb{P}} \left(\frac{\sqrt{\bar{\lambda}_{w,n} d_{\gamma}/n} + J^{-p-1} \mathbf{a}_{\pi,J} + \sqrt{\bar{\lambda}_{w,n}} \mathbf{r}_{\theta,n}^2 + \sqrt{\bar{\lambda}_{w,n} J \mathbf{r}_{\theta,n} \log n/n} + J \sqrt{d_{\gamma} \log n/n}}{\underline{\lambda}_{w,n}} \right).$$

Consequently, the rate condition on $\hat{\boldsymbol{\gamma}}$ in Assumption [SA-HLE](#)(i) holds if there exists a deterministic sequence $\mathbf{r}_{\gamma}$ such that

$$\mathbf{r}_{\gamma,n} = O_{\mathbb{P}}(\mathbf{r}_{\gamma}), \quad \sqrt{d_{\gamma}} \mathbf{r}_{\gamma} = o\left(\sqrt{J/n} + J^{-p-1}\right).$$

□

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